



HVN - Airbus Fleet Review Meeting | This document contains Confidential and Proprietary information

HVN A320Fam Fleet Review Meeting 24 - 25 February 2025

AIRBUS

Agenda - 24 February 2025

Google Meet Link: meet.google.com/nqd-wins-vwt

France GMT+1	Vietnam GMT+7	Duration	Topics	Presenter *Dial-In
08:00h	14:00h	30 mins	Introduction HVN A321 Fleet Performance Summary	HVN Team Thum Kim Hoong Peter Ng
08:30h	14:30h	10 mins	Safety Beyond Standard (SBS)	Peter Ng
08:40h	14:40h	25 mins	ELAC/SEC/SFCC Fault	Marius OLSZEWSKI
09:05h	15:05h	25 mins	ECAM F/CTL SENSOR (SSTU)	Marius OLSZEWSKI
09:30h	15:30h	25 mins	Fuel Inerting System Fault	Tanja RAU*
09:55h	15:55h	25 mins	Air Wing Leak Due To ACM Gasket	Tanja RAU*
10:20h	16:20h	–	End of day 1	

Agenda - 25 February 2025

Google Meet Link: meet.google.com/nqd-wins-vwt

France GMT+1	Vietnam GMT+7	Duration	Topics	Presenter *Dial-In
08:00h	14:00h	45 mins	Hydraulic Leaks Due To Ageing	Marius OLSZEWSKI
08:45h	14:45h	25 mins	BSCU/ABCU NFF	Marius OLSZEWSKI Katia ROULLET*
09:10h	15:10h	25 mins	VHF, DME, ATC Antenna Erosion & Corrosion	Charles MERIC* Amelie SAMUEL*
09:35h	15:35h	20 mins	A320Fam CEO ATA 36 OI	Peter NG
09:55h	15:55h	20 mins	PW1100 NEO FAV Assembly Damage	Peter NG
10:15h	16:15h	20 mins	CIDS DIR Obsolescence	Nurul Aida ABDUL GHAFAR* Leslie BERTHOMIEU*
10:35h	16:35h	—	End of day 2	

Team Organisation

Fleet Efficiency and Engineering



Weiquan XU
Head of Fleet Efficiency & Engineering APAC



Regional Technical Data
Azri IBRAHIM
Manager



Scheduled Maintenance
Romizan Abang Hipni
Manager



In-Service Engineering
Marius OLSZEWSKI
HO In-Service Engineering



Customer Care Centre - Ops
Siti Hazirah OTHMAN
HO AMSB (interim)



Repair Solution & Supply
Engineering Operations
Elaine ELAGO



Working Party Resources
Siti Wahidah MAHAT
HO Working Party Resources



Major Repair Technical Operations
Siti Hazirah OTHMAN
HO Major Repair Technical Operations



Vincent DEBOVE
Fleet Management
Team Lead



Peter NG
Fleet Management
A320 Family



Edwin Cheah
Fleet Management
A220/A320/P2F Family



Esther TAN
Fleet Management
A330/A340



Vincent DEBOVE
Fleet Management A350



Fleet Efficiency

Engineering & maintenance

Asia Pacific In-Service Engineering Organization



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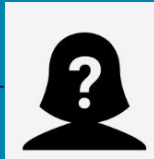
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Weiquan XU
Ho Fleet Efficiency &
Engineering APAC



Marius OLSZEWSKI
HO In-Service Engineering APAC

A300
A320
A330
A350
A380

In-Service Engineering Asia Pacific (AM1S)

- Established in Oct 2021
- Manage technical engineering topics at multi ATA level
- 7 FTE's covering all ATA's except Structure
- Target 2025 10 FTE's

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- 1 A320Fam - Fleet Overview**
- 2 Operational Reliability Trends**
- 3 Operational Reliability vs. Daily Utilization, Daily Cycle**
- 4 Top Ten Drivers - Operational Interruptions**

A320 Family – World Fleet overview

10778

Aircraft in Service (Dec 2024)

359.1M*

Flight Hours since EIS

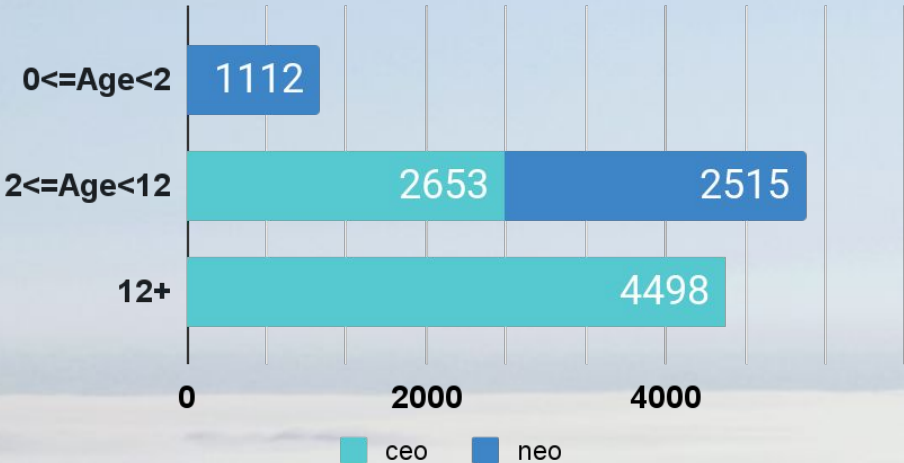
192.4M*

Flight Cycles since EIS

9.2*

Average Daily Utilisation

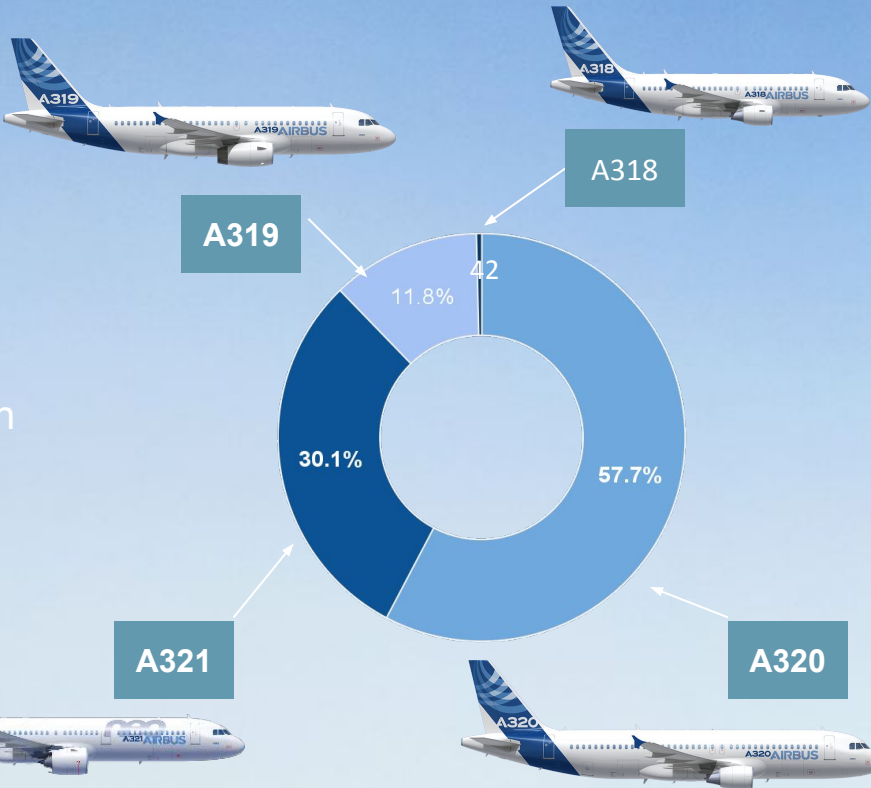
Average Fleet Age 10.8 years



Operators
393

neo Fleet
3627

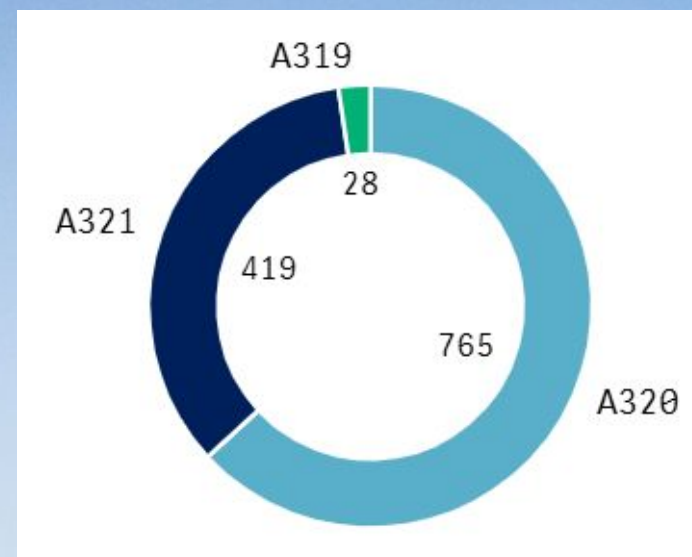
ceo Fleet
7151



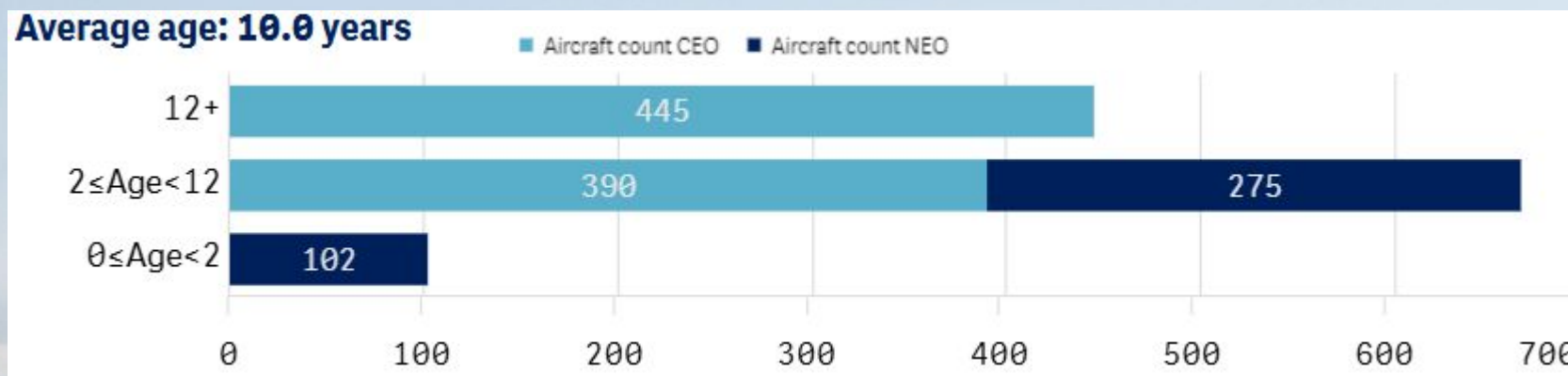
	A/C Nbr	Average Daily utilisation*	Average Flight duration*	Average Daily utilisation*
A320ceo	4744	8.7	1.8	9.2
A320neo	2044	9.7	2.0	
A321ceo	1763	9.4	2.3	10
A321neo (excl ACF/LR)	193	9.8	2.8	
A321neo ACF/LR	1368	10.8	2.6	

A320 Family – APAC Fleet overview

ACIF	CEO fleet	Avg annual FH*	Avg annual FC*
1212	835	2742	1607
Operators**	NEO fleet	FH since EIS*	FC since EIS*
52	377	44.15M	26.74M



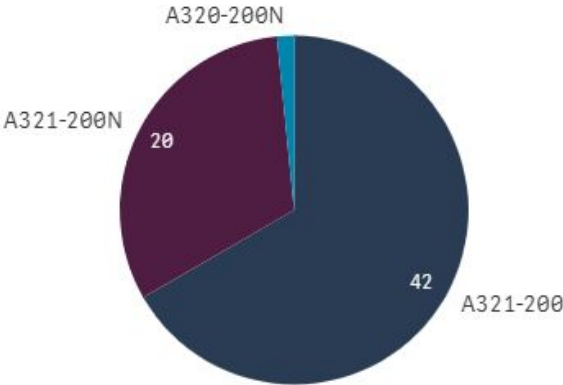
*Data as of 2024-11
 ** Lessors & MRO excluded



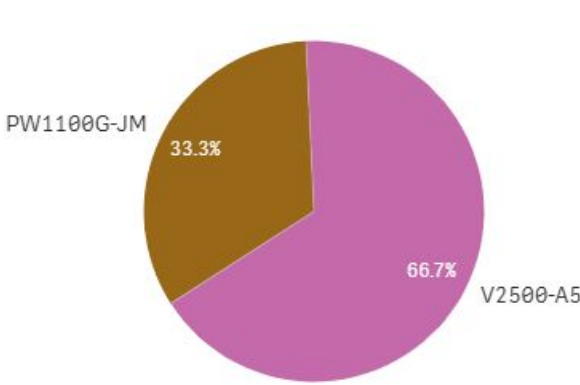
HVN Fleet Overview



Aircraft series

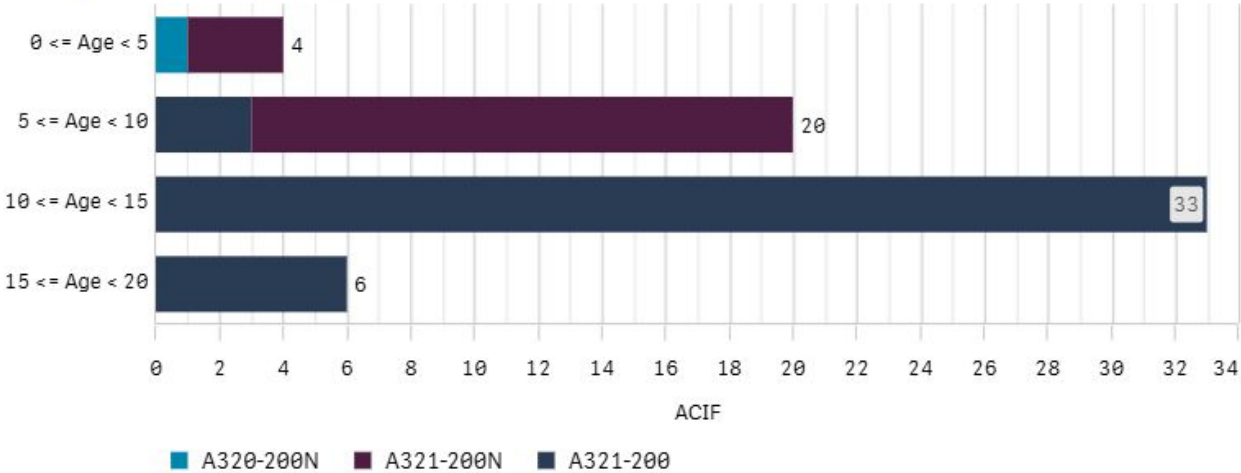


Engine series



*Data as of November 2024

Aircraft age distribution (years)



Definitions:
Rev FC: Revenue Flight Cycles
AFD: Average Flight Duration
DU: Daily Utilization

A/C	63	*10 AC in storage due to PW1100 engine AOG			
FC	100.3k	DC	5.58	Average	10.14
	Rev FC (last 12M)		FC/day (last 12M)	Fleet Age	Years
FH	151.7k	DU	8.44	AFD	1.51
	Rev FH (last 12M)		FH/day (last 12M)		FH/flight (2023-12 - 2024-11)

Revenue Flight-Hour and Revenue Take-Off Trends

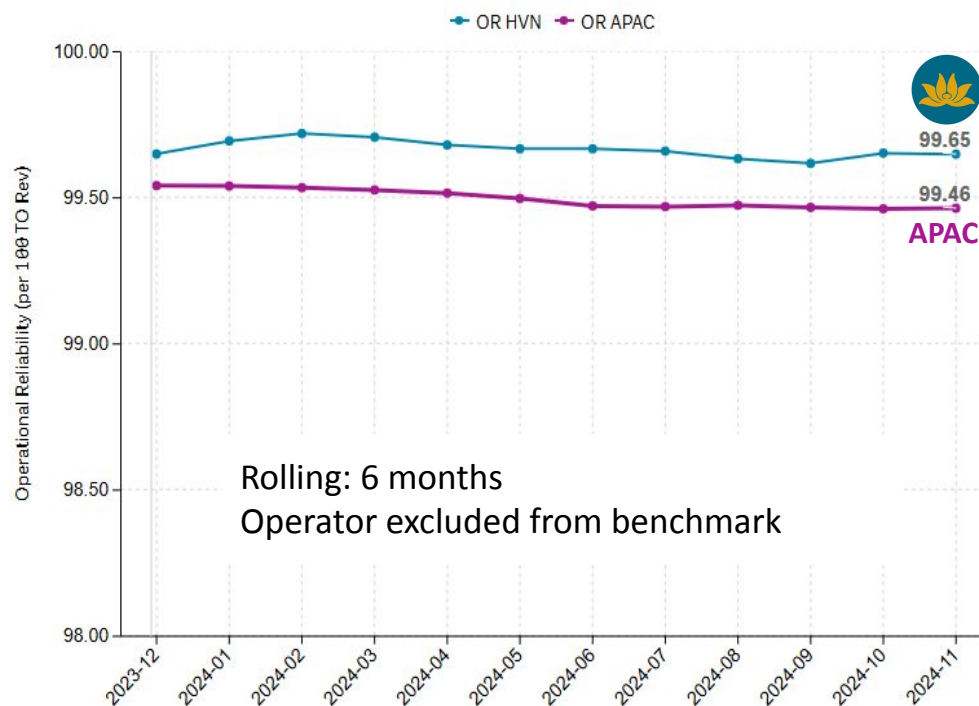
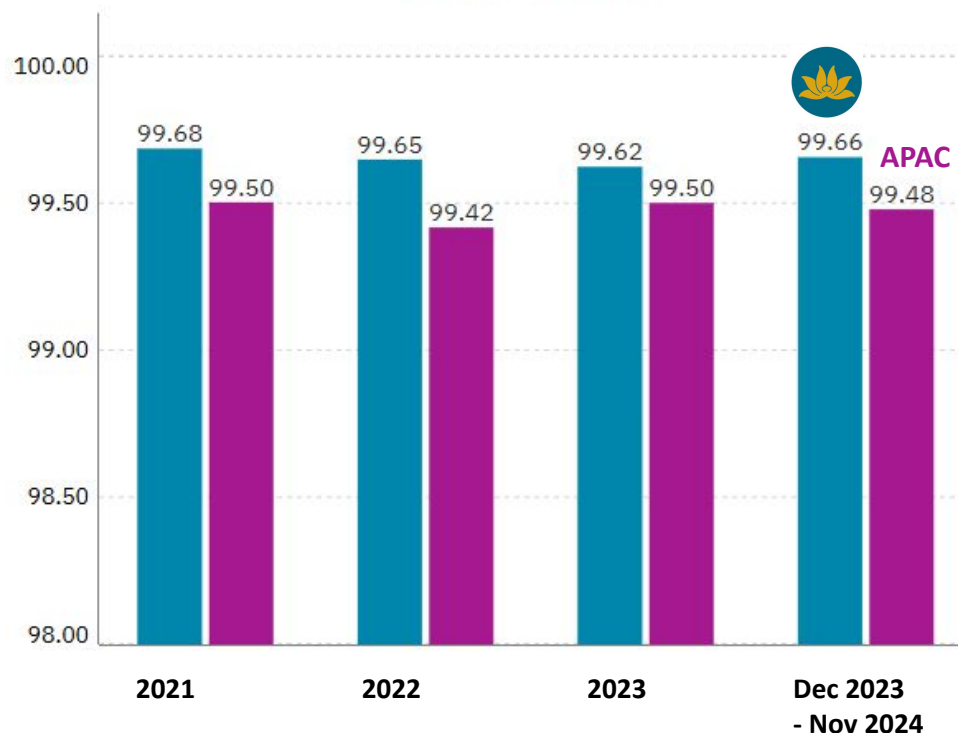


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- 4 Top Ten Drivers - Operational Interruptions**

HVN CEO Fleet Operational reliability trends

■ OR APAC ■ OR HVN



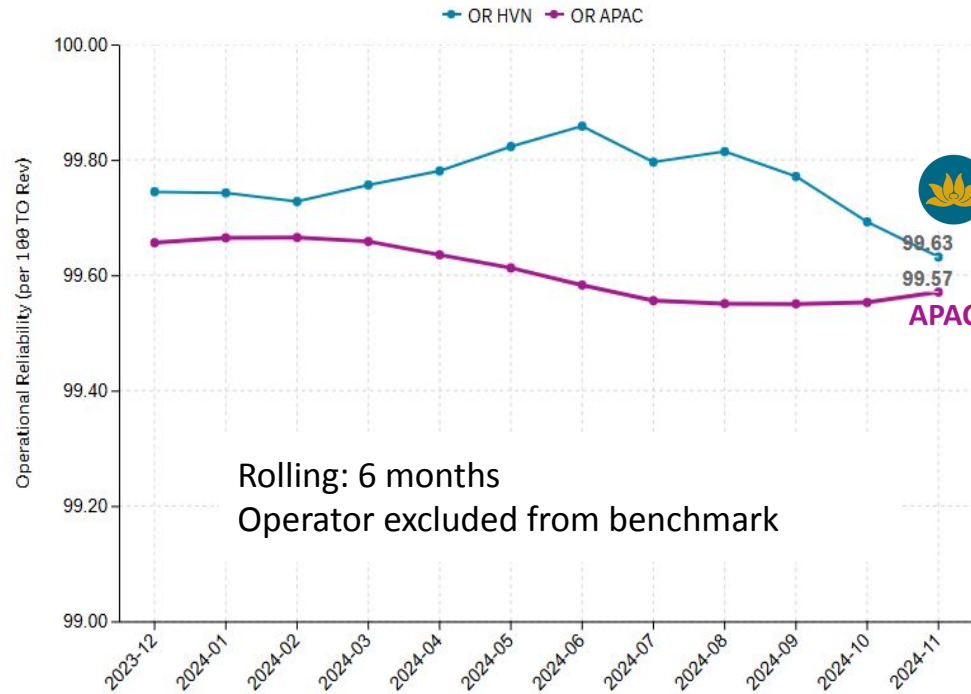
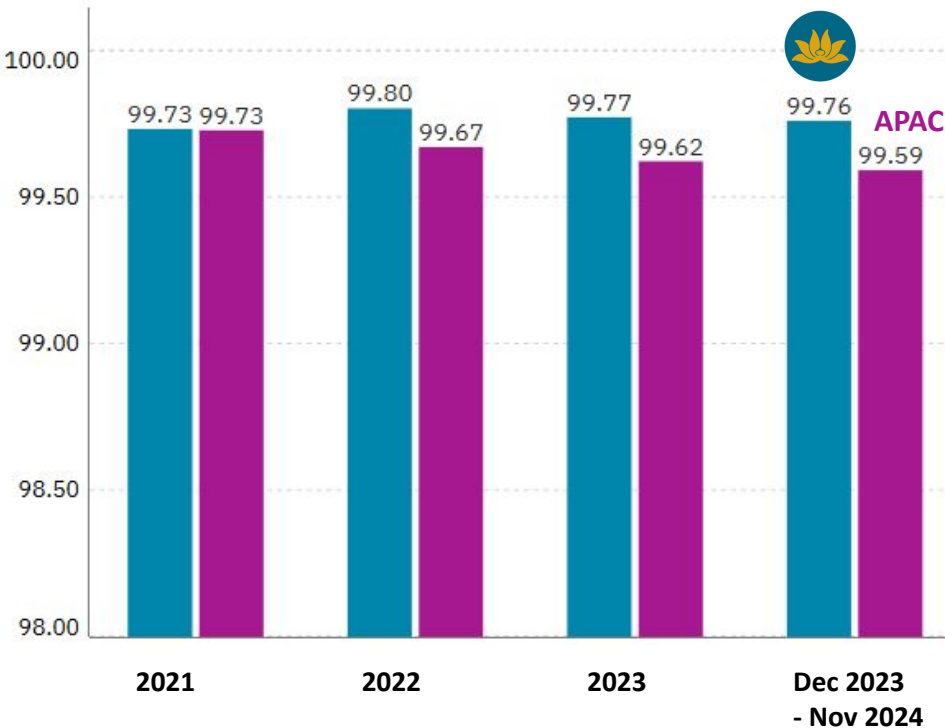
HVN CEO Fleet

Data as of Nov 2024

- CEO Fleet - Rolling 6 months OR 99.65 %
- Above APAC benchmark 99.46 %
- Stable performance

HVN NEO Fleet Operational reliability trends

■ OR APAC ■ OR HVN

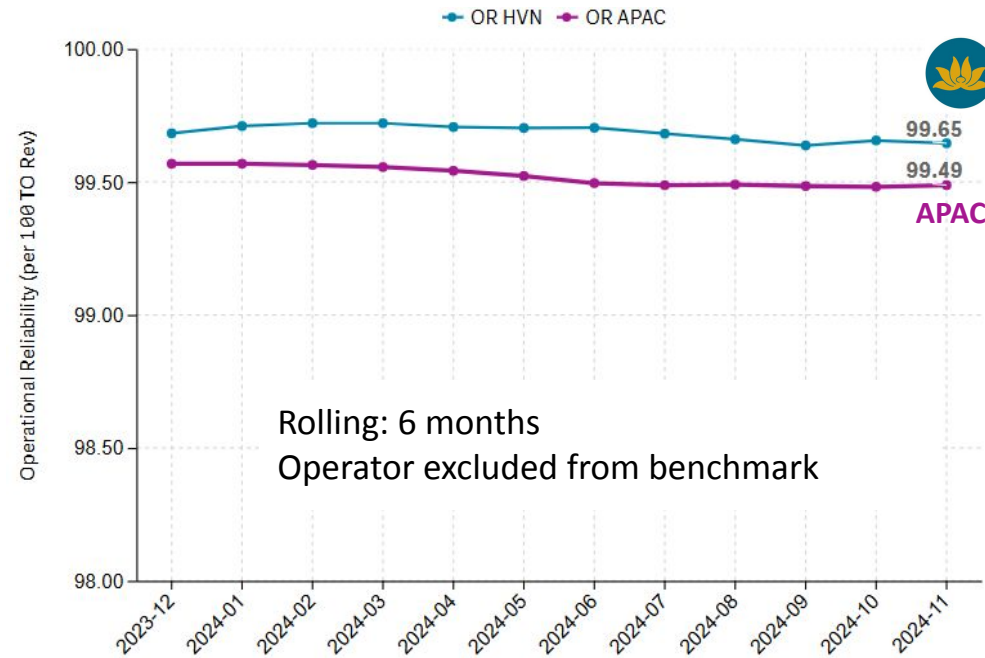
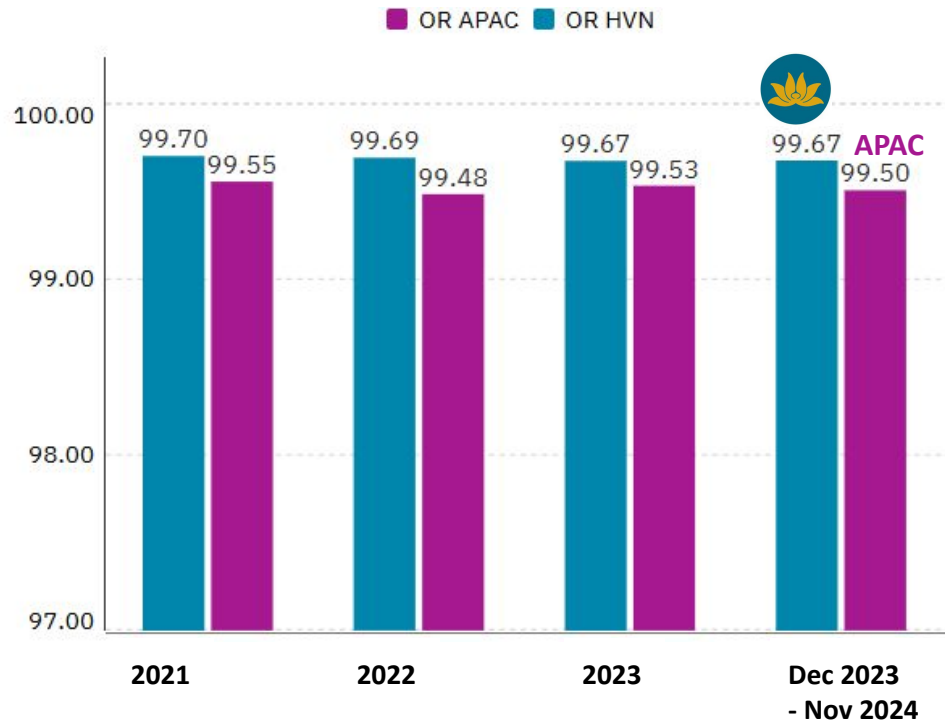


HVN NEO Fleet

Data as of Nov 2024

- NEO Fleet - Rolling 6 months OR 99.63 %
- Above APAC benchmark 99.57 %
- Performance slightly lower since high of June 2024 (NEO daily utilisation lower than CEO fleet by 0.8FC)

HVN Operational reliability trends



HVN Fleet

Data as of Nov 2024

- HVN Fleet - Rolling 6 months OR 99.65 %
- Above APAC benchmark 99.49 %
- Consistent performance last 4 years

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1 A320Fam -Fleet Overview

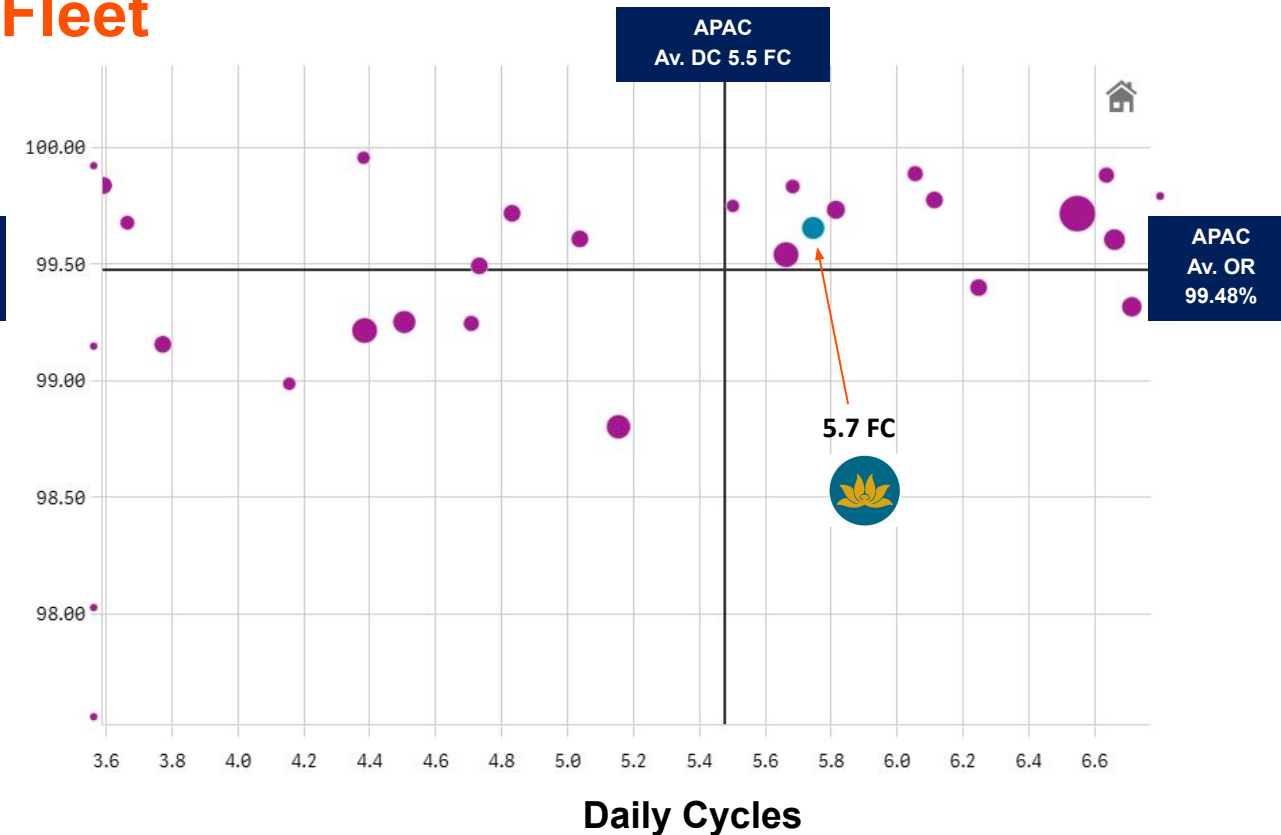
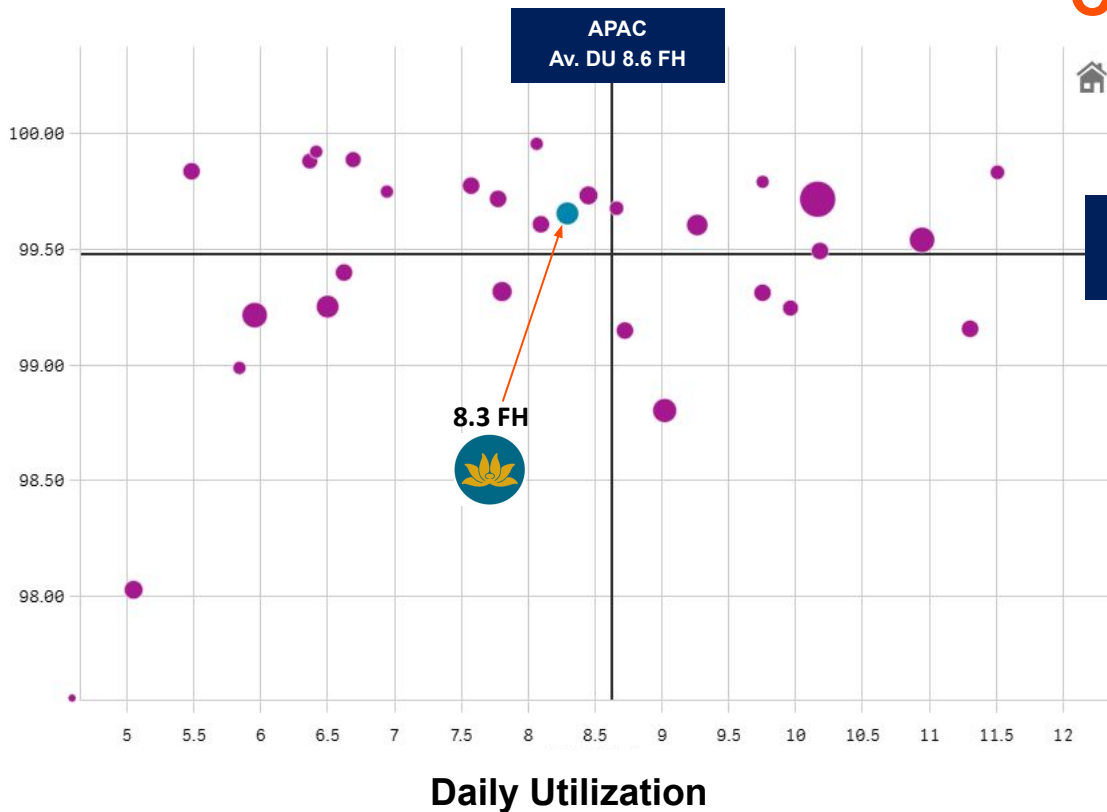
2 Operational Reliability Trends

3 Operational Reliability vs. Daily Utilization, Daily Cycle

4 Top Ten Drivers - Operational Interruptions

HVN CEO Fleet - OR vs Daily Utilization and Cycles

CEO Fleet



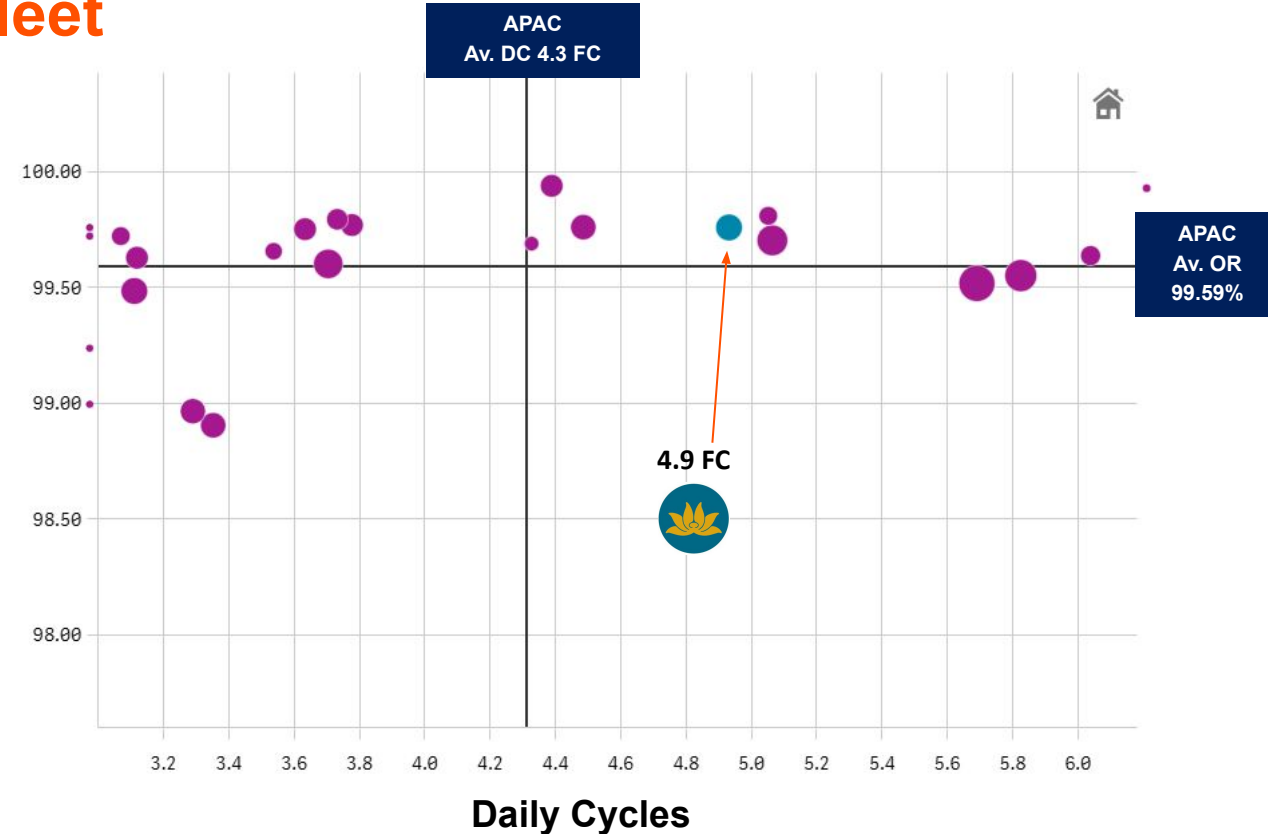
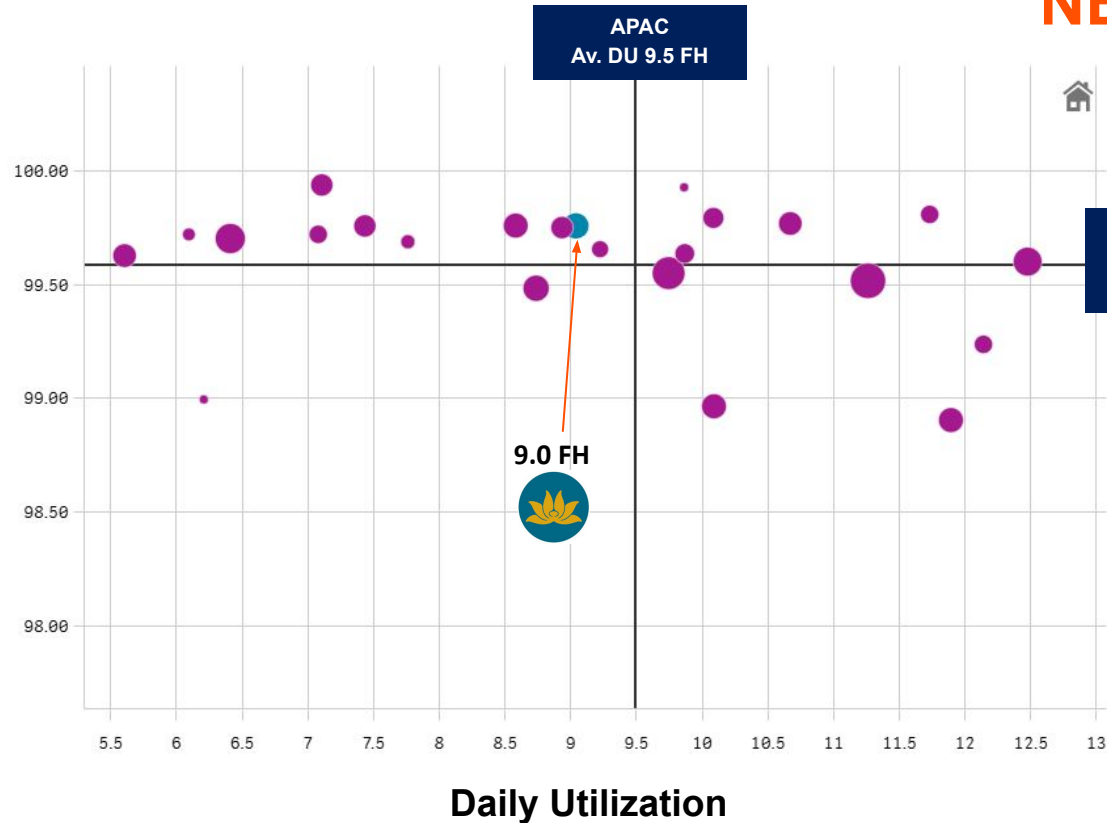
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Key points

- **APAC Fleet Data Used - Period Dec 2023 to Nov 2024 (Operator excluded from benchmark)**
- HVN CEO fleet 8.3 Daily FH is below APAC fleet average / 5.7 Daily FC is above APAC fleet average
- APAC average 8.6 FH Daily / 5.5 FC daily

HVN NEO Fleet - OR vs Daily Utilization and Cycles

NEO Fleet

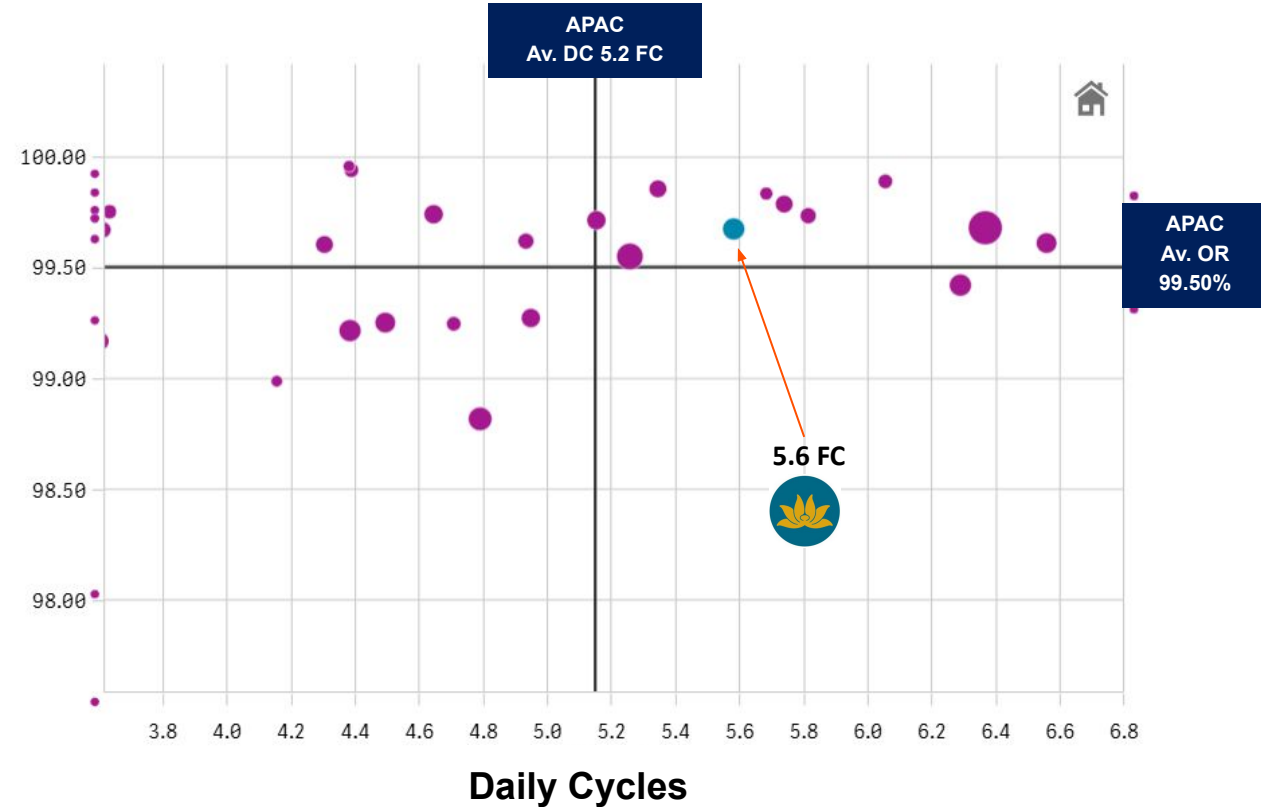
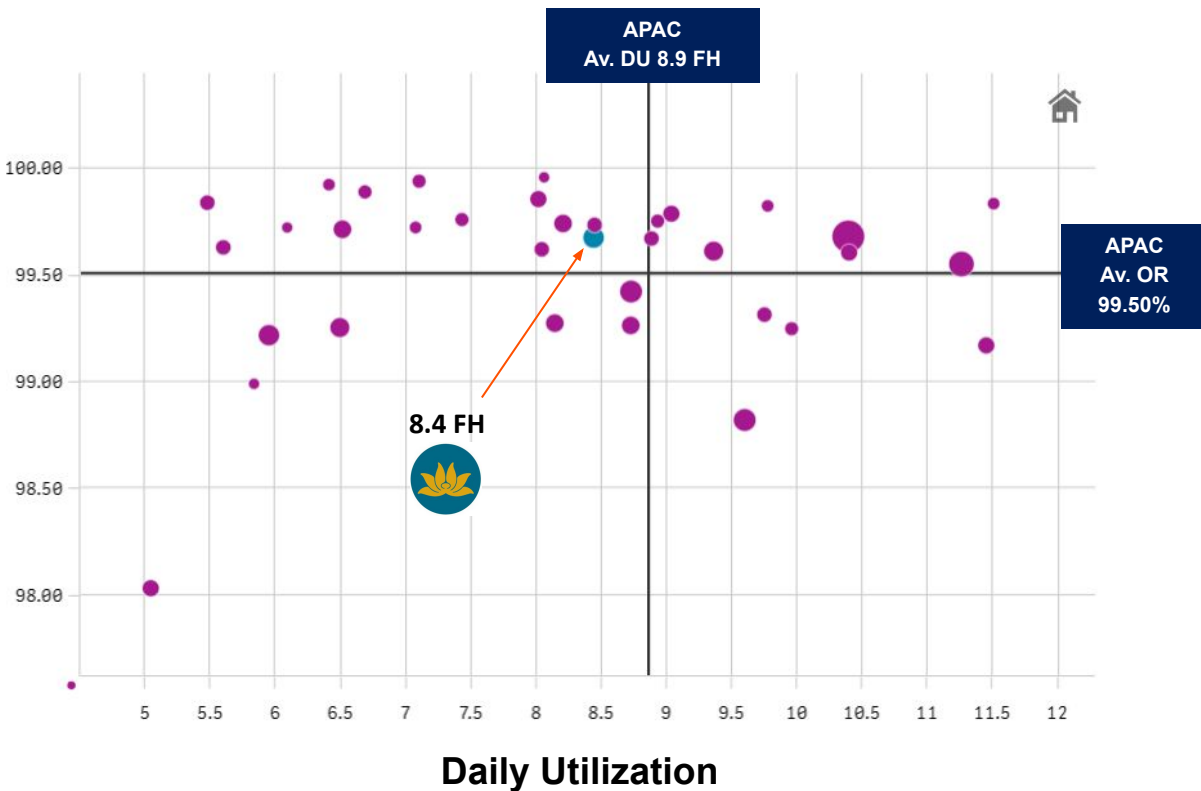


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Key points

- **APAC Fleet Data Used - Period Dec 2023 to Nov 2024 (Operator excluded from benchmark)**
- HVN NEO fleet 9.0 Daily FH is below APAC fleet average / 4.9 Daily FC is above APAC fleet average
- APAC average 7.4 FH Daily / 4.2 FC daily

HVN Fleet - OR vs Daily Utilization and Cycles



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Key points

- **APAC Fleet Data Used - Period Dec 2023 to Nov 2024 (Operator excluded from benchmark)**
- HVN Fleet 8.4 Daily FH is below APAC fleet average / 5.6 Daily FC is above APAC fleet average
- APAC average 8.9 FH Daily / 5.2 FC daily

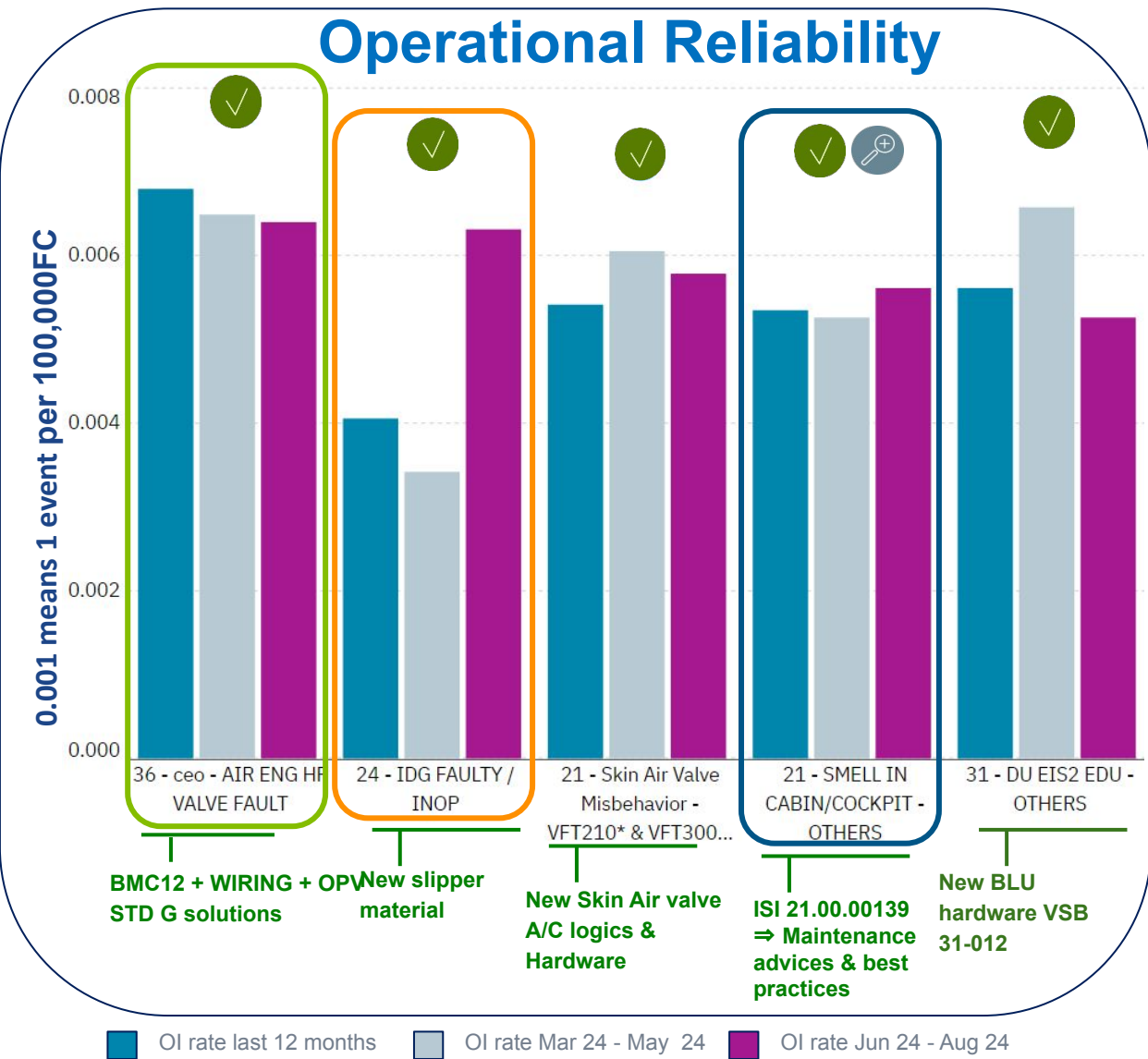
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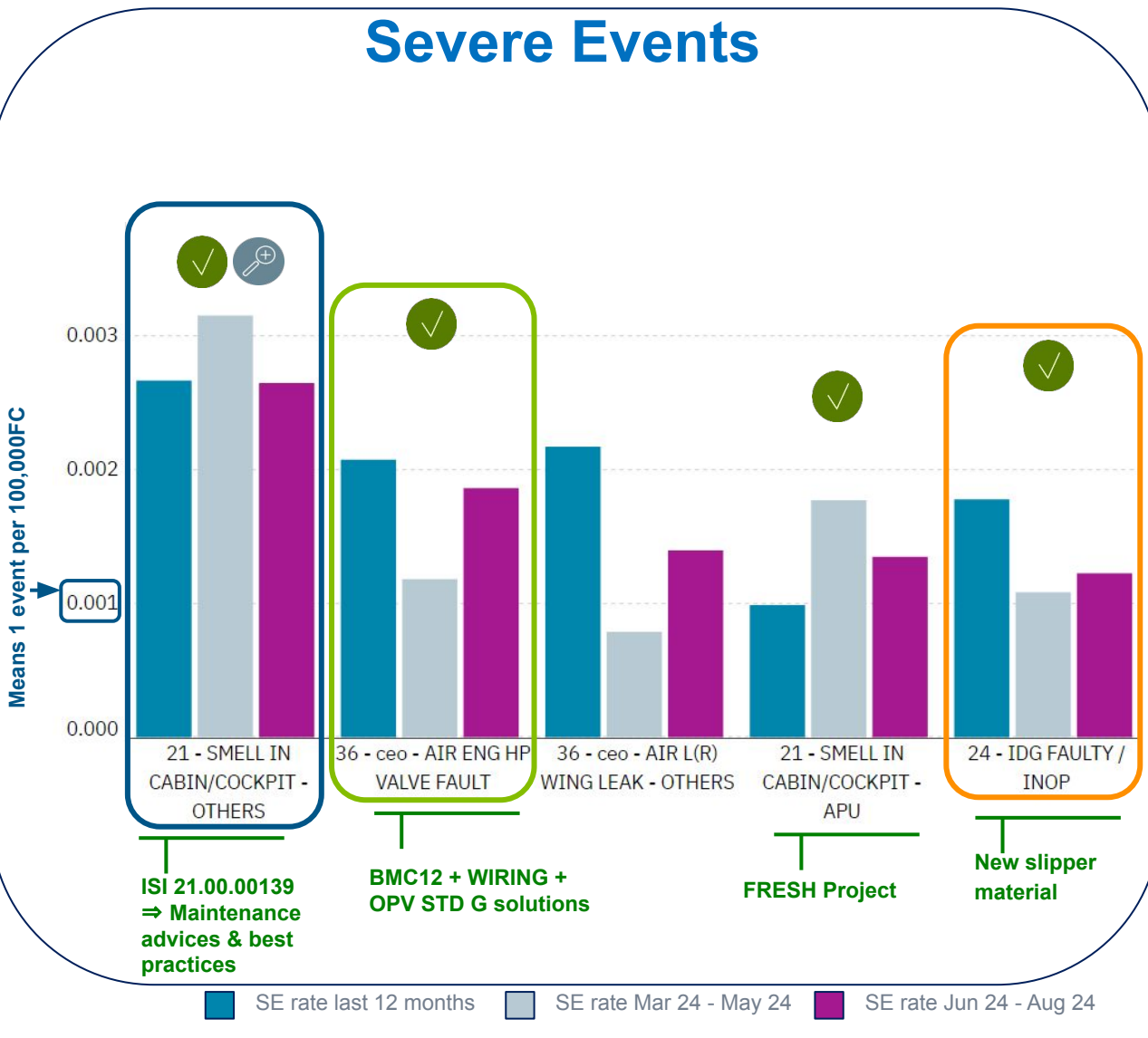
A320ceo Top Fleet Main drivers

(For ceo A/C between 2 and 12 yrs)

Operational Reliability

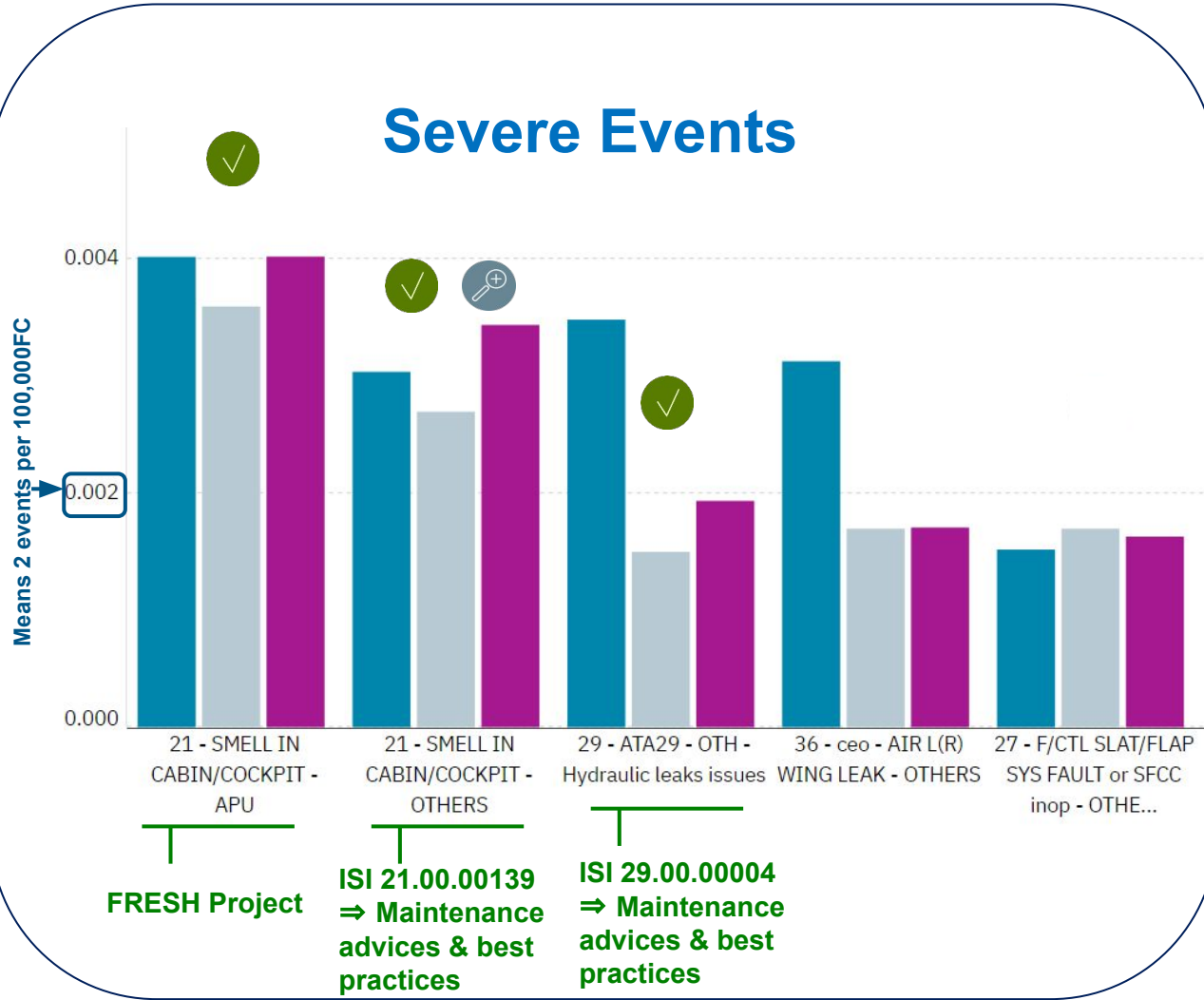
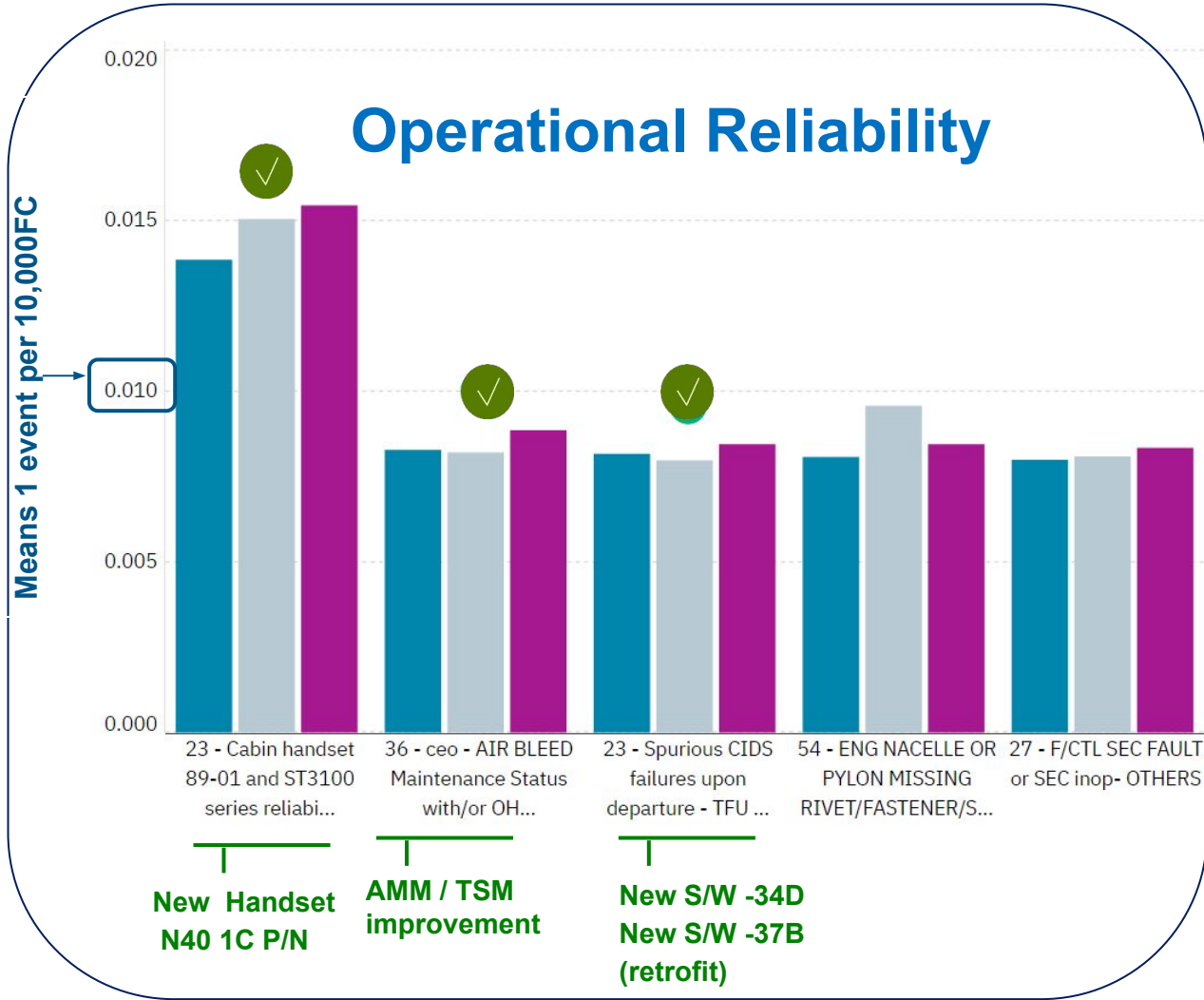


Severe Events



A320ceo Top Fleet Main drivers

(For ceo A/C > 12 years)

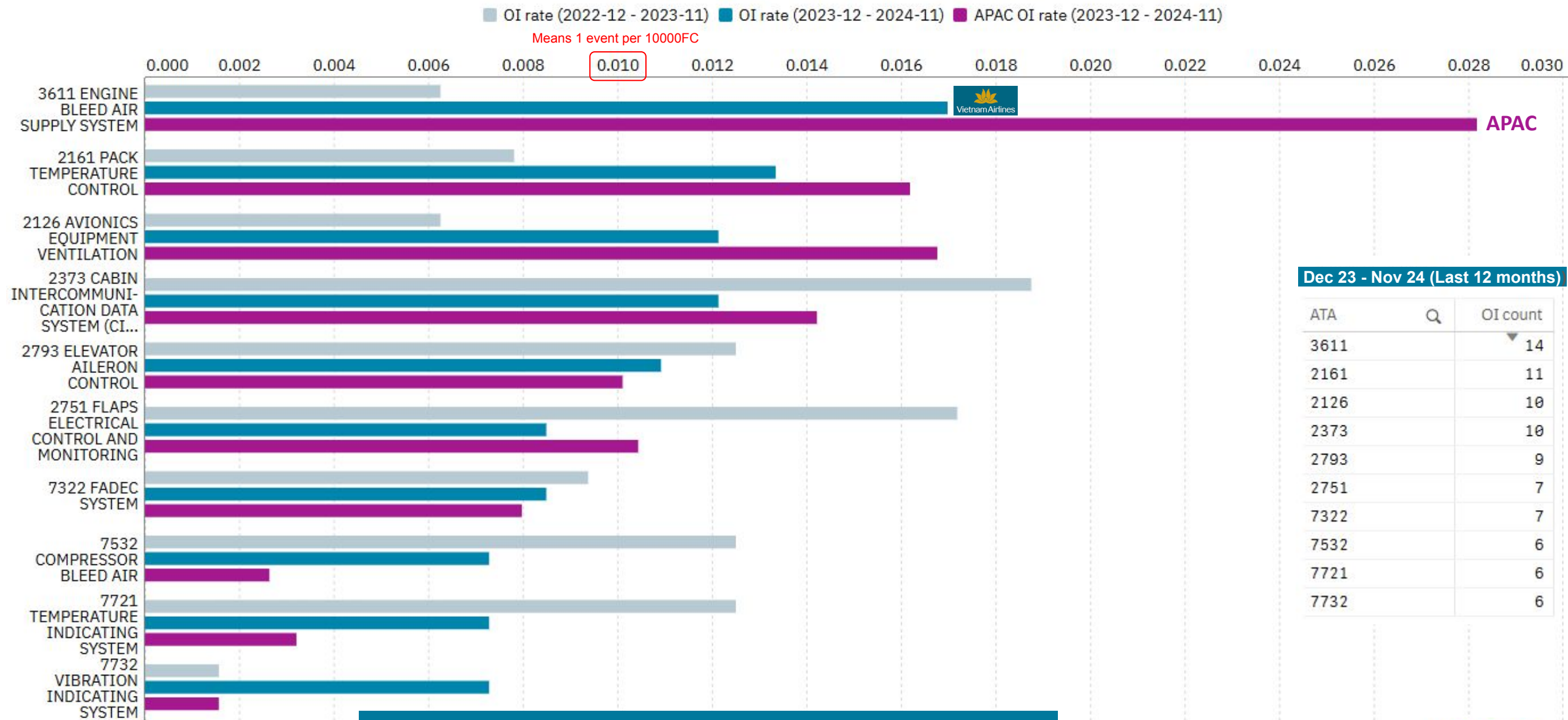


■ OI rate last 12 months ■ OI rate Mar 24 - May 24 ■ OI rate Jun 24 - Aug 24

■ SE rate last 12 months ■ SE rate Mar 24 - May 24 ■ SE rate Jun 24 - Aug 24

✓ Solution available ✓ Solution identified but not available 🔍 On watch / Under investigation

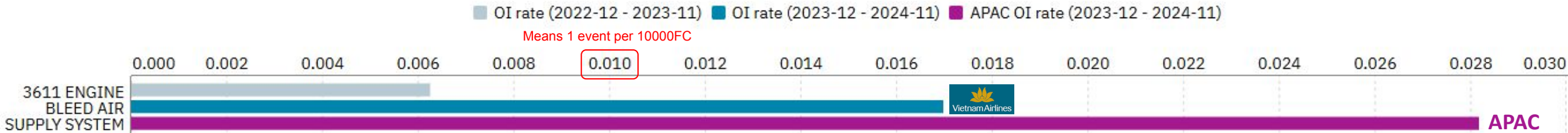
HVN CEO Fleet - Top OI drivers 12 Month (rate per 100FC)



ISI 00.00.00196 A320 Family Major In service Problems status (MISP)

Operator excluded from benchmark

HVN CEO Fleet - Top OI drivers 12 Month (rate per 100FC)



3611 ENGINE BLEED AIR SUPPLY SYSTEM

APAC

Occurrence date	MSN	Event description	Resolution description	Maintenance action	Delay (minutes)
2024 09 02	N03315	CAPT REPORT: IN FLT PRE COOLER TI IS 245C, ENG 2 T BAY GHI S " IN FLT PRE COOLER TI IS 245C, ENG 2. CRS KIM TRA BO TU B C 2 ENG	EVENT CLOSED REPLACE PRE COOLER ENG 1, CLEAN FILTER OF THERMOSTAT 7170HM2 ENG2 ERU SATIS, A/C SERVICEABLE	Replaced	37
2024 08 04	N03355	AIR ENG #1 BLEED FAULT 14:05 AFTER VN1824 CAPT REPORT AIR ENG #1 BLEED FAULT 14:30 MOC VNA GI MAIL TRIN KHAI T/5 T/5GN, OCC CHO TU TM DNG N 19:00 14:35 OCC I TU A396 BAY VN1394 NETO 16:10 THAY A357	FINAL ACTION: REPLACED PRECOOLER, SENS LINE, O RING OF 10HA1 AND 7071HM1 LEAK CHECK AND OPS TEST SATIS. APT REPORT AIR ENG #1 BLEED FAULT, EVENT CLOSED FINAL ACTION: REPLACED PRECOOLER, SENS LINE, O RING OF 10HA1 AND 7071HM1 LEAK CHECK AND OPS TEST SATIS.	Replaced	30
2024 08 11	N03355	AFTER FLIGHT VN1805 CAPT REPORT AIR ENG 2 BLEED FAULT ASOC CHANGE AS05 FLIGHT VN7195	ENG 2 REPLACED PRECOOLER AND SEAL 7170HM2 ENG RUN UP BLEED TEST SATIS	Replaced	19
2024 09 15	N03737	APU BLEED AIR I THNG VO ENGINE CORE. Ttc khi Boarding chuy VN7159 HAN DAD ETD 18:45 v khi APU Bleed ON pht hin APU Bleed Air i thng vo Engine Core.	RPL ENG #1 BLEED AIR PRECOOLER EXCHANGER SATIS, A/C SERVICEABLE, RPL ENG #1 BLEED AIR PRECOOLER EXCHANGER SATIS EVENT CLOSED	Replaced	42
2024 07 22	N04826	AFTER TAKS OFF CAPT INFORM VIA VHF A/C HAD MSG AIR ENG 1+2 BLEED FAULT, RESET NIL HELP, CAPT DECIDE AIR TURN BACK. (TU C ADD ENG BLEED 2 FAULT).	OCC CHANGE A396 BAY THAY A327 00:40 CNG TRIN KHAI KHN NGUY TI CH A327 P BI 73 04:00 MCC IU CHUVN THM OPV HAN SGN/ATA: 07:15. MCC REQUEST THAY OPV ENG#1 V ENG#2 13:00: LP XONG OPV ENG#1 -> CHECK SATIS 17:00: MCC TRIN KHAI ROB NG NG THU LC TU A617 LP LI CHO A327 20:20 TUBE LP LN LEAK CHECK SATIS 21:30: IC C PA BN GIAO 2 24/07 00:30: LP XONG OPV ENG#2 01:30: BT U LP IC; 04:30: TRONG QU TRNH LP NG NG MM NI N P2T2 PHT HIN B TRN REN 04:45: C KH H TR TARO REN; 06:00: LP XONG NG NG NI N P2T2; 06:00 07:00: DD VNG LCH KHAI THC CHA KO A327 N MY C 07:00: SAU KHI N MY CHECK LEAK PHT HIN LEAK NG SENSE LINE ENG#1 10:20: THAY HPV V 10HA1 ENG 1; 13:00: N MY CHECK SATIS TU SER NOTE: ADD AIR BLEED ENG 2 INOP STILL OPEN	MEL applied	70
2024 07 25	N04863	ENG 2 BLEED 2 INTERMITTENT FAULT VN7550 7165/CXR HAN DAD/(10:57 11:55) T/X CHECK ENG 2 bleed 2 intermittent fault (Note: Tu c ADD B1697 ENGINE 1 BLEED AIR SUPPLY SYS INOP) T/S in progres 10:57 t chn 11:07 CRS bo ENG 2 bleed 2 intermittent fault 12:10 Bo i A610 bay VN7165	Engine 2 Bleed RLPD the OPV and PRV2. Engine Run Up Test Bleed 2 System Normal. Note: ADD B1697 ENGINE 1 BLEED INOP STILL ENGINE 2 BLEED RLPD THE OPV AND PRV2, ERU TEST SATIS EVENT CLOSED	Replaced	82
2024 07 29	N05164	HP VAL ENG2 OPENS/CLOSES REPEATEDLY AT THRUST IDLE HP VAL ENG2 OPENS/CLOSES REPEATEDLY AT THRUST IDLE	EVENT CLOSED DEACTIVATE HP VALVE AND RAISE ADD	MEL applied	30
2024 11 30	N05164	ENG 2 BLEED ABNORMAL	ACT: RAISE ADD B2045. VN318 DAD NRT CHANGE A621 DELAY 100MINS Thing tin: B hng hc ny MOC/OCC khng xp tu i QT. pht ch tu khc bay thay RAISE ADD B2045 EVENT CLOSED	MEL applied	100
2024 01 16	N05241	AFTER BOTH ENGS START COMPLETED CAPT REPORT: ENG #2 HP VLV FAULT. A/C RETURN TO GATE	OPEN ADD B1628 & B1629 FOR THIS DEFECT. A/C SERVICEABLE AS ABOVE Final action: ENG#2 HP VLV RPLD. THE ENG#2 BLEED SYS TEST IAW AMM 71 00 00 710 043 & LEAK CHKD SATIS.	MEL applied	30
2024 08 04	N05275	ENG #2 HP VLV FAULT (VLV NOT OPEN)	SECURED REF AMM 36 11 00 040 003. APPLIED MEL 36 11 078 SECURED HP VLV OF ENG 2 AND ADD B1319 IS RAISED EVENT CLOSED	MEL applied	23
2024 05 08	N05297	DURING BOARDING APPEAR MSG BLEED MONIT SYS 1 FAULT.	ACTION PERFORMED BY VAEO0971 ON 09 MAY 2024, 00:43 ENG#1 BLEED ABNOMAL PRESS CO ENG RUN UP IAW AMM 71 00 00 710 043A FOUND ENG BLEED ABNOMAL WITH MSG: OVPRESS V SHA. REF MEL 36 11 01A ADD B2198 RAISED CAT: C DEADLINE: 19/05/2024 PERFORMED SIGN VAEO0971 	MEL applied	40
2024 08 23	N05297	CAPT REPORT PRESSURE ABNORMAL DURING ENGINE RUN UP FOR FLIGHT	EVENT CLOSED C/O CHECK AND RAISED ADD ENG#2 BLEED AIR SUPPLY SYS INOP	MEL applied	35
2024 11 21	N05709	ENG 1 BLEED PRV NOT CLOSE 1.AFTER BOARDING MESS.ENG 1 BLEED PRV NOT CLOSE APPEAR ON ECAM WARN 2.AFTER BOTH ENGINES START CAPTAIN REPORT MESS-A/TR OFF AND AP1 APPEAR ON ECAM	1. Reset satis. 2. Shutdown both ENGS, re start both ENGS, MSG gone. RESET SATIS EVENT CLOSED EVENT CLOSED	Reset	43
2024 09 18	N05994	ENG #1 HP VALVE SHOWS OPENED DURING CRUISE	11:55 CRS BO CAPT REPORT IN TECHLOG: ENG #1 HP VALVE SHOWS OPENED DURING CRUISE, SHOULD BE SECURED CLOSED ACCORDING TO MEL 36 11 078 & amp; ADD B1359. 12:30 TSV ADVISED HPV IS SECURED REF ADD B1359. POSITION INDICATION MAY BE SHOW INTERMITTEN. CAPT AGREE BOARDING. 12:58 DOOR CLOSED TOTAL DELAY: 38 MINS, TECH DELAY: 20 MINS ENG #1 HP VALVE SHOWS OPENED DURING CRUISE EVENT CLOSED	MEL applied	20

ISI 36.22.00047 A320FAM ceo - AIR BLEED Maintenance Status
ISI 36.11.00001 Engine Bleed Health Check Maintenance Program
ISI 36.11.00109 BMC STD12 and Full Kit (BMC STD12 + OPV G + Wiring) Efficiency
TFU 36.11.00.085 "AIR ENG BLEED NOT CLSD" at Engine Start or Engine Shutdown
TFU 36.11.00.109 "AIR ENG HP VALVE FAULT" on Ground when Pack-Off or in Cruise
TFU 36.11.00.120 Premature Failure of OPV 6740G01 Post Manufacturing Process
TFU 36.11.00.103 Premature Failure of OPV PN 6740F01

Dec 23 - Nov 24 (Last 12 months)

ATA	Q	OI count
3611		14
2161		11
2126		10
2373		10
2793		9
2751		7
7322		7
7532		6
7721		6
7732		6



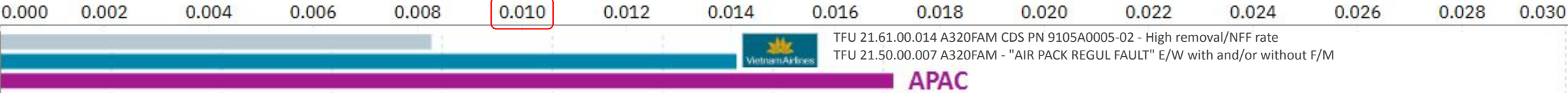
HVN CEO Fleet - Top OI drivers 12 Month (rate per 100FC)



■ OI rate (2022-12 - 2023-11) ■ OI rate (2023-12 - 2024-11) ■ APAC OI rate (2023-12 - 2024-11)

Means 1 event per 10000FC

2161 PACK
TEMPERATURE
CONTROL



Occurrence date	MSN	Event description	Resolution description	Maintenance action	Delay (minutes)
2024-09-26	N03022	AIR PACK 1 OVERHEAT	DEACTIVATED PACK 1 IAW AMM 21-51-00-040-002 AND APPLIED MEL PACK 1 INOP DUE TO OVERHEAT. AIR PACK 1 OVHT EVENT CLOSED	MEL applied	27
2024-07-28	N03315	AFTER VN114 CAPT REPORT: PACK 2 OVHT	C/O DEACTIVATE PACK #2 AND OPEN ADD B2762 FOR THIS DEFECT. A/C SERVICEABLE. PACK #2 OVHT EVENT CLOSED	MEL applied	32
2024-08-08	N03600	PACK 2 REGUL FAULT FOUND PACK 2 REGUL FAULT 	RAISED B1606 AIR CONDITIONING PACK 2 INOP EVENT CLOSED	-	37
2024-02-08	N03737	T/X AT PQC CHK FOUND PACK #2 REGUL FAULT	CLEANED CONNECTOR. OPS TEST SATIS	Cleaned	140
2024-06-26	N04945	PACK#2 OUTLET TEMP IS HIGH PACK#2 OUTLET TEMP IS HIGH	- 17:15: TRONG LC BOARDING GN XONG V CHUN B NG CA. CAPT BO NHIT U RA PACK#2 CAO - 17:20 CRS BO PACK U RA NNG, ANG CH KIM TRA - 17:28 CRS TRAO I VI CAPT PQC C FULL GROUND CART . TU CH CN KH LANH KHI RA PQC, CAPT NG TIP TC BAY TIP RA PQC THEO DI NHIT PACK#2 - 17:30 TU NG CA PACK#2 OUTLET TEMP IS HIGH	-	20
2024-06-09	N04971	T/X FOUND MSG PACK#1 OVH & PACK#2 REGUL FAULT T/X FOUND MSG PACK#1 OVH & PACK#2 REGUL FAULT	EVENT CLOSED - C/O RAISE DEFECT PACK#1 INOP - PACK#2: C/O REPLACED SENSOR 32HH, TEST SATIS. A/C SERVICEABLE AT 15:04 (LT) A/C DELAY 30 MINS	MEL applied	30
2024-08-07	N05340	T/X FOUND MESG: AIR PACK 2 OVHT T/X FOUND MESG: AIR PACK 2 OVHT	EVENT CLOSED ADD B1350 AIR PACK 2 OVHT RAISED	MEL applied	68
2024-03-01	N05427	PACK #1 OVER HEAT	RESET AND CHECK NORMAL AS ABOVE	Reset	27
2024-03-05	N05427	BOARDING COMPLETED, MSG PACK 1 OVHT. RESET BUT NIL HELP. NEED TO BE DEACTIVATED	MEL 21-52-01D APPLIED. PACK 1 DE-ACTIVATED - 17:45 BOARDING GN XONG, A/C C MSG PACK 1 OVHT - 17:55 CAPT GI TU C MSG: PACK 1 OVHT. - 18:10 CRS BO RESET KO HT, XIN DELAY DE-ACTIVATED, RAISE ADD - 18:25 NHIT CABIN CAO, CAPT CHO PAX VO LI NH GA - 18:50 MAINT XONG, BOARDING LI - 19:10 NG CA	MEL applied	55
2024-11-11	N05427	CAPT REPORT PACK 1 REGUL FAULT	C/O RESET AND OPS TEST NIL HELP . C/O DEACTIVATED FCV AND RAISED ADD B1719 PACK #1 INOP CAPT REPORT PACK 1 REGUL FAULT EVENT CLOSED	MEL applied	44
2024-11-20	N06266	PACK #1 OVHT	REF MEL 21-52-01D, AMM 21-51-00-040-002A DEACTIVATE AND RAISE ADD B1075 PACK #1 INOP. EVENT CLOSED	MEL applied	35

Dec 23 - Nov 24 (Last 12 months)

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3611		14
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2126		10
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7532		6
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0.000 0.002 0.004 0.006 0.008 0.010 0.012 0.014 0.016 0.018 0.020 0.022 0.024 0.026 0.028 0.030

2126 AVIONICS
EQUIPMENT
VENTILATION



APAC

TFU 21.26.00.032 A320FAM - Avionics Skin Valves ECAM Alert Leading to OI
ISI 21.26.00064 Recommends skin air valve overhaul at 12K FH
TFU 21.26.00073 A320FAM - Skin Air Valves Failure NFF and Population 3 Failure
ISI 21.26.00023 Recommends avionic fan PN EVT3454HC overhaul at 10K FH
TFU 21.26.00.033 New avionic fan PN AE1819B00 with BBHM function

Occurrence date	MSN	Event description	Resolution description	Maintenance action	Delay (minutes)
2024-08-12	N03022	CAPT REPORT: AVIONIC VENTILATION FAULT. During Start ENG, Capt report: AVIONIC VENTILATION FAULT.	- Ref TSM: 21-26-00-810-804 c/o reset AEVC "10HQ" ops test#160; - IAW AMM: 21-26-00-710-001 rev Aug24, No fault found, - Read last leg report and class 3 of AEVC No fault. - A/c run up both Eng normal. REF TSM: 21-26-00-810-804 EVENT CLOSED	Reset	20
2024-01-21	N03600	FOUND SKIN AIR INLET VLV NOT CLOSE, C/O DEACTIVATED -> NIL HELP	WAITING SEALANT CURE TIME AFTER 12:00 21/01/2024 C/O REPLACED SKIN AIR INLET VLV, OPS TEST AVIONIC VENTILATION SYS FOUND SATIS.	Replaced	35
2024-02-18	N03737	VENT AVNCS SYS FAULT	C/O RAISED ADD FOR VENT AVNCS SYS FINAL ACTION: RPL VALVE -SKIN AIR OUTLET, AVNCS VENT I.A.W AMM 21-26-53-000/400-001-A OPS TEST SATIS	MEL applied	35
2024-05-31	N03737	VENT SKIN VALVE FAULT REJECT TAKE OFF DUE TO APPEAR MSG: - VENT SKIN VALVE FAULT - VENT BLOWER FAULT - VENT EXTRACT FAULT	AEVC TEST FOUND SKIN AIR OUTLET FAULT. C/O DE-ACTIVATE AND RAISE DEFECT REF MEL 21-26-04A	MEL applied	77
2024-07-24	N04783	SKIN AIR INLET ON ECAM SHOW XX DURING TRANSIT CHECK, A/C HAVE ADD VENT AVNCS SYS FAULT& NEED SHOW ANGLE OPEN SKIN AIR INLET ON ECAM BUT ECAM SHOW XX	ROB SKIN AIR INLET VALVE FROM A617 TEST PASS, ADD CLEAR, A/C SERVICEABLE	MEL applied	30
2023-12-05	N05456	A601/BKK VN604 BKK-SGN RTG DUE TO VENT AVNCS SYS FAULT	THAI VJ C/O RESET TEST SATIS	Reset	85
2024-07-12	N05469	SKIN AIR INLET VLV INOP	RAISED MEL AND PERFORM MAINT PROC AS REQ A602/DAD/VN147 DAD-SGN DELAY 38 MINS (TD:22 MINS)/ SKIN AIR INLET VLV INOP/ ACT: ADJUST MANUAL HANDLE SKIN AIR INLET VLV , RAISE MEL AND PERFORM MAINT PROC AS REQ	MEL applied	22
2024-11-28	N05709	A606/ICN ATB VENT SKIN VALVE FAULT	TLP PFR A606 ICN ATB 27NOV2024 EVENT CLOSED 1+2 : RAISED MEL 21-26-04A AVIONICS SKIN AIR OUTLET VLV 3. RAISED ADD WITH DEADLINE 200FC	MEL applied	-
2024-11-30	N06266	CAPT REPORT SKIN AIR OUTLET VALVE FAULT THAI VJC C/O RAISED ADD SKIN OUTLET VALVE INOP	1.DO SUBTASK 21-26-00-710-080-A TO OPS TEST FOR FAULT CONFIRM: -DO RESET OF THE AEVC (10HQ). -DO OPS TEST OF AVIONICS EQUIPMENT VENTILATION FROM THE (MCDU) REF. AMM 21-26-00-710-001 -READ THE LAST LEG REPORT OF THE AEVC REF. AMM 21-26-00-710-009. -READ THE CLASS 3 FAULTS OF THE AEVC REF. AMM 21-26-00-710-008. 2. REF TSM 21-26-00-810-804-A PERFORM RPL VALVE-SKIN AIR OUTLET, AVNCS VENT (22HQ) REF. AMM 21-26-53-04. NIL HELP. ADD STILL OPEN. BACK A/C TO MEL CONFIG (REF MEL 21-26-04A) FOR FURTHER TSM.T/S ADD LINK TO WO 6789466	MEL applied	67
2024-01-16	N06880	CAPT REPORT: SKIN AIR INLET VLV FAULT	SKIN INLET VLV SECURED CLS POS AND AIR CONDITIONING INLET VLV SECURED IN OPEN POS, ADD B1312 IS RAISED IAW MEL 21-26-058. SKIN AIR INLET VLV NOT CLS IN FLT Final action: Replace the VALVE-SKIN AIR INLET, AVNCS VENT (15HQ)	MEL applied	119

Dec 23 - Nov 24 (Last 12 months)

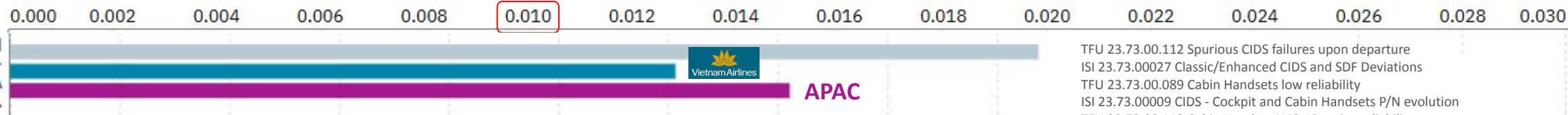
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2373 CABIN
INTERCOMMUNI-
CATION DATA
SYSTEM (CI...



Occurrence date	MSN	Event description	Resolution description	Maintenance action	Delay (minutes)
2024-04-09	N03737	FOUND DEUB 300RH6 INOP	AS ABOVE C/O REPLACED DEUB 300RH6 -> CHECK SATIS	Replaced	40
2023-12-15	N04669	A/C HAD 4 MEL ITEM CAT C, AND NOW STS SMOKE ON MAINT STS INTERMITTENT	- 10:15 B1 INFORM - 10:25 MCC ISSUE WO VN24305638 TO RPL CIDS FOR CLEAR ADD CIDS ON MAINT STS - 11:25 CIDS RPLD, SOFTWARE UPLOADED, OPS TEST SATIS. AND THEN A/C HAVE MSG APPEAR: ANTI ICE L WINDOW AND AIR BLEED - 11:30 : -C/O RAISED ADD ANTI ICE L WINDOW. AIR BLEED (RESET AND TEST SATIS). - 12:15: A/C CLOSE THE DOOR AND DISPATCH	Replaced	30
2024-09-08	N04737	T/X FOUND DEFECT DEU B DOOR 1L	FINAL ACTION: C/O RPLD NEW DEU, TEST FOUND SATIS T/X FOUND DEFECT DEU B DOOR 1L EVENT CLOSED	Replaced	16
2024-06-04	N04863	PA CABIN INTERMITTENT DURING BOARDING FLIGHT CREW REPORT	C/O ISOLATE HANDSET, C/O ISOLATE RESET CIDS, FOUND LOUDER SPEAKER AT PURSER POS STATION NOISE. ISOLATE LOUDER SPEAKER. A/C SERVICE	-	60
2024-11-27	N04945	T/X CHECK FOUND: DEU B 300RH INOP	`- 18:00: CRS BO TU C MSG : TRN BNG FAP "CHECK ATTENDANT RELATED FUNCTIONS " LC ANG BOARDING - 18:30 TU BOARDING XONG . C/O RESET "DEU B" NHNG KHNG C, CRS B2 TEST CIDS LY FAULT CODE, V BO CN SWAP DEU B DOOR 1L VI DOOR 4 - 18:30: MCC ISSUE WO SWAP DEU B 300RH1 AT DOOR 1L WITH DEU B 300RH8 DOOR 4 - 19:00: SWAP XONG DEU B , V RAISED ADD DEU B (300RH8) INOP (DOOR 4), BO TU SER FINAL ACTION; C/O WO SWAP DEU B 300RH1 AT DOOR 1L WITH DEU B 300RH8 DOOR 4, CHECK SATIS. A/C SERVICEABLE T/X CHECK FOUND: DEU B 300RH INOP EVENT CLOSED	MEL applied	62
2024-06-18	N05297	CABIN HANDSET DOOR 2 INOP	SWAP DEU B DOOR 4R WITH DOOR 2. RAISE ADD B2159 A/C READY ON 18.Jun.2024 AT 18:30 EVENT CLOSED	MEL applied	58
2024-10-11	N05427	PRE-FLIGHT FOUND PA AT DOOR 1LH INOP	EVENT CLOSED	-	50
2024-03-10	N05438	INFORM DEU B AT DOOR 2 FAULT	- 07:28 A/C ARRIVAL BAY 22 - 07:42 CRS INFORM DEU B AT DOOR 2 FAULT - 07:43 MCC ISSUE WO SWAP DEU B AT DOOR 4L - 08:07 SWAP SATIS. BOARDING - 08:38 ALL DOOR CLOSE	Swapped	18
2024-01-02	N05555	Preflight check found INTERPHONE AT DOOR 2R INOP	C/O SWAP DEU-B(300RH4) WITH DEU-B(300RH8) RAISED: B1563 CAB DEU-B(300RH8) INOP B1564 CHECK SLIDE PRESS B1565 CHECK DOOR PRESS B1566 SMOKE LAV DET FAULT B1567 CAB HANDSET AT 4L INOP	MEL applied	25
2024-11-10	N06810	HANDSET AT DOOR 3 INOP VN242-512 SGN-HAN-PEK (08:35-10:00): Before boarding Cabin crew report handset at door 3 inop	C/O reset CIDS 1-2 nil help C/O swap handset and support bracket at door 4 with door 3nil help C/O Clean connector DEU B (300RH6) reset CIDS and DEU B then Ops test found satis HANDSET AT DOOR 3 INOP EVENT CLOSED	Reset	18

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2793 ELEVATOR
AILERON
CONTROL



APAC

Occurrence date	MSN	Event description	Resolution description	Maintenance action	Delay (minutes)
2024-09-04	N03022	A353+GTB+F/CTL AIL SERVO FAULT DURING PUSHBACK AND START ENGS; CAPT CALL VHF AND INFORM: F/CTL AIL SERVO FAULT AND ELAC#2 FAULT ; DUE TO SGN RAIN HEAVY CAPT DECIDED RETURN TO GATE . A/C GTB AT PARKING 74 	- 15:00: CAPT GI VHF : F/CTL AIL SERVO FAULT. TSV ANG H TR TH TRI MA TO , CAPT QUY NH TU QUAY LI BI. -15:20: TU QUAY LI BI 74. CRS B BO CAPT GHI S : F/CTL AIL SERVO FAULT AND ELAC#2 FAULT - 15:50: C/O OPS TEST FAIL. BO AOG I TU - 16:20 : OCC, ASOC BO I TU A356 BAY THAY - 16:30: MCC ISSUE WO SWAP ELAC 2 VI ELAC 1. - 17:20: BO TU SER FINALACTION: C/O SWAP ELAC#1 WITH ELAC#2 . DO TEST ALL AND ERU . CHECK SATIS. A/C SERVICEABLE F/CTL AIL SERVO FAULT AND ELAC#2 FAULT EVENT CLOSED FINALACTION: C/O SWAP ELAC#1 WITH ELAC#2 . DO TEST ALL AND ERU . CHECK SATIS. A/C SERVICEABLE	Swapped	60
2024-06-29	N04737	DURING PAX BOARDING A/C HAS MSG ELAC 2 FAULT DURING PAX BOARDING A/C HAS MSG ELAC 2 FAULT	- 07:00: CRS BO TRONG LC PAX BOARDING GN XONG TH A/C HIN MSG ELAC 2 FAULT. - 07:30: C/O SWAP ELAC#1 WITH ELAC#2 , AND APPLIED MEL 27-93-01A FOR ELAC#1 INOP . A/C SERVICEABLE. - 07:31: TU NG CA C/O SWAP ELAC#1 WITH ELAC#2 AND RAISED ADD	MEL applied	30
2024-04-13	N05340	CAPT REPORTED ON TLP: F/CTL DISPLAY ON STS PAGE	SWAP ELAC 1 AND 2 FOUND MSG MOVE POS 1 ELAC 2 TEST OK ALSO ELAC1 IS OPENED ADD B1290 FOR ELAC 1 INOP - 08:00 MCC ISSUE WO RPL ELAC 2 - 11:00 LNH KHI ELAC LN THAY NHNG KHNG C DO KHC SOFTWARE (112 V 115) - 11:15 MCC NGHI SWAP ELAC T A616 BQDB, TU OCC LI VN1372 NETD: 12:40 - 11:50 MCC ISSUE WO SWAP ELAC 1 & ELAC 2, M ADD J - 12:15 SWAP ELAC 1 AND 2 FOUND MSG MOVE POS 1 ELAC 2 TEST OK ALSO ELAC 1 IS OPENED ADD B1290 FOR ELAC 1 INOP. - 12:37 ALL DOORS CLOSED	MEL applied	55
2024-06-30	N05392	DURING 02 ENGS RUNNING. MSG: ELAC 1 PITCH FAULT. AC GTB	RAISED MEL AND PERFORMED MAINT PROCEDURE EVENT CLOSED EVENT CLOSED	MEL applied	60
2024-05-06	N05418	PITCH TRIM SYSTEM NOT WORKING PRECISELY	REF TR 81397740 FROM AIB MCC REQ: PERFORM THE FOLLOWING TSM- TASK 27-93-00-810-818-A, REPLACE ELAC-2 AND KEEP IT ON HOLD AND MONITOR IN THE FOLLOWING FLIGHTS IAW AMM 27-93-34-000-001-A AND AMM 27-93-34-400-001-A ITEM	Replaced	45
2024-02-21	N05709	AFTER START ENG#1 AT PARKING AREA (APU INOP) FOUND ECAM WRN F/CTL ALTN LAW AND F/CTL ELAC 1 PITCH FAULT.	FOUND FAILURE MSG 27-34-51 LG ELEV MODE VLV 34CE1. REF TSM 29-93-00-810-860A DO OPS TEST SIDE STICK ASSY IAW AMM 27-96-00-710-020 AND BITE TEST EFCS IAW AMM 27-96-00-740-001. ALL TEST OK. SATIS AS ABOVE	Inspected, checked	33
2024-04-25	N05709	AFTER ENGINE RUN UP COMPLETED APPEAR MSG ELAC 2 FAULT	C/O REPLACED SERVO CONTROL 34CE -> CHECK SATIS	Replaced	40
2024-05-10	N06748	1. AFTER START: ELAC#1 PITCH FAULT (RESET BACK TO NML) 2. AFTER LANDING: F/CTL AIL SERVO FAULT	1. REF TSM 27-00-00-810-801A REV MAY 24 NO FAULT MSG. SATIS 2. REF TSM 27-10-00-810-814A REV MAY24 RPL ELAC 2 IAW AMM 27-93-34-00/400/001A AND OPS TEST SATIS	Replaced	39
2024-02-28	N06880	T/X CHK FOUND F/CTL ELAC 1 PITCH FAULT	REF TSM 27-93-00-700-132A OPS TEST OF THE SIDE STICK ASSEMBLY. SATIS. DO THE BITE TEST FOUND SATIS.	Inspected, checked	20

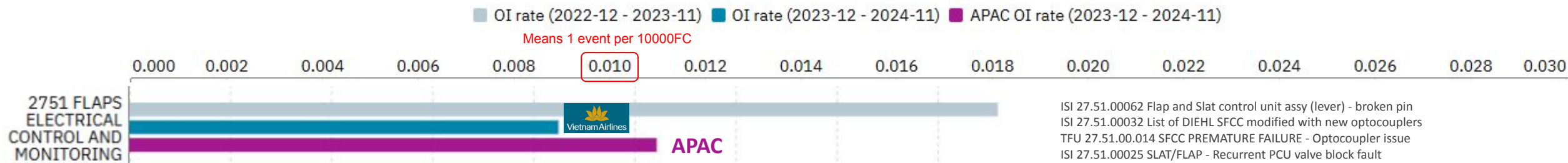
TFU 27.92.41.002 Faults Related to Sidestick Sensor Faults
ISI 27.00.00087 ELAC/SEC/FCDC/Wiring Interface T/Shooting tips
ISI 27.90.00002 ELAC and SEC reliability
ISI 27.93.00024 How to identify ELAC and SEC reset Vs Genuine issue
TFU 27.93.00.051 ELAC ACCELEROMETER FAULT MESSAGES

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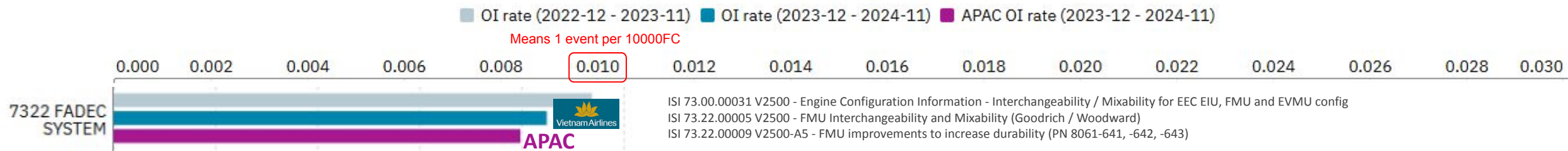


Occurrence date	MSN	Event description	Resolution description	Maintenance action	Delay (minutes)
2024-09-20	N03737	C/O WO 6600601: Capt report FLAP/SLAT control lever stiff operation.	C/o replaced Control Unit (114VU), OPS test satis. 55 C/O WO 6600601 EVENT CLOSED	Replaced	45
2024-06-05	N05469	F/CTL SLAT SYS 1 FAULT.	ACTION: C/O RPLD SFCC COMPUTER..NIL HELP . ADD B1757 ;SFCC ;MAINT STS ; . ; TOTAL DELAY: 44mins. TECH: DELAY 30mins C/O WO T/S MSG: F/CTL SLAT SYS 1 FAULT.	-	30
2023-12-08	N05709	Capt informs MSG F/CTL FLAP SYS 1 FAULT	CRS (on call) support from BKK (TVJ) c/out Swapt SFCC1 & SFCC2, OPS TEST SYSTEM FOUND NORMAL	Swapped	50
2024-06-18	N05709	B1660 FLAP SYS 2 FAULT AND FLAP SYS 1 FAULT.	REPLACED SFCC 1, OPS TEST SATIS B1660 STILL OPEN REPLACED SFCC 1, OPS TEST SATIS A/C READY ON 18.Jun.2024 AT 07:20 EVENT CLOSED	MEL applied	30
2024-03-28	N05994	READY FOR TAXI: FLAP SYS 2 FAULT, AC GTB	RAISE ADD AND REFUEL MORE FUEL RAISED ADD B1242 SFCC FLAP CHANNEL #2 INOP AND ADD B1243 MINIMUM IDLE ON GROUND INOP. C/O M PROC IAW AMM 27-5-00-040-001A	MEL applied	60
2024-04-01	N05994	ECAM F/CTL: FLAP FAULTS AFTER ENG 2 START.	REF TSM 27-51-00-810-851-A, BITE TEST AMM 27-51-00-740-002-A, NO FAULT RECORD. A/C SERVICEABLE AS ABOVE REF TSM 27-51-00-810-851-A, BITE TEST AMM 27-51-00-740-002-A, NO FAULT RECORD. A/C SERVICEABLE	Inspected, checked	97
2024-04-12	N05994	+ A/C arrived late 30Mins. + After first engine start, ECAM message: Flap System fault.	SHUTDOWN ENGINE TO CRS GET ACCESS THE COCKPIT - RESET FLAP SYSTEM 1 OK	MEL applied	20

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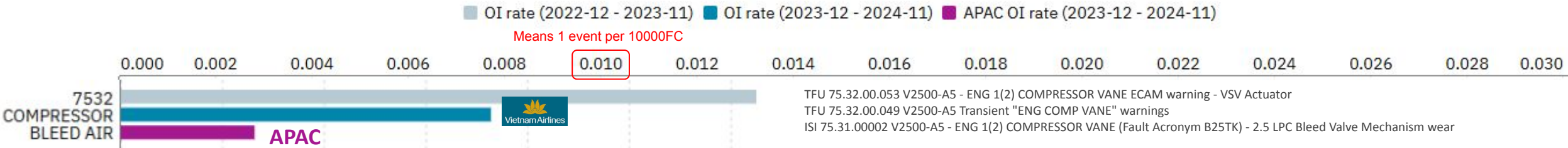


Occurrence date	MSN	Event description	Resolution description	Maintenance action	Delay (minutes)
2024-06-21	N04703	ENG#1 OVER SPEED ENG#1 OVER SPEED PROT FAULT	REF TSM 73-22-00-810-851B: C/O REPLACED EEC & FMU OF ENG#1, RESET FADEC THEN ENG #1 RUN UP TEST SATIS.	Replaced	90
2024-07-19	N04826	IGN A + B FAULT ENG 2 IGN A + B FAULT ENG 2	EVENT CLOSED C/O REPLACED FMU ENG 2 -> ENGINE RUN UP SATIS.	Replaced	95
2024-04-05	N04945	CAPT REPORT: ENG #2 AIR EXCHGR FAULT / AIR VLV CLOSED	FOUND FAULT : ACRONYM AVOPTF REF TSM 79-20-00-810-806B , FOUND FAULT ACRONYM AVOPTF RESET FADEC THEN ENG #2 RUN UP, NO FAULT FOUND , MSG DISAPPEARED	Reset	59
2024-04-29	N04945	ERU AFTER EA12 FOUND LEAK AT FMU ENG#2	REPLACED FMU REF AMM 73-22-52-000/400-010-B FMU TEST AMM: 80-13-51-710-040 FADEC TEST AMM: 73-22-00-710-040 IDLE LEAK CHECK: AMM 71-00-00-710-012	Replaced	40
2024-09-13	N05251	A338 RTO/GTB/ MANY MSG SHOWS Webdrive "A338 UIH 13SEP24" . A338/UIH/VN1622 UIH-HAN/AOG/ RTO/GTB/MANY MSG SHOWS + DATA LINK ATSU FAULT + ENG 2 FADEC + ENG 2 SENSOR FAULT - Sau khi lm cc bc kim tra ban u ca AOG (test fadec, Pack...), 15:10 CRS i walk around kim tra pht hin cc lp #2, #3 b cut out limit. 	ACT: - ADD B1209 ALREADY (DATALINK ATSU FAULT) - C/O TSM 77-00-00-810-806A RPLD EEC AMM 73-22-34-000-010A/73-22-34-400-010A. ENG RUN UP FOR LEAK CHK. SATIS. - C/O REPLACED BOTH M/W #2 & #3 CHECK ALL SATIS. A/C SERVICEABLE 02:00+ Thng tin: + K hoch ban u VN1622 i li lc 13h15 + Bo AOG: Lc 14h20 + 02 crs B1, B2 t SGN i VN1394 h UIH lc 17h23 + Nhn s v kh ti i cu tu A338 xut pht t DAD lc 17h18 n UIH lc 00:07+ ngy 14/9 ENG #2 FADEC , ENG #2 SENSOR FAULT EVENT CLOSED	Replaced	645
2024-04-28	N05456	DURING TAXI SHOW MSG ENG #2 EPR MODE AND FADEC A.	C/O RPLD EEC OF ENG# 2 AND ENG 2 RUN UP FOUND SATIS.	Replaced	135
2024-05-06	N05495	ENG 1 OVDP PROT FAULT	Ref TSM 73-22-00-810-855A -> eng run up check found no fault mess. Note: take 30 mins to tow A/C to eng run up pos.	Inspected, checked	30

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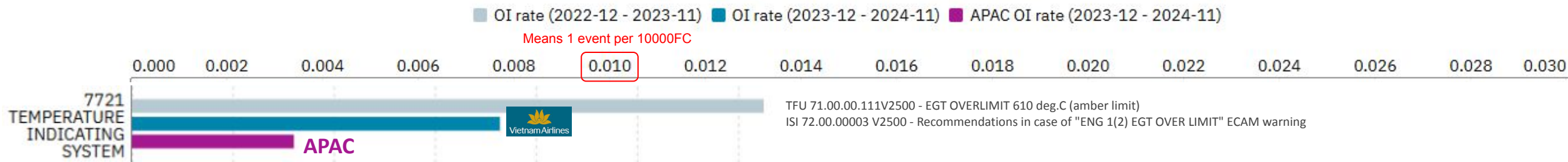


Occurrence date	MSN	Event description	Resolution description	Maintenance action	Delay (minutes)
2024-04-25	N03355	CAPT REPORT ENG 2 COMPRESSOR VALVE AFTER LANDING.	REF TSM 75-30-00-810-850 C/O ENGINE RUN UP -> CHECK SATI	Inspected, checked	20
2024-09-08	N04669	ENG 2 COMPRESSOR VANE	Eng#2 Run up Satis. A/c Serviceable	Inspected, checked	55
2024-03-24	N04703	CAPT REPORT ON TECHLOG ENG#1 COMPRESSOR VANE MSG NO GO	- 09:15: TU VO BI. CAPT REPORT ON TECHLOG ENG#1 COMPRESSOR VANE MSG NO GO - 09:45: C/O OPS TEST AND ERU. MSG GONE. A/C SERVICE - 10:10 : C/O RELEASE A/C . BOARDING PASSENGER - 10:30: A/C CLOSED THE DOOR AND DISPATCH	Inspected, checked	20
2024-03-28	N04703	A/C HAS MSG: NO GO : ENG#1 COMPRESSOR VANE	C/O WO LUB 2.5 XONG, TU KO LN KHAI THC - 10:10: MCC ISSUE WO VN 24357018 ENG#1 COMPRESSOR VANE - 13:00 TU KO V HG1 - 14:45 B1 BO KHONG 30 PHT NA XONG LUB 2.5 - 15:30 FINISH, NG NG C, TUY NHIN LATCH B KT, CHA NG C - 16:30 FINISH NG ENG, TU KO LN KHAI THC - 16:55 TU KO TI BI 24 - 17:43 A/C CLOSED THE DOOR AND DISPATCH	Inspected, checked	23
2024-03-11	N05306	DUE TO A390 : T/X CAPT REPORT ON TECHLOG: ENG2 COMPRESSOR VANE DURING FLT AND TAXI INTERMITTENT. A/C NEED MORE TIME TO T/S. A/C CHANGED TO A392 FLT SGN-PNH/VN920.	A390 C/O T/S AND ERU . CHECK SATIS A/C SERVICEABLE . 15:05: TU A392 VO BI 15:20: A392 FLT SGN-PNH/VN920 INSTEAD OF A390 15:48: BT U BOARDING 16:39 : KT THC BOARDING V TU NG CA	Inspected, checked	30
2024-01-10	N06810	After VN1805 found ENG 2 COMPRESSOR VANE	REF TSM 75-30-00-810-846A & 75-30-00-810-844A C/O REPLACED ACTUATOR VSV OF ENG#2, ENG RUN UP -> CHECK SATIS.	Replaced	40

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7732		6

HVN CEO Fleet - Top OI drivers 12 Month (rate per 100FC)



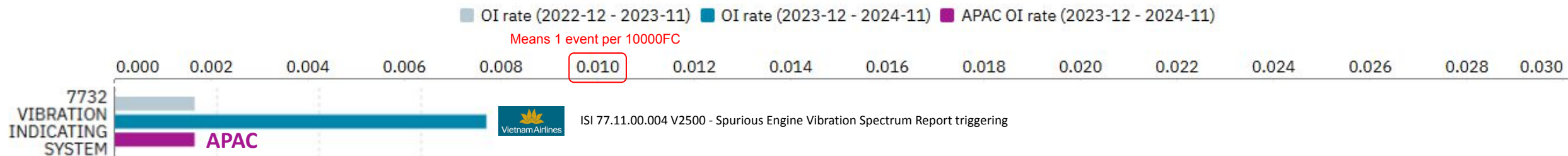
Occurrence date	MSN	Event description	Resolution description	Maintenance action	Delay (minutes)
2024-02-08	N03315	ENG 2 EGT OVER LIMIT	AS ABOUT REF TSM 77-20-00-810-802B, FOUND THERMOCOUPLE 4012KS3 LOOSEN. C/O RPL NEW THERMAL COUPLE IAW AMM 77-21-15-00-010A/77-21-15-400-010A. PER IDLE LEAK CHECK IAW AMM 71-00-00-710-012. SATIS	Replaced	162
2024-08-10	N05275	A339/GTB DUE TO ENG#1 EGT OVER LIMIT AFTER START A339 VN581 KHH-SGN RTG DUE TO ENG#1 EGT OVER LIMIT AFTER START	C/O HARNESS DRY OUT PROCEDURE; EGT IND CHK ON ECAM NML - VN581 KHH-SGN DELAY 100MIN	Inspected, checked	100
2024-01-24	N05306	AFTER PUSHBACK, PREPARE FOR ENG 2 START, MSG ENG 1 EGT OVERLIMIT SHOW ON ECAM. A/C RETURN TO GATE FOR T/S.	SIAECs ENGINEER C/O TSM, FOUND SATIS. A/C SERVICEABLE Final action: RPL A THERMOCOUPLE POS 4012KS3 ERU 1 SATIS NO MSG 647AC	Replaced	80
2024-05-13	N05306	AFTER VN7151 HAN-DAD, CAPT REPORT ENG 1 EGT OVERLIMIT	C/O REPLACE 01 THERMOCOUPLE EGT FIN 4012KS1. SATIS - ENG 2 IDLE RUN FOR LEAK CHECK AND CONFIRM EGT IND SATIS. A/C SERVICEABLE	Replaced	480
2024-08-01	N05306	SUPPLY POWER APPEAR MSG ENG 1 EGT OVER LIMIT	C/O replaced 01 EGT harness and 01 Thermocouple I.A.W AMM 77-21-43 and AMM 77-21-15 Run up check satis Action: C/O replaced 01 EGT harness and 01 Thermocouple I.A.W AMM 77-21-43 and AMM 77-21-15 Run up check satis	Repaired	25
2024-05-26	N06748	Hn ch khai thc do ENG 1 EGT HIGH	0	-	42

Dec 23 - Nov 24 (Last 12 months)

ATA	Q	OI count
3611		14
2161		11
2126		10
2373		10
2793		9
2751		7
7322		7
7532		6
7721		6
7732		6



HVN CEO Fleet - Top OI drivers 12 Month (rate per 100FC)



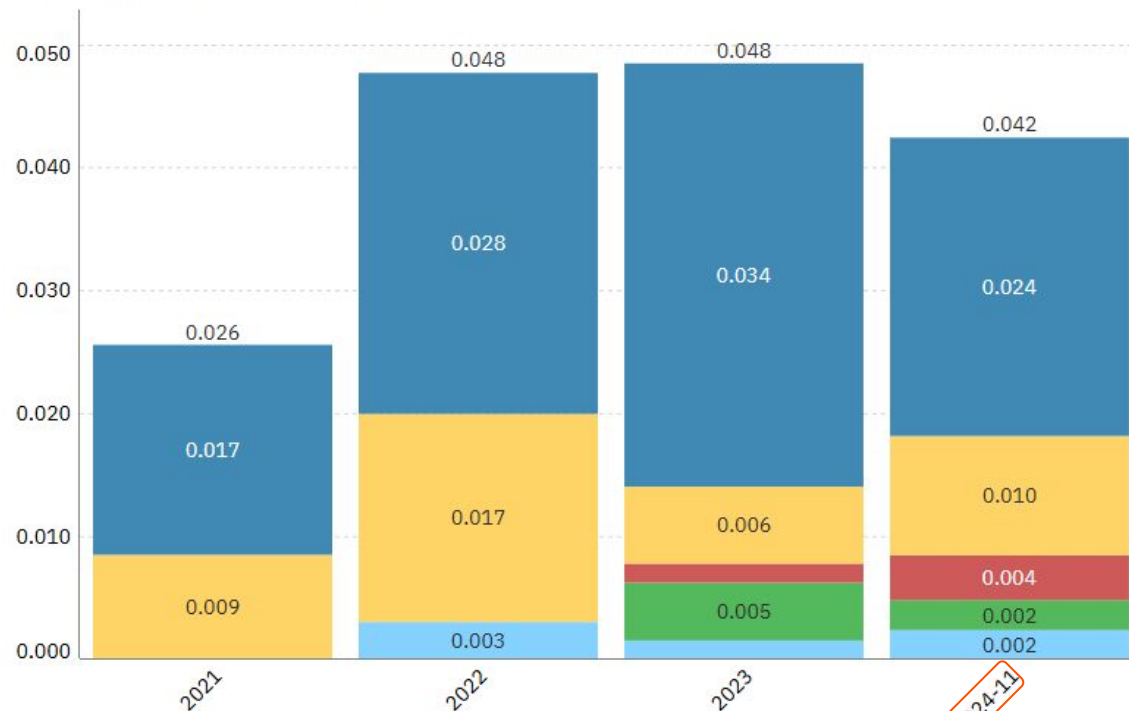
Occurrence date	MSN	Event description	Resolution description	Maintenance action	Delay (minutes)
2024-11-01	N04826	AFTER VN660 CAPT REPORT IN TECHLOG ENG 1 HIGH VIBRATION AFTER VN660 CAPT REPORT IN TECHLOG ENG 1 HIGH VIBRATION	REF TSM 77-30-00-810-826-A: C/o replaced VIB transducer of ENG#1. Eng run up satis & VIB survey test between in limit. A/c serviceable REF TSM 77-30-00-810-826-A EVENT CLOSED	Replaced	17
2024-04-18	N04863	Eng 2 N1 vibration, sau VN650 SIN-SGN REF WO VN 24367649: PER ENGINE CONDITION MONITORING, WE FOUND THAT VN-A329 WITH ENGINE#2 GRADUAL INCREASE IN N1 VIBRATION. (4.6 CU)	- Ch bay c 01 cp SGN//DAD ri N/S ti SGN khc phc, nh hng i tu VN146/147 SGN//DAD - C/O FAN TRIM BALANCE WITH THE EVMU ONE SHOT METHOD, FOUND N1 VIB PARA IS 0.4 UNIT AT 85% N1	Inspected, checked	30
2024-04-27	N04945	engine 1 vibration (rung 5.0)	Tu sau khi h cnh HAN c kim tra, tn timer tt, msg bo gi.	-	40
2024-06-20	N05247	CAPT REPORT ENG 1 N1 HIGH VIBRATION ABOUT 6 UNITS CAPT REPORT ENG 1 N1 HIGH VIBRATION ABOUT 6 UNITS	EVENT CLOSED REFER TSM 77-32-00-810-806-A REV 01/05/24. C/O ENGINE 1 RUN UP VIBRATION SURVEY IAW AMM 71-00-00-700-023-B SATIS. MAX N1 VIB=1.2 CU. A/C SERVICEABLE.	Inspected, checked	35
2024-06-17	N05456	ENG 1 N1 VIB 5.8MAX DURING CRUISE FL340	A/C READY ON 17.Jun.2024 AT 09:45 EVENT CLOSED REF TSM 77-32-00-810-806A ENG 1 N1 VIB 5.8MAX DURING CRUISE FL340. ACTION : VISUAL ENG VIB HARNESS FOUND NML . PERFORM ENG #1 VIB SURVEY TEST FOUND MAX ENG #1 VIB : 0.5	Inspected, checked	155
2024-08-16	N05994	CAPT REPORT: ENG 2 HIGH VIB CAPT REPORT: ENG 2 HIGH VIB	EVENT CLOSED KLM ENGINEER C/O TSM TASK 77-32-00-810-807-A REV AUG/24 FOLLOWED. INSPECTIONS C/OUT FAULT ISOL. SUB TASK A.1.a to A.1.e . NO ABNORMALITIES FOUND. EVMU TEST VIA CFDS WAS NORMAL	Inspected, checked	50

Dec 23 - Nov 24 (Last 12 months)

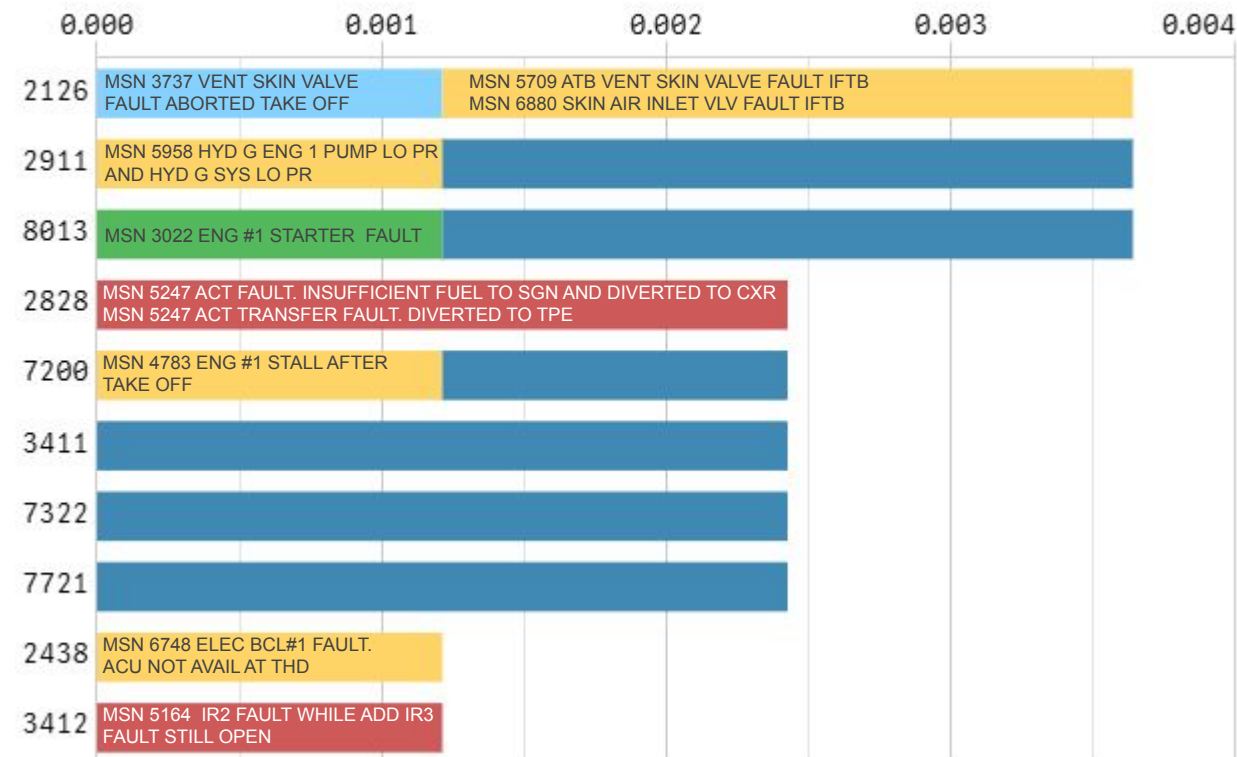
ATA	Q	OI count
3611		14
2161		11
2126		10
2373		10
2793		9
2751		7
7322		7
7532		6
7721		6
7732		6

HVN CEO Fleet - Top Severe Event Rate

Severe event rate per 100 FC (last 4Y)



Top ATA 4D severe event rate per 100 FC (2023-12 - 2024-11)

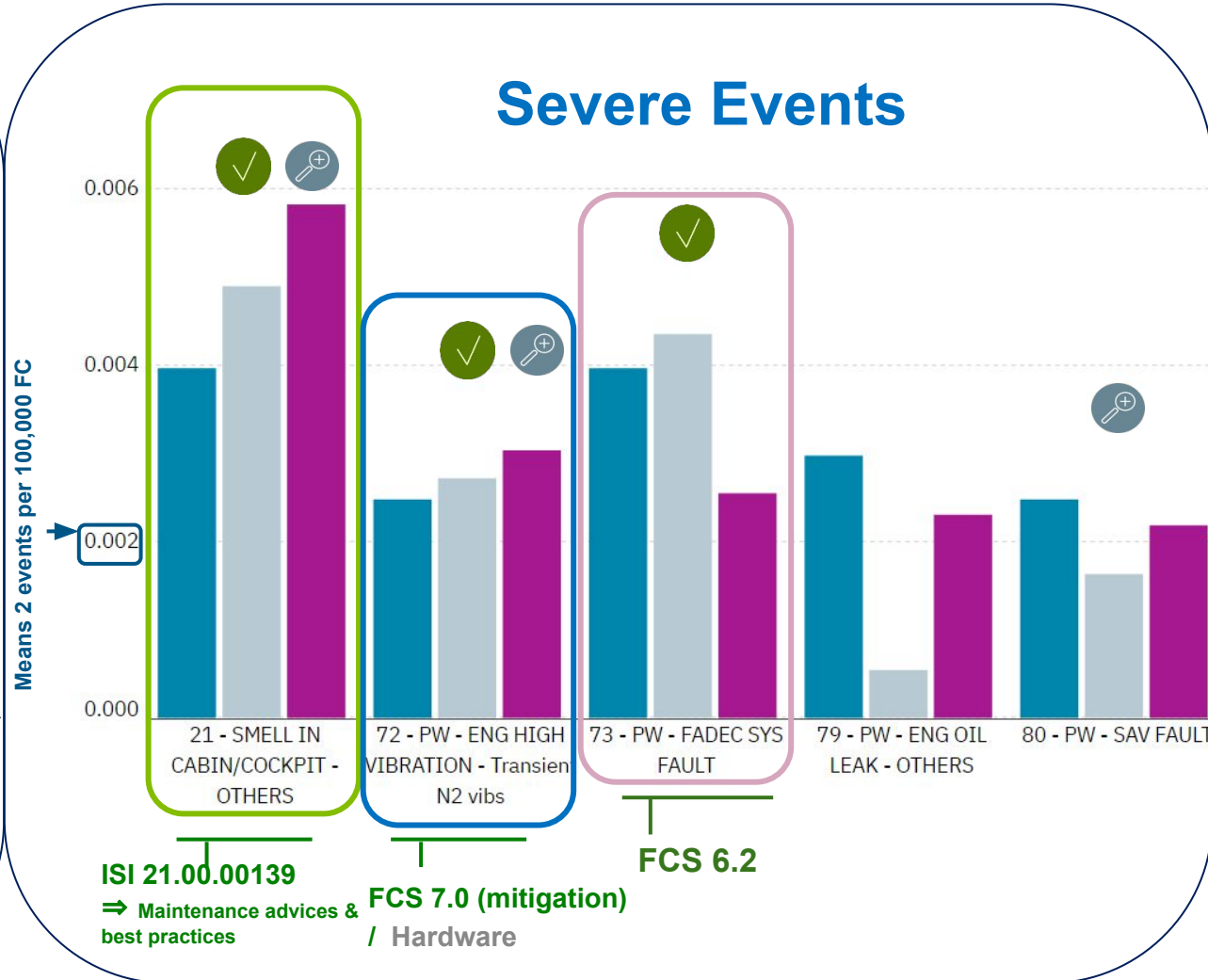
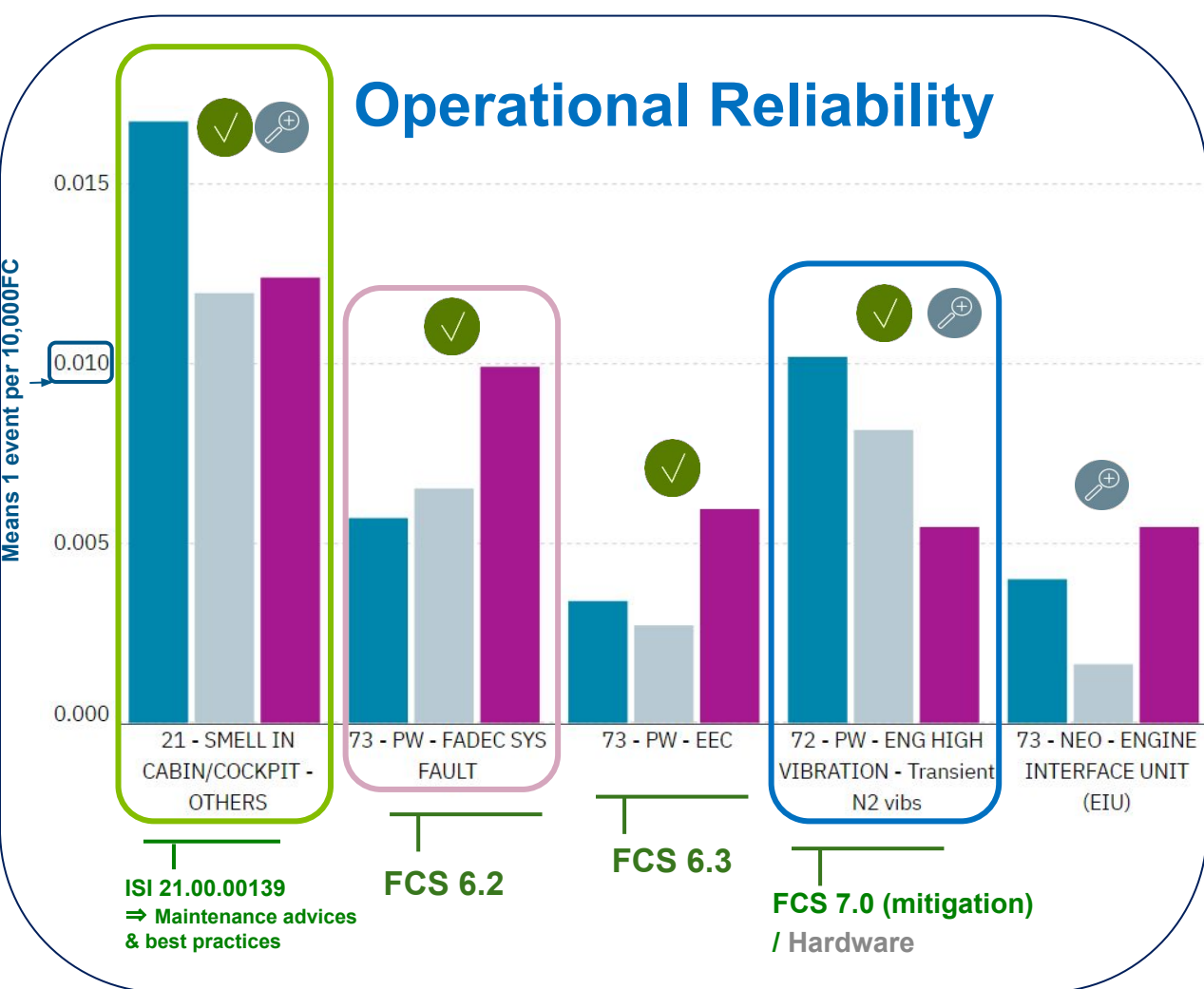


Severe event count (2023-12 - 2024-11)



A320neo Top Fleet Main drivers

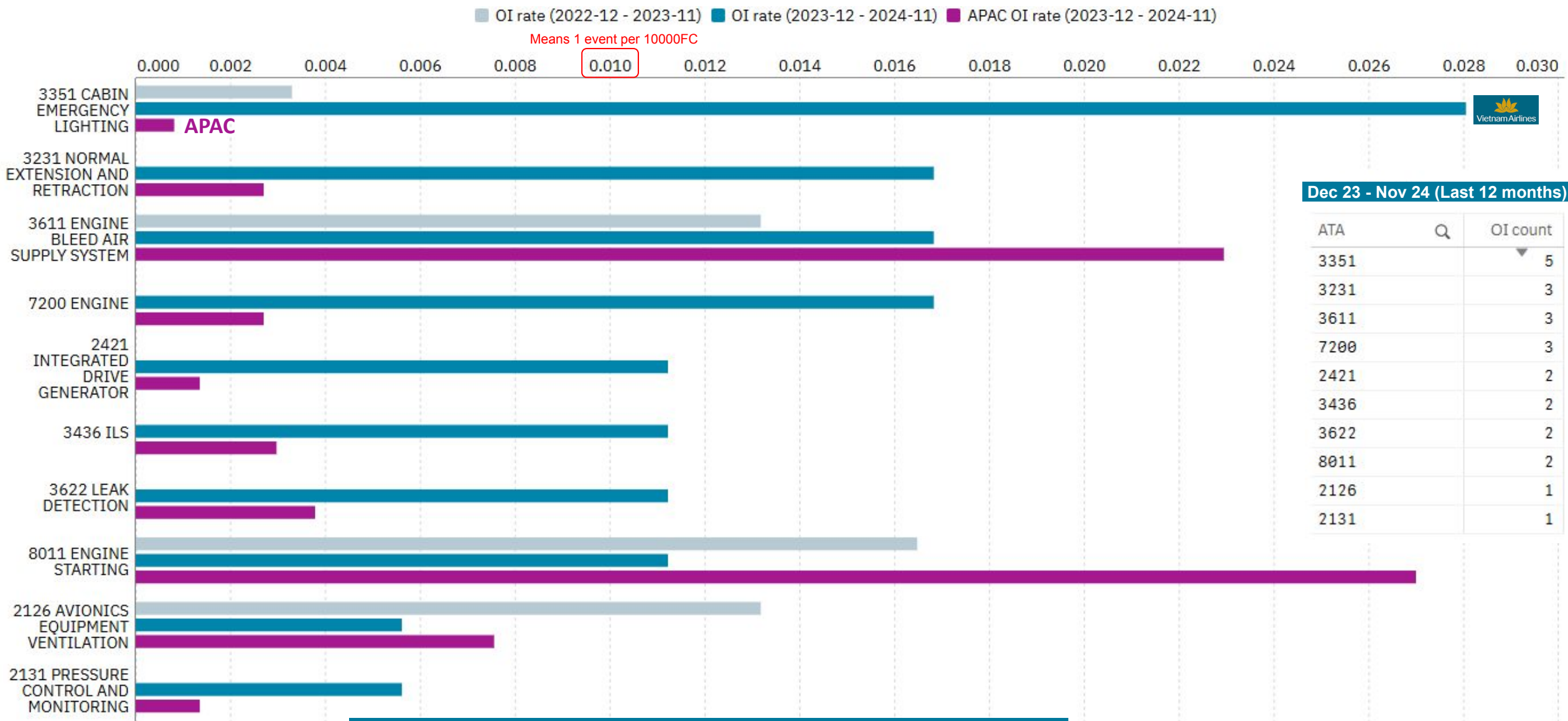
(For neo A/C > 2 years) - (Fleet equipped with PW engines/Collins nacelles)



■ OI rate last 12 months ■ OI rate Mar 24 - May 24 ■ OI rate Jun 24 - Aug 24

✓ Solution available ✓ Solution identified but not available ⚙ On watch / Under investigation

HVN NEO Fleet - Top OI drivers 12 Month (rate per 100FC)



ISI 00.00.00196 A320 Family Major In service Problems status (MISP)

Operator excluded from benchmark

HVN NEO Fleet - Top OI drivers 12 Month (rate per 100FC)

■ OI rate (2022-12 - 2023-11) ■ OI rate (2023-12 - 2024-11) ■ APAC OI rate (2023-12 - 2024-11)

Means 1 event per 10000FC

3351 CABIN
EMERGENCY
LIGHTING

APAC

TFU 33.51.00.022 Power Supply Unit (EPSU/ ASPSU/ RSPSU) Battery Reliability issues (Update on how to retrieve age of the batteries)

Battery capacity test performed on aircraft as per AMM 33-51-00-710-003-A will be possible only until battery age reaches 2 years. After that, battery capacity test will have to be performed only in shop as per Ref. 5 / Ref. 6/ Ref. 7/ Ref. 8 and Ref. 9 revision 13 or higher.

Additionally, in order to ease the monitoring of battery age, Airbus recommends the operators to perform one of the following as communicated via Ref. 10:

Take the “next test date” given on the EASA Form 1 of a new battery and subtract 6 months in order to have the actual Date of Manufacturing (DMF) of the battery available.

Contact DIEHL Customer Service Centers and provide the list of batteries with their S/Ns for which DMF information is required.



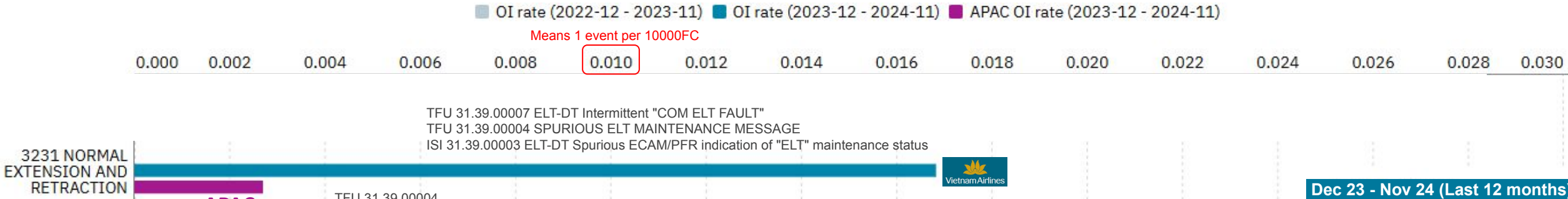
Dec 23 - Nov 24 (Last 12 months)

ATA	Q	OI count
3351		5
3231		3
3611		3
7200		3
2421		2
3436		2
3622		2
8011		2
2126		1
2131		1

Occurrence date	MSN	Event description	Resolution description	Maintenance action	Delay (minutes)
2024-01-27	N08649	AFTER FLT VN705 FOUND 3 EMER LIGHT AT DOOR #2 INOP	AS ABOVE OPEN ADD FOR THIS DEFECT. DUE TO THIS DEFECT, VNA440 CHANGE AIRCRAFT TO FLIGHT BY A613 Final action: replace 02 EPSU 14WL, 15WL	MEL applied	192
2023-12-08	N08669	PRE CHECK FOUND ALL SEATMOUTED CONTROL BY 12WL NOT ILLUMINATED	REF TSM 33-51-00-810-841 WIRING CHECK SATIS. CLEAN CONNECTOR OPS TEST EPSU IAW AMM 33-51-00-710-001 SATIS. A/C SERVICEABLE AS ABOVE	Inspected, checked	53
2024-07-14	N08769	03 RED EMERGENCY LIGHTS AT DOOR #2 INOP	RPLD EPSU. OPS TEST OK EVENT CLOSED	Replaced	126
2024-07-24	N08769	03 RED EMERGENCY LIGHTS AT DOOR #2 INOP 	REPLACED EPSU 14WL AND LIGHT TERMINAL, OPS TEST SATIS. EVENT CLOSED	Replaced	50
2024-07-29	N08769	T/X FOUND EMER LT AT SEAT 31BC, 32ABC INOP. WAITING FOR RPL EMER LT.	VN979 SGN-BOM STD/1755 Dla/0125.	Repaired	85

Operator excluded from benchmark

HVN NEO Fleet - Top OI drivers 12 Month (rate per 100FC)



- A Fix with the Standard S3.2 has been defined with the supplier
- Fix will be under SB 31-1691 [released](#).
- SB Conditions:
 - SB type: Standard Recommended (A320NEO a/c only)
 - ICY: 100 ("One-way interchangeable" with the Old P/N (S1865501-06))
 - Chargeability: FOC (parts and noria) without MH
 - Chargeability Note:
 - FOC conditions applicable for the 2 years after first SB release
 - FOC chargeability already negotiated by Airbus with the Supplier
 - Operators are required to provide FOC purchase order to Supplier (FAU027) in accordance with SB 31-1691"Planning Information" conditions
- **Airbus decided to monitor SB 31-1691 and a RIL (Retrofit Information Letter) will be available by end JAN 2025.**
 - **RIL will contain logistics information for the retrofit and Supplier VSB ref. S1865501-25-01 for the retrofit.**
- **Airbus recommends to launch purchase order for retrofit, upon RIL's release.**

Dec 23 - Nov 24 (Last 12 months)

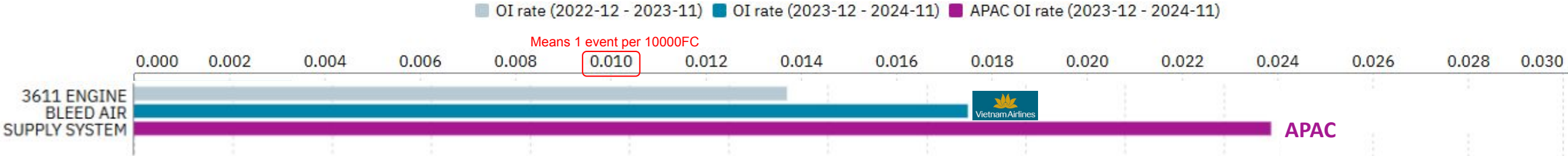
ATA	Q	OI count
3351		5
3231		3
3611		3
7200		3
2421		2
3436		2
3622		2
8011		2
2126		1
2131		1

Occurrence date	MSN	Event description	Resolution description	Maintenance action	Delay (minutes)
2024-11-10	N09051	AFTER VN1381 DLI-SGN CAPT REPORT L/G DOORS NOT CLOSED	FINAL ACTION: FINDING 01 BOLT ATTACHED SENSOR LOOSEN, C/O RE-TIGHTEN BOLT AND OPS TEST FOUND SATIS	Tightened, secured	20
2024-07-24	N12148	MSG: <u>ELT EMITTING</u> AND L/G SYS DISAGREE	OPS TEST OK.MSG GONE	Inspected, checked	35
2024-08-06	N12148	MSG L/G SYS DISAGREE AND COM <u>ELT EMITTING</u> MSG: L/G SYS DISAGREE AND AFTER, FOUND MSG: COM ELT EMITTING	RESET AND TEST FOUND SATIS	Reset	48

Operator excluded from benchmark



HVN NEO Fleet - Top OI drivers 12 Month (rate per 100FC)



Occurrence date	MSN	Event description	Resolution description	Maintenance action	Delay (minutes)
2024-02-14	N08703	AFTER BOARDING APPEAR MSG ENG #1 BLEED NOT CLOSED	AS ABOVE C/O RAISED ADD ENG#1 AIR BLEED INOP, PRV SECURED CLOSED POSITION. FINAL ACTION: RPL ACE PRV + ILDE LEAK CHK ENG 1 SATIS	MEL applied	35
2024-02-28	N08753	BEF PUSHBACK, MSG APPEARED - AIR APU BLEED LEAK - ENG 2 LO START AIR PR	T/S IN PROGRESS - C/O LEAK CHECK OF THE APU DUCT BTW APUCV AND APUBV SATIS. - C/O INSPECT CHECK VALVE 7260HM SATIS. - C/O BMC1 WIRING CHECK: PIN AA/4C TO GRND: THE RESISTANCE VALUE GREATER THAN THE TSM REF VALUE: BMC1 RPLD AND OPS TEST SATIS. - ENG 12 RUN FOR OPS TEST SATIS.	Replaced	205
2024-09-25	N08985	AIR ENG #2 BLEED FAULT MCC BO DNG TU N 18H T/S	- 04:05 CAPT INFORM VIA VHF MSG: AIR ENG #2 BLEED FAULT - 04:25 MCC BO DNG TU T/S - 05:00 OCC GI A506 BAY VN7346 (STD:12H30, NETD: 18H00), KHNG I TU - 06:00 TU KO V HG1 - 06:35 FOUND ROD FAN AIR VALVE ENG 2 DAMAGED - 08:10 ROB BOLT T A616 CHO A506, RPLD ROD ASSY, VNG 874 BASE N MY - 08:40 KO TU RA N MY TEST - 09:00 FINAL ACTION: C/O RPLD ROD ASSY OF FAN AIR VALVE, C/O ERU FOR OPS TEST ENG #2 BLEED SYS SATIS, A/C SERVICEABLE. - 09:30 TU KO LN SN - 11:00 DOOR CLOSED AIR ENG #2 BLEED FAULT EVENT CLOSED	Replaced	210

3622 LEAK DETECTION

Occurrence date	MSN	Event description	Resolution description	Maintenance action	Delay (minutes)
2024-02-29	N08753	TRANSIT CHECK, AFTER PUSHBACK (02 ENG RUNNING) CAPT REPORT: APU BLEED & ENG 1 BLEED FAULT. A/C GTB	AS ABOVE - REPLACED ELEMENT SENSING 17HF SATIS - ENG START AND RUN SATIS - USING BLEED 2H NIL MSG, REMEASURE SATIS A/C SERVICEABLE	Replaced	45
2024-06-15	N09051	MSG AIR L WING LEAK AFTER BOARDING FOUND MSG ;AIR L WING LEAK;	C/O RESET C/B FOUND SATIS. MSG DISAPPEARED RESET A/C READY ON 15.Jun.2024 AT 02:10 EVENT CLOSED	Inspected, checked	30

Dec 23 - Nov 24 (Last 12 months)

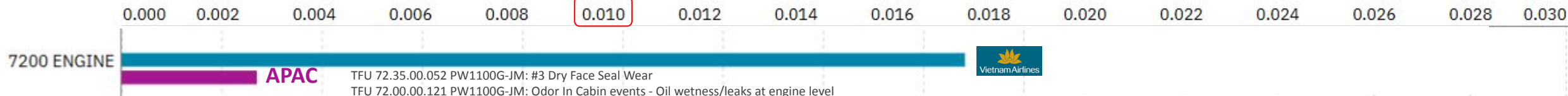
ATA	Q	OI count
3351		5
3231		3
3611		3
7200		3
2421		2
3436		2
3622		2
8011		2
2126		1
2131		1

Operator excluded from benchmark

HVN NEO Fleet - Top OI drivers 12 Month (rate per 100FC)

■ OI rate (2022-12 - 2023-11) ■ OI rate (2023-12 - 2024-11) ■ APAC OI rate (2023-12 - 2024-11)

Means 1 event per 10000FC



Occurrence date	MSN	Event description	Resolution description	Maintenance action	Delay (minutes)
2023-12-17	N08699	CAPT REPORT: DURING APPROACH IN VVNB, BAD SMELL IN COCKPIT AND CABIN	T/S INPROGRESS WAITING TAR2300733 TU BQDB T? 22/12/2023-21/01/2024 ER-2023-1150: A621_ENG 1_ ESN P770994: + Oil wetting at LPC2, LPC3 and HPC 2ND stage at 6h + Oil wetting at 2.5 bleed, weep holes + No leak at the Fan shaft cover. ENG 1 must be replace.	Replaced	35
2023-12-23	N08729	REF HVN-20231222-771685-00025 FROM PW: THE HIGH N2 VIBRATION SUGGEST POTENTIAL SHIFT IN IMBALANCE OF THE HIGH ROTOR. A/C MUST BE C/O REQ FR PW: - PROVIDE FULL 5 FLT DATA FROM 22DEC23, MINIMUM 5 LAST FLT. - HIGH ROTOR PEAK COCKPIT UNIT (CU) RECORDED - ASSOCIATED N2 SPEED IN THE GROUND RUN, (%). - A/C ON GRN WAITING SOLUTION FROM PW	IN PROGRESS - E.R.U AND DOWNLOAD QAR, DATA SENDED TO PW - A/C ON GRN WAITING SOLUTION FROM PW	Inspected, checked	40
2023-12-10	N09051	TX CHK -> Capt report bad smell after engine start (even when the packs are off and APU bleed off), bad smell during approach.	C/O TSM 71-00-00-810-847-A check found oil puddle at ENG 1 FAN SHAFT cover, replace new seal satis, C/O ERU no bad smell.	Repaired	35

* SB 73-1154 FCS 7.0 FADEC SOFTWARE on NEO PW partially embodied for HVN fleet

INTRODUCE FCS 7.0 FADEC SOFTWARE ON NEO PW

		Scheduled	Fully Embodied
8501	VN-A616	✓	✓
8545	VN-A617	✓	✓
8589	VN-A618	✓	✓
8649	VN-A620	✓	✓
8669	VN-A501	✓	✓
8678	VN-A619	✓	✓
8699	VN-A621	✓	✓
8703	VN-A622	✓	✓
8729	VN-A623	✓	✓
8747	VN-A624	✓	✓
8753	VN-A625	✓	✓
8769	VN-A503	✓	✓
8924	VN-A505	✓	✓
8985	VN-A506	✓	✓
9051	VN-A507	✓	✓
9108	VN-A508	✓	✓
9161	VN-A509	✓	✓
9241	VN-A510	✓	✓
9242	VN-A511	✓	✓
9294	VN-A512	✓	✓

Dec 23 - Nov 24 (Last 12 months)

ATA	Q	OI count
3351		5
3231		3
3611		3
7200		3
2421		2
3436		2
3622		2
8011		2
2126		1
2131		1

8011 ENGINE STARTING



Occurrence date	MSN	Event description	Resolution description	Maintenance action	Delay (minutes)
2024-03-29	N09294	DURING START ENG, CAPT REPORT: ENG#2 START OK, BUT ENG#1 STARTER VLV NOT CLOSED.	C/O RESET FADEC SATIS, C/O START ENG OK,	Reset	60
2024-09-02	N09294	ENG 1 START VAL FAULT. START VALVE NOT CLOSE	Ref TSM 80-11-00-810-828 C/O rpl Eng #1 starter vlv , ops test found satis.	Replaced	95

Operator excluded from benchmark

HVN NEO Fleet - Top OI drivers 12 Month (rate per 100FC)



Occurrence date	MSN	Event description	Resolution description	Maintenance action	Delay (minutes)
2024-01-26	N08669	ELEC IDG#2 OIL LOW LVL	Waited CRS K6 (on call) raise ADD REF MEL 24-09-03C	MEL applied	38
2024-02-25	N08669	ECAM MSG : ELEC IDG #2 LOW OIL LVL .	AS ABOVE C/O CHECK AND APPLIED MEL 24-09-03C.	MEL applied	16



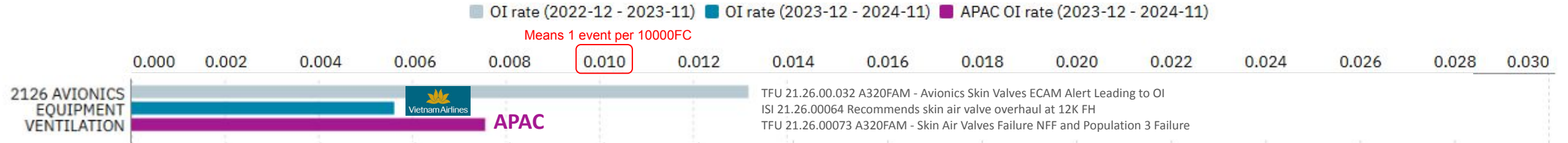
Occurrence date	MSN	Event description	Resolution description	Maintenance action	Delay (minutes)
2024-01-04	N08703	CAPT REPORT: 1. NAV LS TUNING DISAGREE 2. NAV ILS 1 FAULT	1. DO OPS TEST LAND CAT III IAW AMM 22-97-00-710-001 AND BITE TEST MMR1 TSM 36-00-810-896A SYS OK. 2. C/B RECYCLE AND BITE TEST MMR1 TEST NML. A/C SERVICEABLE. AS ABOVE	Reset	21
2024-03-14	N08753	After boarding found GPS SYS fault, ADS1+2 fault	Reset many time nil help Raised 02 B defect: ADD B380 GPS SYS INOP ADD B831 ADS-B OUT FUNCTION INOP	MEL applied	65

Dec 23 - Nov 24 (Last 12 months)

ATA	Q	OI count
3351		5
3231		3
3611		3
7200		3
2421		2
3436		2
3622		2
8011		2
2126		1
2131		1

Operator excluded from benchmark

HVN NEO Fleet - Top OI drivers 12 Month (rate per 100FC)



Occurrence date	MSN	Event description	Resolution description	Maintenance action	Delay (minutes)
2024-09-01	N09051	DURING T/X APPEAR MSG VENT AVIONIC SYS FAULT	T/S MSG VENT AVIONIC SYS FAULT - 11:50 A/C ARIVED BAY 11 - 12:40 BOARDING COMPLETED - 12:45 APPEAR MSG VENT AVIONIC SYS FAULT - 12:50 RESET AND TEST CB OK - 13:00 MSG APPEARED AGAIN - 13:00 CALL TECH SGN FOR SUPPORT - 14:00 STILL T/S (CAN RAISED DEFECT BUT STILL DIDN'T DEACTIVATE VLV) - 15:00 A/C SERV - 15:10 REFUELING - 16:00 START BOARIDNG (DELAY DUE TO IT'S RAINNING) - 16:20 A/C CLOSED DOOR TOTAL DELAY 210 MINS , TECH DELAY 120 MINS	MEL applied	120

2131 PRESSURE CONTROL AND MONITORING

Occurrence date	MSN	Event description	Resolution description	Maintenance action	Delay (minutes)
2024-01-12	N09241	CAPT REPORTED: CAB PR SYS 1 + 2 FAULT. A/C DIV TO SGN.	T/S IN PROGRESS REF TSM 21-31-00-810-807-A OPS TEST FOUND FAULT CODE 26, CHK OUTFLOW VLV FOUND HINGER OF OUTFLOW VLV BROKEN. C/OUT RPLD OUTFLOW VLV. OPS TEST SATIS.	Replaced	-

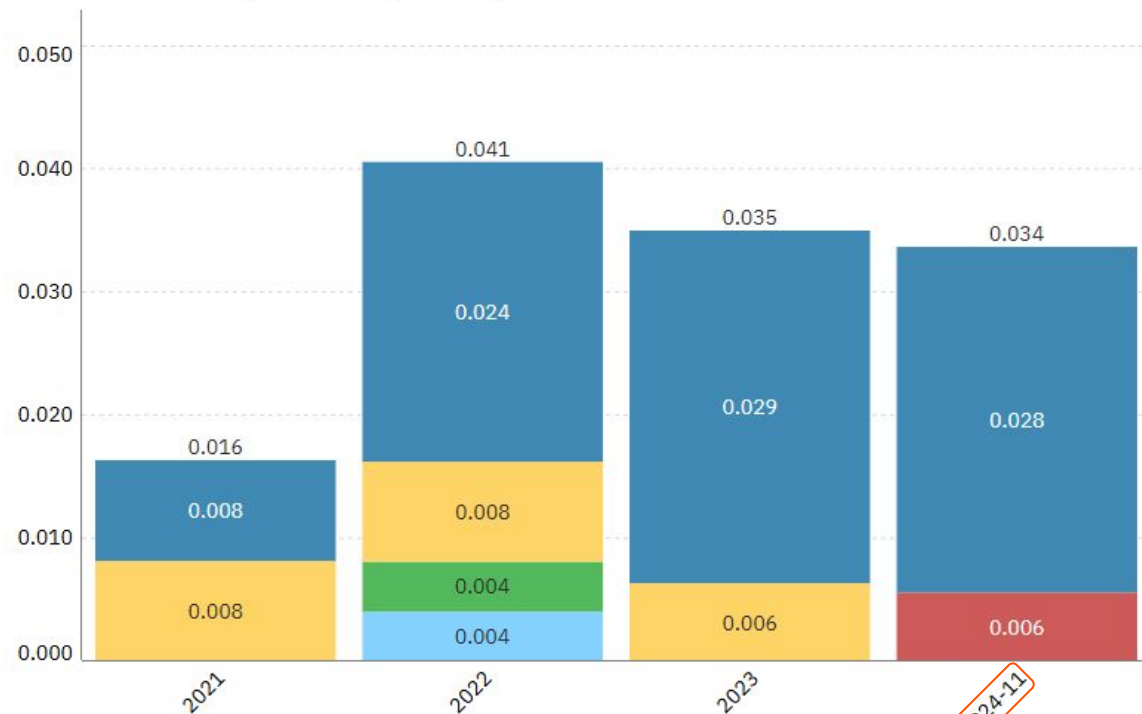
T/S found hinge of outflow vlv broken. Outflow vlv replaced.
Otherwise useful - ISI 21.31.00041 A320FAM - Cabin Depressurization Troubleshooting Advices
- ISI 21.30.00010 A320FAM - Cabin Pressure Control System Main Enhancements

Dec 23 - Nov 24 (Last 12 months)

ATA	Q	OI count
3351		5
3231		3
3611		3
7200		3
2421		2
3436		2
3622		2
8011		2
2126		1
2131		1

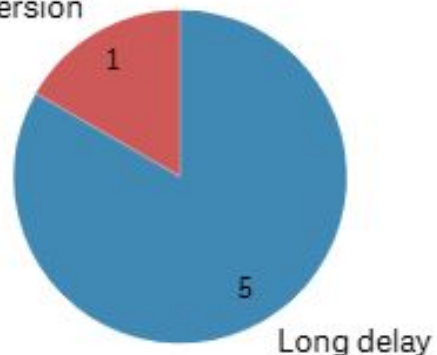
HVN NEO Fleet - Top Severe Event Rate

Severe event rate per 100 FC(last 4Y)



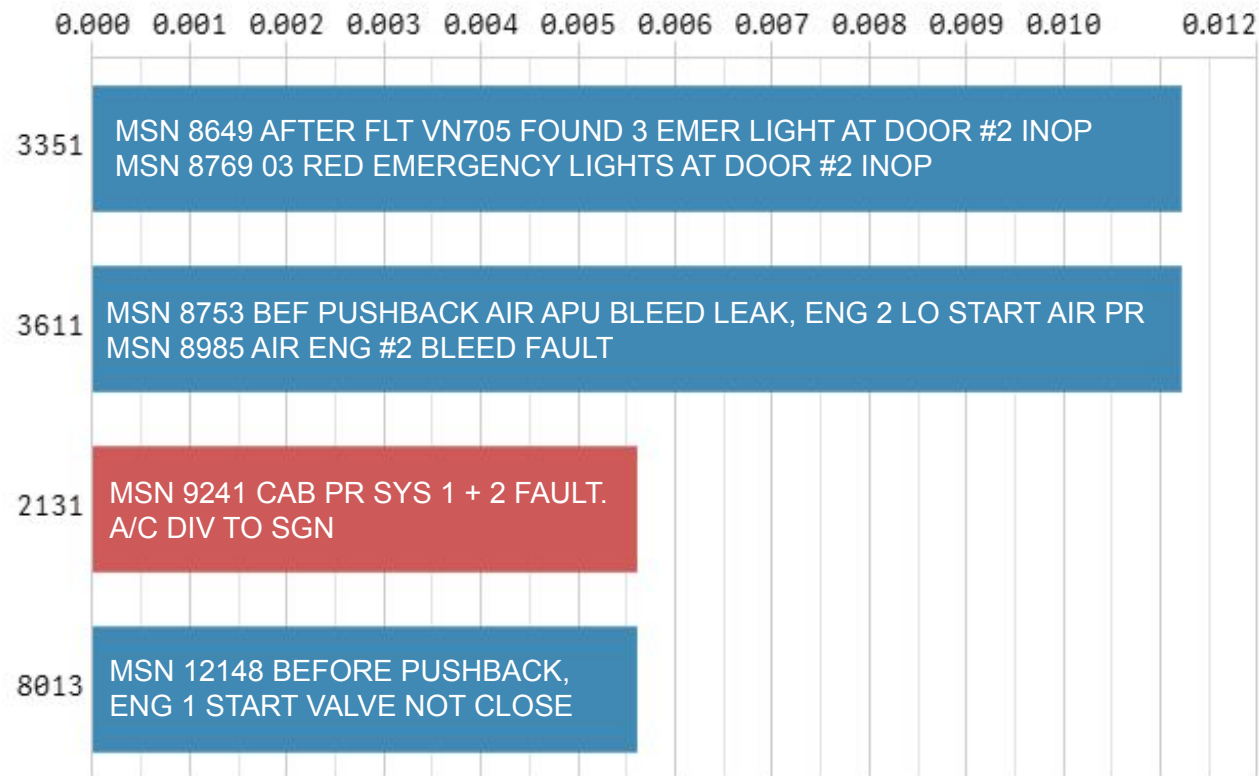
Severe event count (2023-12 - 2024-11)

Diversion



- Aborted TO
- Cancellation
- Diversion
- IFTB
- Long delay >120 mins

Top ATA 4D severe event rate per 100 FC (2023-12 - 2024-11)





A320 Fam Safety Beyond Standard

Peter Ng

AIRBUS

Why Safety Beyond Standard (SBS) ?

Safety Beyond Standard (SBS) initiative aims to further enhance the safety of the fleet beyond requirements, by reducing exposure to 8 identified safety threats

→ Focusing on the top 3 safety threats:



Loss Of Control In-flight

SW FOC



- Enhanced AP / FD / ATHR availability
- Enhanced flight envelope protection availability
- Enhanced flight crew interface

→ Supported by new standards of EIS2, FWC, ELAC, FAC C, FMGC 3G, HUD



Controlled Flight Into Terrain

Attractive price



- Enhanced guidance during landing
- Enhanced trajectory & energy management during go-around

→ Supported by new functions (FLS FMS Landing system, SLS SBAS Landing System, EGPWS upgrade, SGA soft go around)

Contact CSD for further information



Runway Excursion

Attractive price



- Enhanced runway overrun risk awareness
- Enhanced runway condition awareness
- Takeoff performance surveillance

→ Supported by new functions (ROPS, BACF, TOS)

Contact CSD for further information

→ Implementation in retrofit:

- SB standard recommended or RFC/RMO → all released
- Attractive commercial policy available as per (Retrofit Information Letter) conditions

SBS is a continuous improvement initiative ; regular updates via SBIT/RIL/SB/ISI

Key points

- Safety Beyond Standard is a priority for the A320Fam community
- Airbus supports its deployment in retrofit through Free of Charge conditions for software upgrade (LOC-I)
- For A/L status, or question, please contact Airbus monitored retrofit monitored.retrofit@airbus.com



Refer to ISI article

22.00.00104

00.00.00438

AIRBUS

Implementation of Safety Beyond Standard on HVN fleet (Focus on prevention of Loss Of Control in Flight)

HVN fleet size A320Fam	65 A/C (42 CEO / 23 NEO) (at date 17-FEB-2025)
------------------------	--

Computer / Standard / Function	Sub-fleet requiring SW upgrade only	Delivery rate for the sub-fleet	Embodiment rate for the sub-fleet
EIS2 S16	42 CEO 20 NEO	100% 100%	100% 100% Completed 
FWC F13	42 CEO 20 NEO	100% 100%	95% 100% IN PROGRESS
FAC CAA13	5 CEO 20 NEO	100% 100%	100% 100% Completed 
FMGC FMS H4 FG PI18	20 NEO	100%	100% Completed 
FMS H2C FG PI18	5 CEO	100%	100% Completed 
ELAC L104	42 CEO 20 NEO	100% 100%	100% 80% IN PROGRESS
Digital Back Up speed (UAMM step 2) = Nbr of a/c which have Step1 applied and can do Step2	5 CEO 20 NEO	0% n/a	0% 100% To be planned

Key points

- FWC F13: 2 CEO A/C remaining
- ELAC L104: 4 NEO A/C remaining

Appendix - Applicable SBs and RILS

Computer / Standard	SBs reference	RILs reference	Commercial policy end date for the sub fleet requiring SW upgrade only
EIS2 S16 (and S17)	EIS2 S16 NEO 31-1577, CEO 31-1576 EIS2 S17 NEO 31-1595, CEO 31-1596 (EIS2 S17 not monitored)	SA31M19006813	FOC for A/C with EIS2 until 30-JUNE-2025
FWC H2F13	NEO 31-1609, CEO 31-1608	SA31M21004114	FOC until 31-DEC-2025
FAC CAA13	NEO 22-1798, CEO 22-1797	SA22M21000009	FOC for A/C with FAC C until 31-MAR-2024
FMGC FMS H4 FG PI18	NEO 22-1819	SA22M21001923	FOC until 31-MAR-2024
FMGC FMS H2C FG PI18	CEO 22-1896	SA22M23001404	FOC until 31-DEC-2025
FMGC FMS H2C FG I16	CEO 22-1B15 → in preparation		
ELAC L104	NEO 27-1307 CEO 27-1305, 27-1306	SA27D22021674	FOC until 28-FEB-2027
Digital Back Up Speed (UAMM step 2)	NEO 22-1739, CEO 22-1738	SA22M21002261	Material : SB A320-22-1738 – FOC until 30-JUNE-2026 for conf 002 SB A320-22-1739 – Not applicable

Digest of available safety-related enhancements

For your A320 fleet



A320 Family Digest Edition 2024

A selection of valuable safety/operational enhancements

- Considering their potential benefits
- To reduce the number of OEBs
- Covering Safety Beyond Standard initiative
- A complement to other operational recommendations

Version 11	
Table of contents – ATA chapter	
ATA 21 UPDATED Modification of label for precautions during ground air conditioning.....09 UPDATED Introduction of new CPV, CPD and Flight Deck Monitoring.....09 UPDATED Introduction of new skin air valves logic and monitoring.....10 Introduction of new Avionics Equipment Ventilation Computer (PEVC).....11 Introduction of Ball Bearing Health Monitoring on avionics fan.....12	ATA 24 Upgrade to Generator Control Unit standard 5.2.....26 Enhancement of Electrical Network Reconfiguration.....27 ATA 24 ATA 25 Modification of static inverter.....28 ATA 25 New cockpit storage box robust to Portable Equipment Devices lithium battery runaway.....29 NEW Introduction of new probe covers.....14 NEW Introduction of a probe cover on cockpit seats.....13 UPDATED Introduction of FG standard PC20/P18.....14 NEW Introduction of FG standard C1516.....15 Introduction of FMS2 Release 1A Standard 37B (Thales).....17 UPDATED Introduction of FMS2 Release 2 standard H3 (Honeywell).....16 Introduction of FMS2 Release 1A Standard 37B (Thales).....17 UPDATED Introduction of FMS2 Release 2 standard SBA (Thales).....20 Introduction of FMS2 Release 2 standard H4 (Honeywell).....19 UPDATED Introduction of FMS2 Release 2 standard SBA (Thales).....20 Activation of the Automatic WVR mode engagement at Go Around.....21 UPDATED APFD logic enhancement.....22 Tail strike indication on PFD and YRCH PITCH call out at landing.....23 Introduction of FAC BGS.....24 UPDATED Introduction of FAC CAA13.....25
ATA 24 Upgrade to Generator Control Unit standard 5.2.....26 Enhancement of Electrical Network Reconfiguration.....27 ATA 24 ATA 25 Modification of static inverter.....28 ATA 25 New cockpit storage box robust to Portable Equipment Devices lithium battery runaway.....29 NEW Introduction of new probe covers.....14 NEW Introduction of a probe cover on cockpit seats.....13 UPDATED Introduction of FG standard PC20/P18.....14 NEW Introduction of FG standard C1516.....15 Introduction of FMS2 Release 1A Standard 37B (Thales).....17 UPDATED Introduction of FMS2 Release 2 standard H3 (Honeywell).....16 Introduction of FMS2 Release 1A Standard 37B (Thales).....17 UPDATED Introduction of FMS2 Release 2 standard SBA (Thales).....20 Introduction of FMS2 Release 2 standard H4 (Honeywell).....19 UPDATED Introduction of FMS2 Release 2 standard SBA (Thales).....20 Activation of the Automatic WVR mode engagement at Go Around.....21 UPDATED APFD logic enhancement.....22 Tail strike indication on PFD and YRCH PITCH call out at landing.....23 Introduction of FAC BGS.....24 UPDATED Introduction of FAC CAA13.....25	ATA 29 Activation of the PTU inhibition logic to prevent potential loss of hydraulic systems.....41 ATA 30 NEW Introduction of new WHC standard.....42 ATA 31 Stat Warning awareness enhancement.....43 Activation of Aural indication on Engine Warning Display EWD.....44 Activation of ATC Transponder mode.....45
Version 11	
Introduction of FG standard PC20/P18 UPDATED	
High pitch protection during Go Around New safety functions capability	
A320 family A/C fitted with FG 3G	
> Description of safety topic FG standards PC20 & P18 introduce several enhancements, some Safety Beyond Standard (see ISI 22.00.00104): - Enhance autopilot robustness in case of FMS failure and in case of Avoid excessive pitch during Go Around with AP engaged on A320 and A319 CEO; - Avoid FMGC reset during flare in crosswind conditions on A321 NEO; - TCAS Alert Prevention function (TCAP), which allows to avoid unnecessary resolution advisories when capturing flight level; - FMGC hardware pin-prog display on MCDU; - A/THR auto-engagement in overspeed situation to automatically re-protection against secondary glide slope capture. > Available enhancements The following SBs are subject to Airbus monitored retrofit campaign: - SBAP18 HW4: NEO A320-22-1827; CEO A320-22-1826 (RIL SA22M21003904); - SBAP20 HW4: NEO A320-22-1828; CEO A320-22-1829 (RIL SA22M21003904); - SBAP18 HW1: NEO A320-22-1824 (RIL SA22M21003904); CEO A320-22-1823 (RIL SA22M21003904); - SBAP20 HW1: NEO A320-22-1822 (RIL SA22M21002334); CEO A320-22-1821 (RIL SA22M21002334); - H4P118: NEO A320-22-1819 (RIL SA22M21001923); CEO A320-22-1818 (RIL SA22M21003964); - H4P20: NEO A320-22-1821 (RIL SA22M21003901); CEO A320-22-1820 (RIL SA22M21003901); - H2CP20: CEO A320-22-1895 (standard compatible with ES1) (RIL SA22M21003904); - H2CP18: CEO A320-22-1896 (standard compatible with ES1) (RIL SA22M21003904); > Other references - ISI 22.83.00003 > Contacts For more information on the monitored retrofit campaign please contact Operations at the following email address: monitored.retrofit@airbus.com	

Version 11

25

Safety Beyond Standard

Introduction of FAC CAA13

UPDATED

Enhancement of Characteristic Speeds

Enhancement of ROW/ROP function

Enhancement of aircraft attitude crew awareness

ATA 22

A320 family A/C fitted with FAC C

Cancel OEB 54

> Description of safety topic

Improvement of ROW/ROP function availability and accuracy. FAC and FMGC pin-programming status could only be checked by wiring continuity test with multimeter that led to maintenance mistakes during pin-programming modification.

> Available enhancements

FAC CAA13 introduces the following enhancement part of Safety Beyond Standard:

- Display of a digital estimated air speed display on the PFD in case of Unreliable Airspeed Situation (UAMM Step 2).

The installation of FAC CAA13 introduces the following safety enhancements:

- Improvement of FAC gross weight and VLS/S/F/Green dot speed computations in case of invalid FMS data;
- ROPS step 2+ availability (runway condition selector + ROPS OFF P/B);
- Fine tunings of the Runway Overrun Protection (ROP) stop distance and the Runway end Overrun Warning (ROW) ground distance;
- Address undue SURV ROW/ROP LOST messages in phase 9;
- FAC and FMGC pin-programming status is available on MCDU LRU IDENT page.

This modification requires installation of FAC CAA13 standard as follows:

- **SB A320-22-1797** (CEO);
- **SB A320-22-1796** (NEO).

These SBs are subject to a monitored retrofit through RIL SA22M21000009.

> Other references

- ISI 22.60.00004 - FAC - Evolution & interchangeability

> Contacts

For more information on the monitored retrofit campaign, please contact Retrofit Operations at the following email address: monitored.retrofit@airbus.com

Safety Beyond Standard
items identified with the following tag



Digest Available
enhancement_S/

[Click here](#)
to get the digest
in AirbusWorld

Appendix - Useful links and references

ISI article	22.00.00104/00.00.00438
Safety Digest	In Airbus World: Flight Ops > Operational Expertise > Digest of available enhancements



ELAC/SEC/SFCC Fault

Marius OLSZEWSKI

AIRBUS

Description

- Ol's due to flight computer (ELAC,SEC,SFCC) failure
- Electronic board, relay failures
- High number of NFF's
- FCOM does not cover all messages related to FC's.
- HVN is requesting one time reset procedure

HVN request:

FCOM doesn't cover all msg related to flight control. HVN seeks AIB to provide a one time reset procedure for EW related to flight control



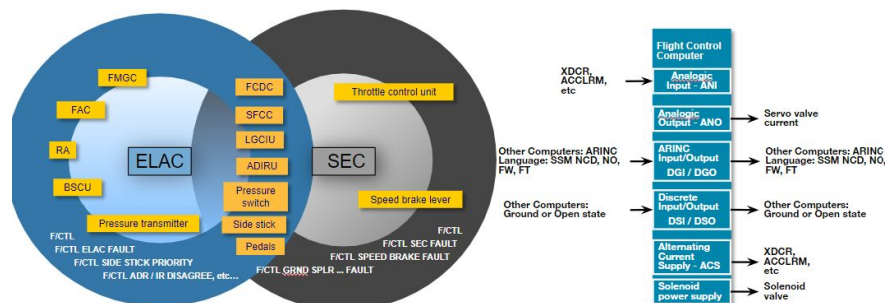
Root cause

EFCS complex environment



Multiple monitorings, and thus any discrepancy detected will lead to:

- ECAM with **failure messages linked with ELAC/SEC** computers (e.g. detected by the ELAC or SEC, which does not mean that it is failed)



- TSM architecture can propose computer removal



Operators complex environment

Operators also within a complex environment:

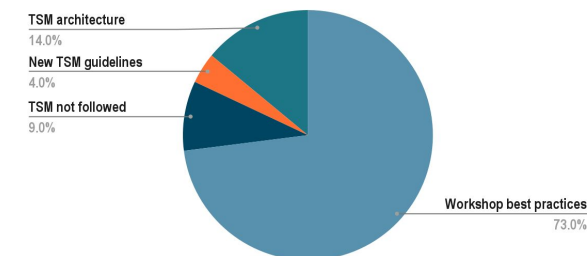
- ❑ very **short time to T/S** (dispatch pressure)
- ❑ sometimes **not a complete/clear knowledge of ATA27 system** (which can be complex)

Note: It is important to differentiate between:

- **ECAM Warning & status display:** intended to the flight crew
- **Failure messages:** intended to the maintenance staff

⇒ Unnecessary ELAC/SEC removals leading to MTBUR decrease and NFF rate increase. This was demonstrated thanks to **2010 EFCS Working group (ISI 27.90.00002)**

Repartition of NFF avoidable by T/S



ATA27 FLIGHT CONTROLS COMPUTER REMOVALS



Reliability



⇒ Worldwide

OEM (Thales) data from mid-2023 to mid-2024:

- ❑ **Good and stable reliability** above GMTBUR
- ❑ Overall stable NFF rate during the analysis period
- ❑ No new ELAC/SEC in-service driver

Detailed figures (Dated Dec 2023)	P/N	GMTBUR (FH)
ELAC B-DL	3945129100	6 000
ELAC B	3945128XXX	6 000
SEC BB'	B372BAM05XX B372BBM01XX	3000
SEC C	B372CAM01XX	7 500

Reliability ELAC B & B-DL

> From July 2023 to June 2024

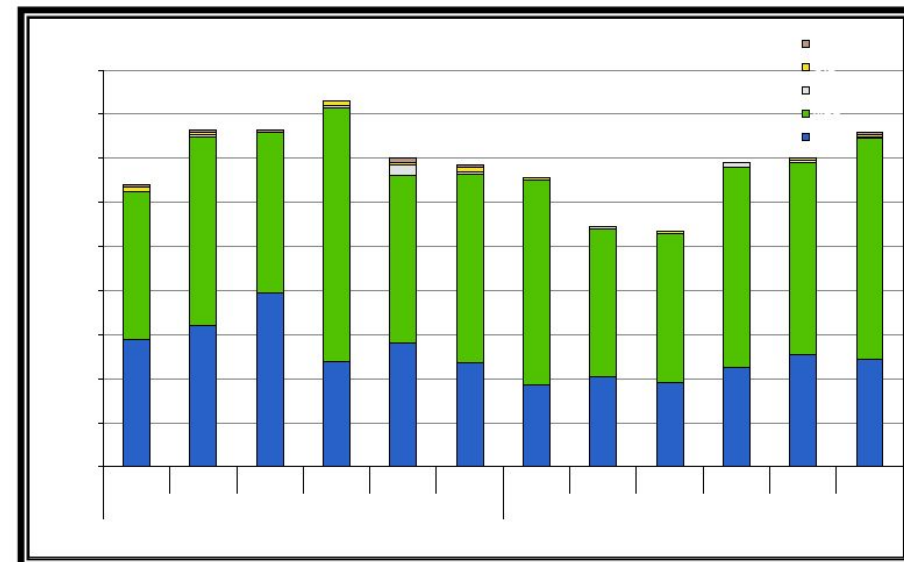
> 1654 in Service Removals

- 626 FF 38%
- 1010 NFF 61%
- 18 NED 1%

> Quantity per aircraft : 2

MTBUR warranty: 6000 FH

Detailed figures (dated Feb-2021)	MTBUR (Flight Hours)	MTBF (N) (Flight Hours)	NFF
ELAC-B	10 959 FH	26 969 FH	59 %
ELAC-BDL	42 795 FH	127 989 FH	65 %



RELIABILITY SEC B/B

From July 2023 to June 2024

1158 in Service Removals from July 2023 to June 2024

- 649 FF 54,6%
- 509 NFF 44%
- 16 NED 1,4%

Quantity per aircraft : 3

NFF Rate 39 % (WW)	MTBUR	MTBF
Warranty Target	3000	-
THALES reliability (12 months sliding)	15201FH	25098FH



RELIABILITY SEC C

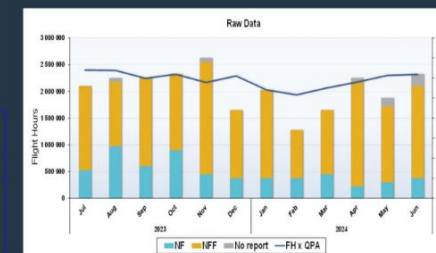
From July 2023 to June 2024

583 in Service Removals from July 2023 to June 2024

- 119 FF 21%
- 462 NFF 79%
- 2 NED

Quantity per aircraft : 3

NFF Rate 79 % (WW)	MTBUR	MTBF
Warranty Target	7500	-
THALES reliability (12 months sliding)	81163FH	328762FH





Reliability



⇒ Worldwide

OEM (Thales) data from mid-2022 to mid-2023:

- ❑ **Good and stable reliability** above GMTBUR
- ❑ Overall stable NFF rate during the analysis period
- ❑ No new ELAC/SEC in-service driver

Detailed figures (Dated Dec 2023)	P/N	GMTBUR (FH)
ELAC B-DL	3945129100	6 000
ELAC B	3945128XXX	6 000
SEC BB'	B372BAM05XX B372BBM01XX	3000
SEC C	B372CAM01XX	7 500



Airbus advice

How can reliability be improved?

⇒ Focus on the **troubleshooting**

Available workshop dedicated to EFCS to support operators with the objectives to:

- Prevent unnecessary removals of ELAC & SEC thanks to a better understanding of:
 - ELAC & SEC system (EFCS)
 - ATA 27 failure message and ECAM warning generation
- Provide guidelines and tips to the Maintenance Team for a quicker and more efficient fault resolution
- For info refer to AIRBUS WORLD



FCDC 60 available, allows to save troubleshooting time and reduce undue ELAC removal:

- Addition of specific maintenance messages for push button reset: "ELAC PB SW", "SEC PB SW" with ELAC FAULT, SEC FAULT
- Creation of TSD for "ELACx" fault
- Specific ECAM display for SSTU/SSU faults
"F/CTL ELACx FAULT" ECAM warning message to be displayed associated with "SIDESTICK SENSOR FAULT" subtitle



Technical description

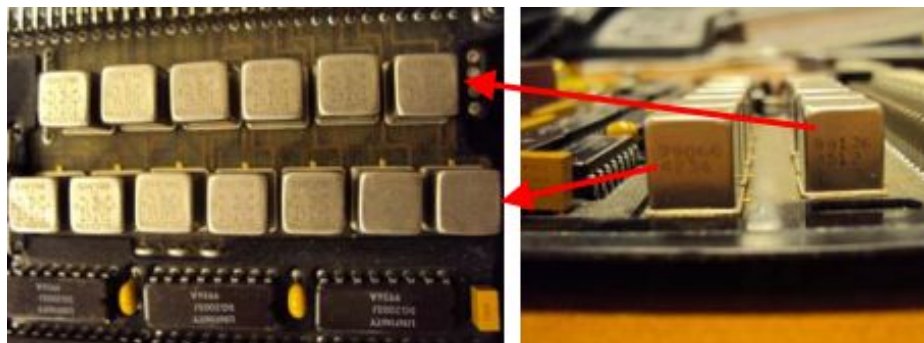
ELAC relays “lifetime” has been estimated by the supplier to **10Y**

⇒ This concerns 5 ELAC internal relays for **Hardware B & B-dl**

Relay issue can lead to **NFF**

SIGNATURE

⇒ ECAM / failure messages linked with elevator mode valve monitoring
LG(B) ELEV MODE VLV 34CEx // RY(B) ELEV MODE VLV 34CEx
With
F/CTL ELAC x PITCH FAULT



Solutions

- ⇒ Incriminated ELAC relays **replacements** proposed by THALES repair shop, following repetitive failure triggering on a particular unit
- ⇒ ELAC relays **preventive replacements** proposed by THALES repair shop, after 10 years of operation



References

- **THAV/SIL-1376** (general preventive maintenance operations for ELAC NFF on ATEC)
- **SXT/A320-064** (preventive maintenance action on relays of the power MON and COM boards)

Maintenance advice:

- ★ To send the PFR to repair shop with the removed unit

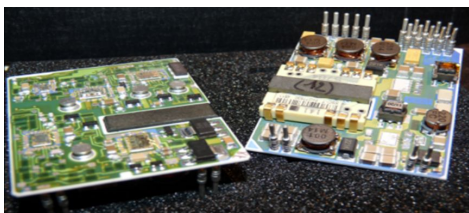
Technical description

Intermittent failure of the ELAC **ERACLE** supply module (soldering issue)
Issue leading to High ATP **NFF** rate (but confirmed by THALES' BITE decoding tool)

SIGNATURE

⇒ ECAM “F/CTL ELAC 1(2) PITCH FAULT” and / or “F/CTL ELAC 1(2) FAULT” associated with failure message (mainly during **cruise**):

- “CHECK PITCH CHANGE OVER OF ELAC1(2)”
And / Or
- “ELAC1(2) COM(MON) BUS1”



MAINTENANCE POST FLIGHT REPORT			A320	DB/N
A/C ID	DATE	GMT	FLTN	CITY PAIR
	26DEC	0919/1106		SBGL SBSV
WARNING/MAINT.STATUS MESSAGES				
GMT	PH	ATA		
0927	06	77-11	ENG 2 FADEC	
1005	06	27-00	F/CTL ELAC 2 FAULT	
FAILURE MESSAGES				
GMT	PH	ATA		
0927	06	75-32-41	CHA VSV ACT/HC/EEC2	
1005	06	27-93-34	ELAC2-COM -BUS1	
1005	06	27-93-34	CHECK PITCH CHANGE OVER OF ELAC2	
			SOURCE	IDENT.
			EIU2FADEC	
			EFCS 1	EFCS 2/F
			EFCS 1	EFCS 2/F

Solutions

- ⇒ **Removal of modules** with soldering manufacturing issues from production/repair since 2013 (WELWYN supplier)
- ⇒ **Replacement of ERACLE module** even if NFF during ATP test:
 - Replacement of the whole concerned module (COM/MON)
 - Replacement of the faulty component based on OEM BITE decoding outcome



References

- **THAV/SIL-1379**: ELAC intermittent ERACLE failure detection via the BITE decoding
- **THAV/SIL-1293**: Chronic Unit process/ reliability recommendations, including related ERACLE module issue
- **TFU 27.93.34.001**

Maintenance advice:

- ★ To send the PFR to repair shop with the removed unit

Technical description

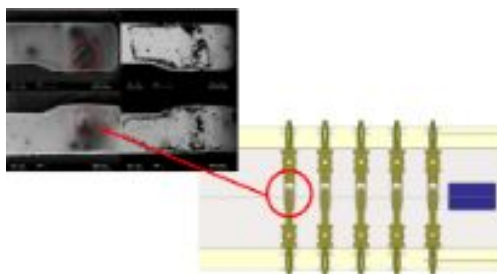
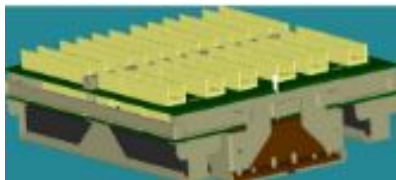
Pollution of ARINC / Interconnection board connectors
Issue leading to High ATP **NFF** rate

SIGNATURE

⇒ “F/CTL” maintenance status (class 2) associated with “ELACx-COM (MON) OR WIRING FROM ACCLRM y” failure message.

“ELACx OR WIRING FROM x HYD PRESS XMTR y ” could also be reported

⇒ Some “**pitch up**” tendency in manual mode observed



Solutions

Thales

- ⇒ **Cleaning** of ELAC boards - **VSB 394512B-27-021**
- ⇒ **Improve the housing** of the computer- **VSB 394512B-27-022**
- ⇒ Install **elastomeric** seals in the ELAC connectors - VSB 394512B-27-027

Airbus

- ⇒ revise ELAC airflow from the avionics rack by relocating the position of ventilation holes - **SB 27-1209**
- ⇒ Decrease the humidity in the avionics bay by regulating the airflow coming in and out of the avionics bay - **SB 21-1216 & 21-1217**

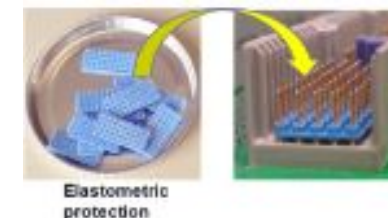


References

- TFU 27.93.34.001
- ISI 21.26.00029

Maintenance advice:

- ★ To send the PFR to repair shop with the removed unit



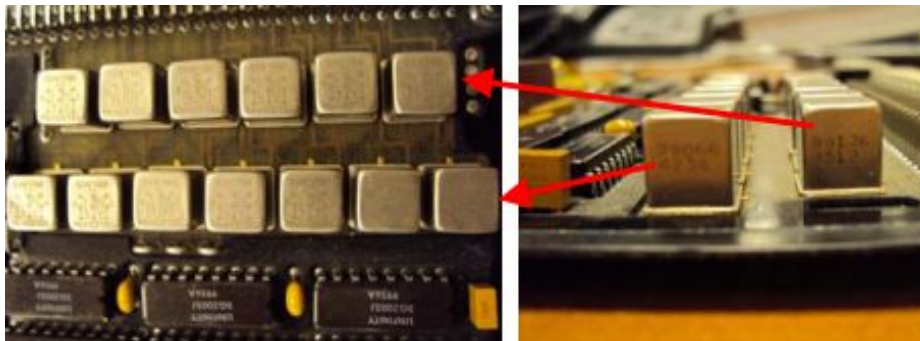
Technical description

SEC relays “lifetime” has been estimated by the supplier to **10Y**
⇒ This concerns 25 ELAC internal relays for **Hardware B & B’**
⇒ Due to different technology, SEC Hardware C is not concerned

Relay issue can lead to **NFF**

SIGNATURE

⇒ ECAM / failure messages linked with elevator, spoilers, THSA
(depending on the relay failed)



Solutions

- ⇒ Incriminated SEC relays **replacements** proposed by THALES repair shop, following repetitive failure triggering on a particular unit
- ⇒ In case of detected failed relay, **preventive replacements** of all the relays on the same board older than 10 years proposed by THALES.



References

- THALES SIL-1185: SEC – Preventive maintenance action
- THALES SIL-1162: SEC – Preventive maintenance action

Maintenance advice:

- ★ To send the PFR to repair shop with the removed unit

- QRH

 A318/A319/A320/A321 QUICK REFERENCE HANDBOOK	ABNORMAL AND EMERGENCY PROCEDURES	02.01A 25 NOV 24
SYSTEM RESET - GENERAL		

- ¹ ● To know if a system reset is authorized, the flight crew applies the following guidelines:

In Flight	
- If ECAM/OEB/FCOM/QRH procedure requests a reset, or - If the ECAM alert or the system malfunction is listed and authorized in the system reset table	Reset authorized
On Ground	
- If ECAM/OEB/FCOM/QRH procedure requests a reset, or - If the ECAM alert or the system malfunction is listed and authorized in the system reset table, or - If the affected system is NOT listed in the system reset table.	Reset authorized

- > ENG DUAL COOLING - NOT AVAIL
- > ENG 1(2) EIU FAULT
- > ENG - MAINTENANCE ENG 2 EIU m
- > F/CTL AIL SERVO FAULT
- > F/CTL ELAC 1 PITCH FAULT
- > F/CTL ELAC 2 PITCH FAULT
- > F/CTL SPLR FAULT
- > FUEL FQI CH 1 + 2 FAULT or Loss o
- > FWS FWC 1(2) FAULT
- > F/CTL 1(2) FAULT

AIRBUS answer:

To maintain the safety margins, and based on the system experience, the possibilities for further reset procedure are exhausted



ECAM F/CTL SENSOR (SSTU)

Mariusz OLSZEWSKI

AIRBUS

FAIR 13.0112 - SIDE STICK TRANSDUCER UNIT (SSTU) FAULTS

Description

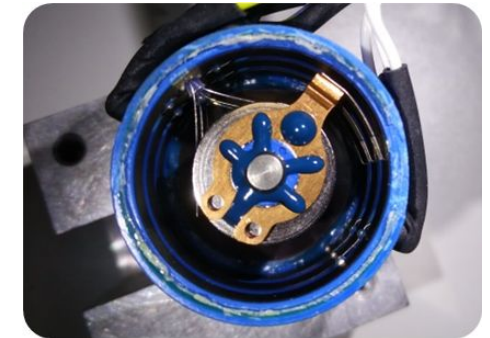
- Maintenance messages related to SSTU detected by ELAC or SEC computers. → Applicable to old generation side stick (SSA)
- **Classic failure signature:** *ELACx (SECx) OR INPUT OF CAPT (F/O) ROLL (PITCH) CTL SSTU 4CEy* associated with *F/CTL ELAC x FAULT, F/CTL...*
But can also lead to the triggering of other monitorings such as elevator/aileron servo valve monitoring, or ELAC internal failure.
- Potential **NO GO** if detected by ELAC2

Root Cause

- Suspected **contact degradation** at interface between the wiper and the track leading to an increase of the contact resistance, at the interface between the potentiometers track and the wiper
- Failure confirmed at potentiometer level but very **difficult to reproduce**

Reliability

- **High NFF** rate in repair shop
- But **MTBUR** remains above **100 000 FH** >> 50 000 FH GMTBUR



TFU 27.92.41.002

FAIR 13.0112 - SIDE STICK TRANSDUCER UNIT (SSTU) FAULTS

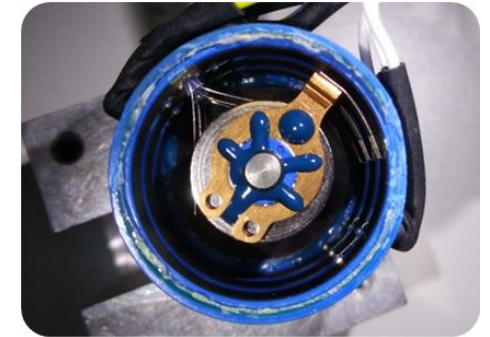
Maintenance advices

In case of “clear SSTU signature” in PFR/PLR/Airman history, perform:

- Early SSTU replacement on A/C as soon as issue is detected by the system
- Potentiometer active replacement in repair shop even if NFF

For operators currently highly impacted, first actions to do are:

- Identify any SSTU repeater units (removed several times on a short period)
- Ensure that potentiometers active replacements is requested to the repair-shop / MRO ... and performed



TFU 27.92.41.002

Airbus improvements completed to support impacted operators

TSM improvement:

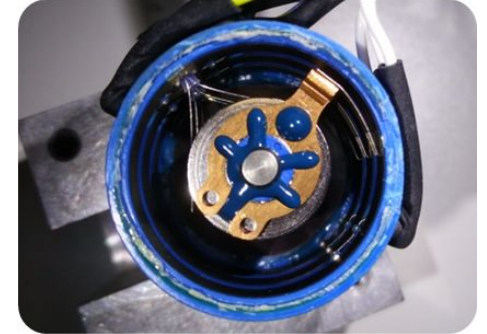
- Several TSM tasks have been updated to focus on SSTU when applicable (ref TFU 27.92.41.002)
 - Available within AirnavX since **Aug-2021** revision
- Creation of a new TSM task in case of simultaneous LH and RH elevator servovalve monitoring
 - Targeted within AirnavX **Feb-2025** revision

FAIR 13.0112 - SIDE STICK TRANSDUCER UNIT (SSTU) FAULTS

Airbus improvements completed to support impacted operators

ECAM display improvement:

- Operators can be confused by the triggering of the *F/CTL ELAC x FAULT* ECAM, and think that there is an issue with the ELAC
 - Since **FCDC 60** with **FWC F14** (available since 2022):
“*F/CTL ELACx FAULT*” ECAM warning message is displayed associated with “***SIDESTICK SENSOR FAULT***” subtitle
FCDC 60 : VSB 115370-27-013
FWC F14: VSB 350E0-31-039 and Airbus SB A320-31-1631/32



TFU 27.92.41.002

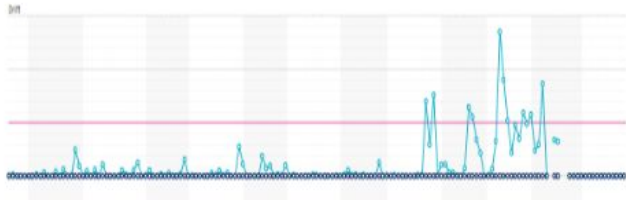
FAIR 13.0112 - SIDE STICK TRANSDUCER UNIT (SSTU) FAULTS

Airbus improvements completed to support impacted operators

Skywise Predictive model - Side Stick Transducer Monitoring:

- The purpose of the **model** is to predict a **mechanical degradation** of the side stick transducer internal potentiometers, which could lead to unexpected roll and/or pitch value of the side stick.

Roll(deg) Pitch(deg)

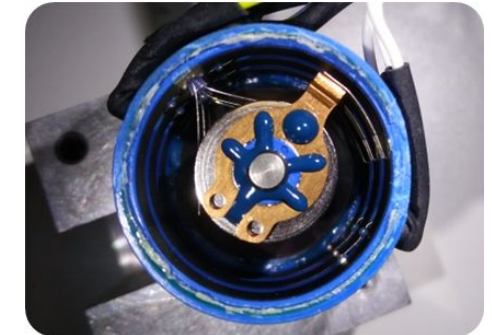


SSTU drift above a threshold

SPM alert

Operator

- The model triggers a CAPT or F/O side stick **SPM alert**:
 - Recommendation to **replace** the related SSTU
- The Extended data collection (**FOMAX/RMAX**) enabler is required to perform this model in the frame of Skywise Predictive Maintenance Services offer.
- Today the model is still in “observation” phase, but candidate for **full maturity** due to good detection rate based on operators feedback.



TFU 27.92.41.002



Fuel Inerting System Fault

Tanja RAU

AIRBUS

E/W Fuel Inerting Sys Fault

Voice of the customer

- ❖ HVN experienced repetitive **Fuel Inerting Sys Fault**
=> replacement of several components (TR 81476326)

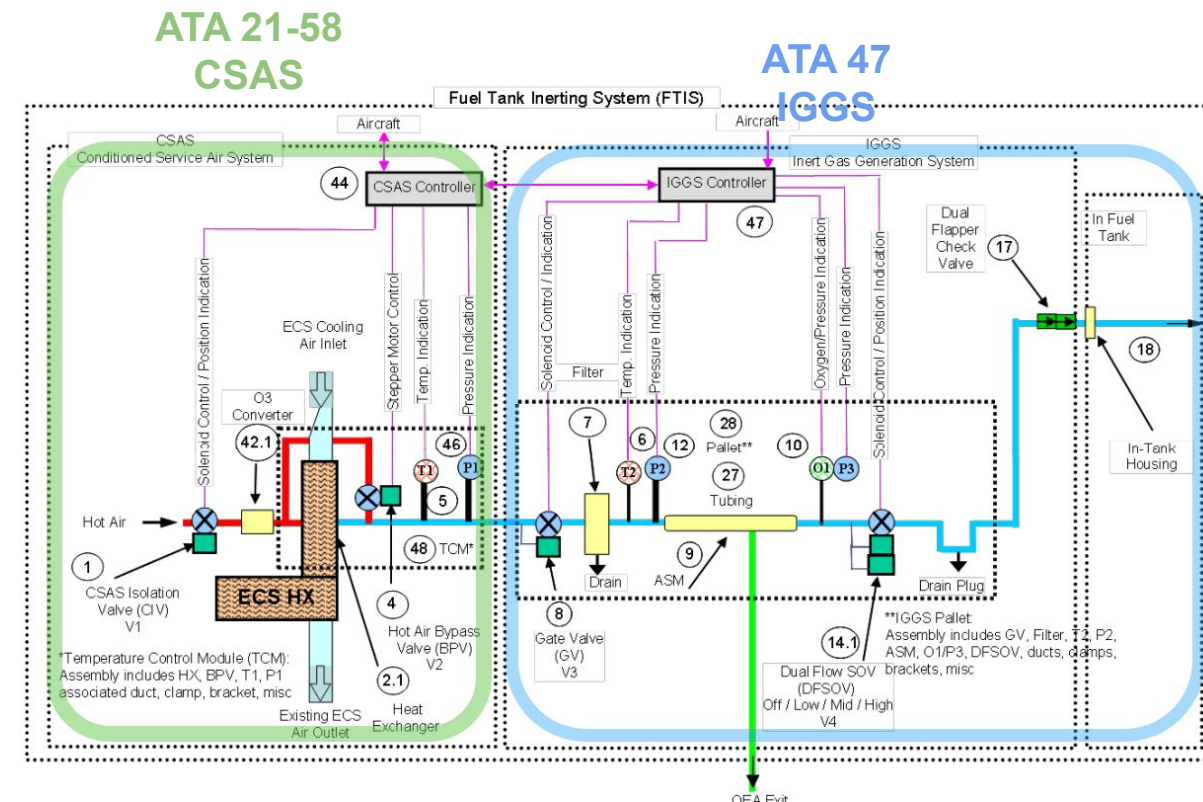
HVN request: A large number of components must be replaced due to E/W Fuel Inerting System Fault. AIB to provide troubleshooting action.

Description - Fuel Tank Inerting System

Flammability reduction mean - supplies inert gas to fuel tanks:

- ❖ CSAS => cools hot bleed air and supplied the IGGS
- ❖ IGGS => removes oxygen and provides nitrogen enriched air into centre fuel tank
- ❖ In case of issue => system latched => E/W + fault message triggered => MMEL 47-10-01A

CSAS = Conditioned Service Air System
IGGS = Inerting Gas Generation System



E/W Fuel Inerting Sys Fault - TR 81476326

Root cause/ Investigation

- ❖ In 2019 Parker records high NFF rate of ASM and ICU
=>Investigation launched via TFU **47.10.00004**
- ❖ Several contributors identified:
 - Critical OVTEMP fault margins
 - CCU - ICU interaction
 - Human factor. Removal of the ICU when it is not the accused LRU
- => Challenging to allocate correct distribution

Mitigation

- ❖ ATA 47 TSM tasks modified => TSM Feb 2025 revision
- ❖ ATA 21-58 => TSM May 2025 revision

PFR data of HVN A320FAM fleet

From JAN to DEC 2024, top

- ❖ CRITICAL OVTEMP
- ❖ BYP VLV (113HW)
- ❖ IGGS CTL UNIT (1YA)
- ❖ ASM PERFORMANCE DEGRADED

Mitigation

- **BYP VLV (113HW)** =>TSM 21-58-00-810-804-A
- **CRITICAL OVTEMP** => TSM 21-58-00-810-815-A
- **IGGS CTL UNIT (1YA)** => TSM 47-11-00-810-801-A
- **ASM PERFORMANCE DEGRADED** => TSM 47-11-00-810-806-A
- **E/W orphan** =>TSM 21-58-00-810-824-A

In case of repetitive failure, ensure:

- Hard reset completed correctly
- No leakage existing
- Wiring check as per dedicated TSM

When TSM exhausted:

- Printout FTIS TSD
- Attach TSD with unit before sending it to supplier

Note: CSAS BYP VLV => FAIR 23.0211 => share TSD when BPV removal



Air Wing Leak Due To ACM Gasket

Tanja RAU

AIRBUS

Air wing leak due to ACM gasket issue

Voice of the customer

- ❖ Many cases of ECAM Air Wing Leak due to ACM torn gasket
- ❖ TR 81230949&TR 81232315 => Pressurization issue on MSN 3737

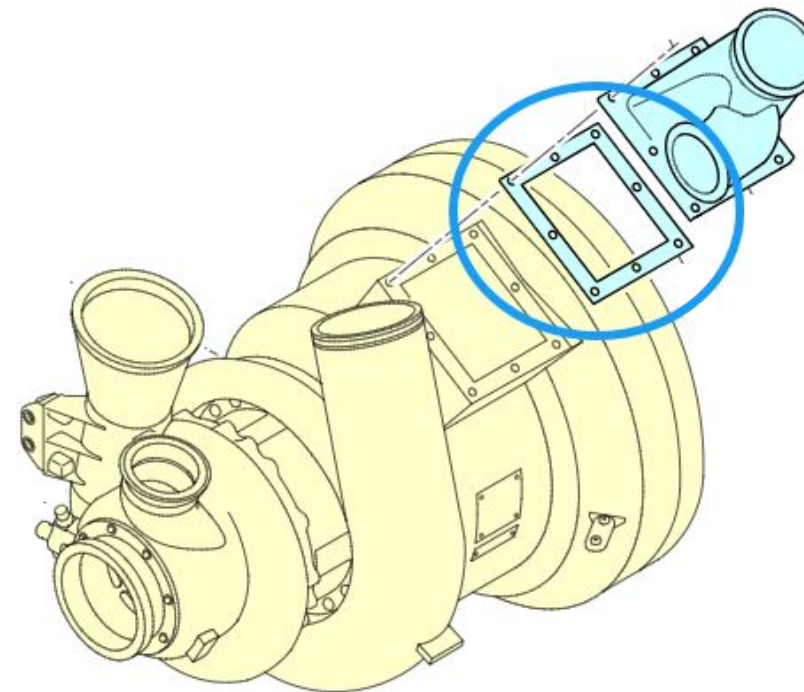
HVN request: AIB to provide mitigation or terminating action

Background - ACM gasket

- ❖ Few reports of broken ACM gasket PN 3455-71 leading to leak at PACK Level => no increase for past 5 years
- ❖ Gasket wear issues reported on high time units
- ❖ Depending on the leakage rate => E/W AIR WING LEAK fault
- ❖ FAIR 14.0159 and FAIR 21.0204 (closed)
- ❖ Reliability of ACM PN 1263* (status NOV2024)
A320FAM **110931 FH** vs. GMTBUR 9000 FH

Root cause

- ❖ ACM gasket exposed to high temperature cycling => deterioration of gasket material => leak
- ❖ Repetitive gasket replacement could damage thread inserts => inserts tightening performances decrease => inserts & screws get loosen => leak



Recommendation

- ❖ Leak test as per AMM Task 21-52-21-400-002-A, Subtask 21-52-21-710-052-A as preventive action
- ❖ Adaptation of interval based on HVN's In-Service experience
- ❖ Insert repair as per vendor CRM 21-52-16 Repair #6 (replacement of thread inserts by twin inserts)

Air wing leak due to ACM gasket issue

In-Service occurrence on MSN 3737

Event description:

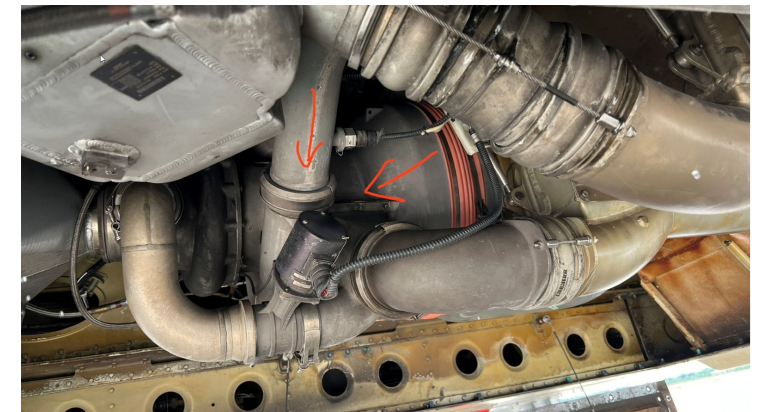
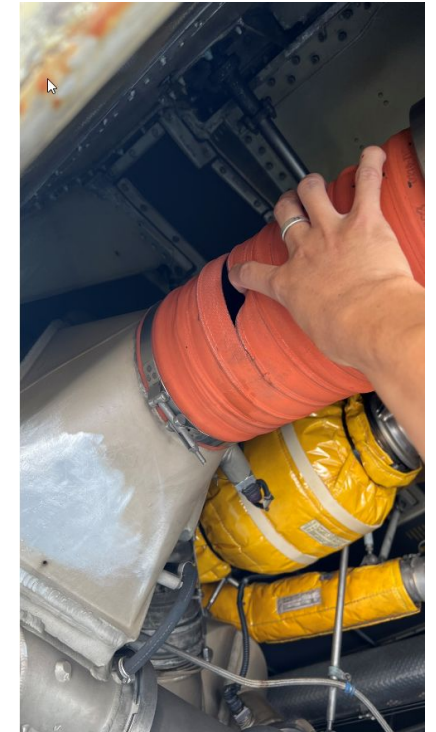
- ❖ On 28th of May, MSN 03737 experienced a loss of cabin pressure during flight VN165.
- ❖ Crew reported when A/C flew over 27000 ft, Cabin Altitude increased to 9000 ft. Captain decided to reduce flight level to 20000 ft then Cabin altitude was stable. A CAB PR EXCESS CAB ALT warning was not triggered.

Maintenance finding

- ❖ Pack 1: ACM gasket damage
- ❖ Pack 2: Outlet bellow cracked

Investigation outcome:

- ❖ Combination of both findings could create an important leakage rate
=> Correct pressurization in higher altitude is impacted
- ❖ After repair of the damages no further issues reported





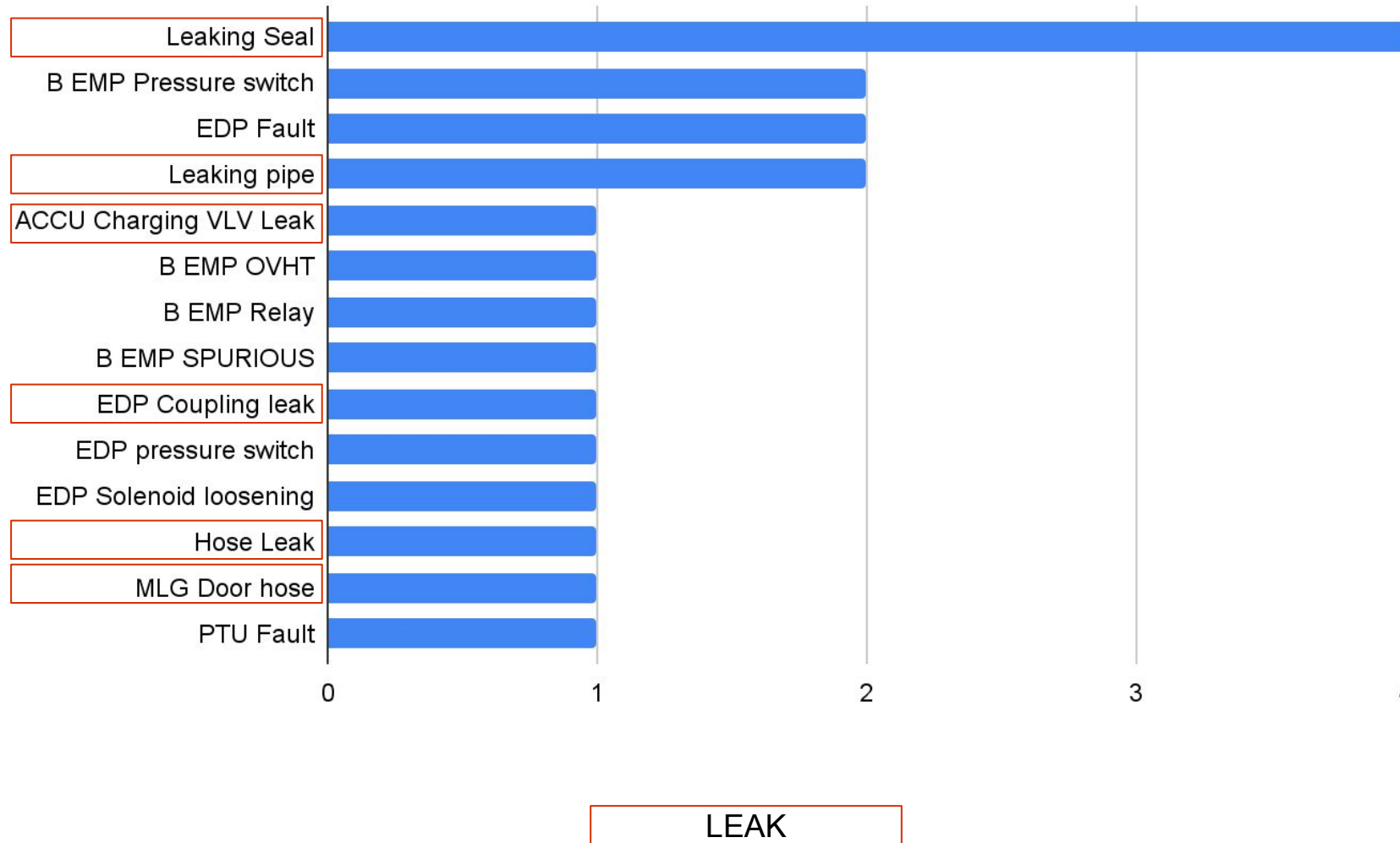
Hydraulic Leaks Due To Ageing

Marius OLSZEWSKI

AIRBUS

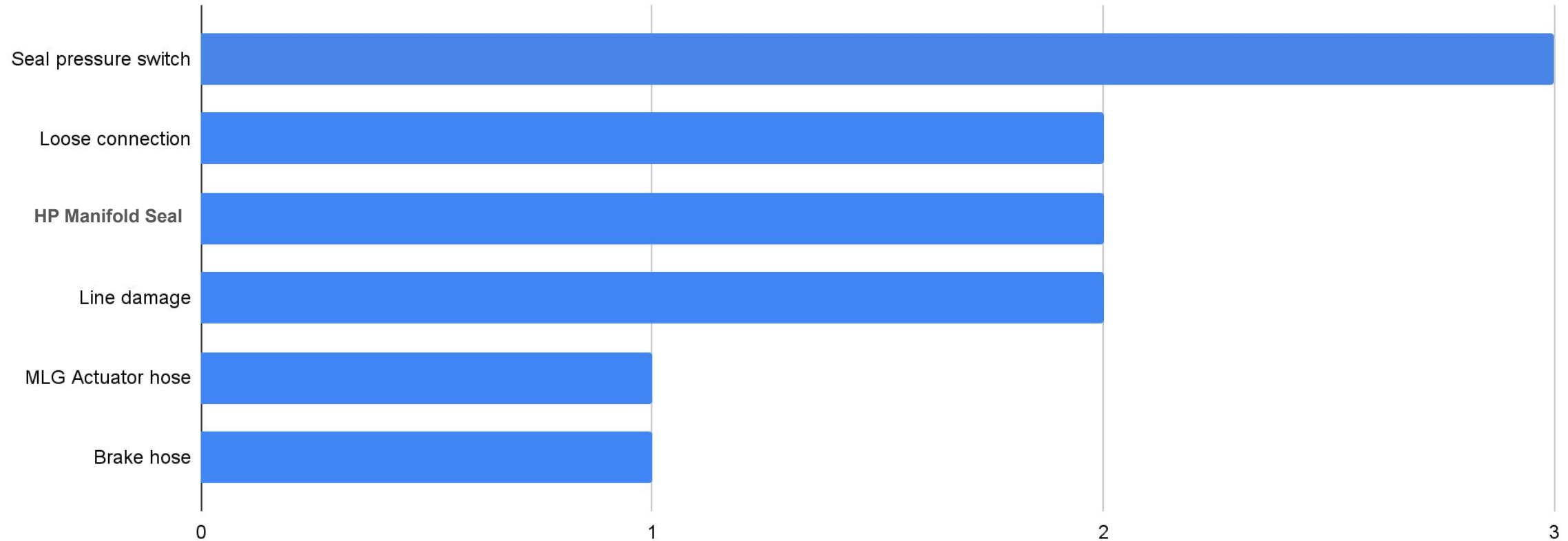
Hydraulic OI's

Events reported by HVN from Oct 2022 till today



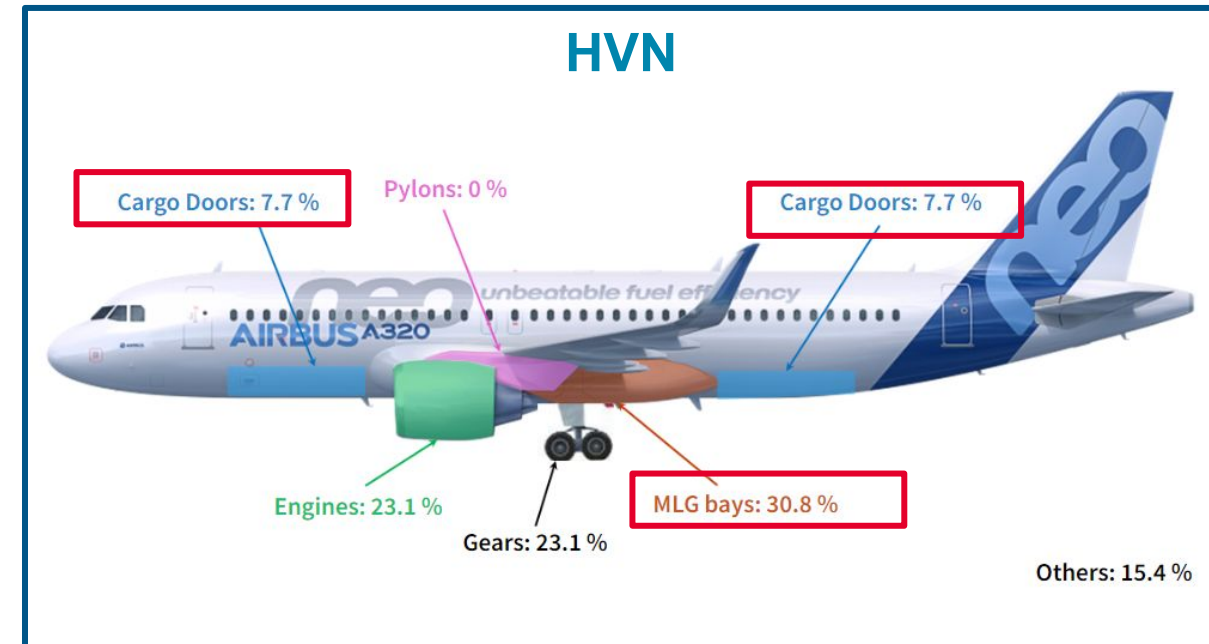
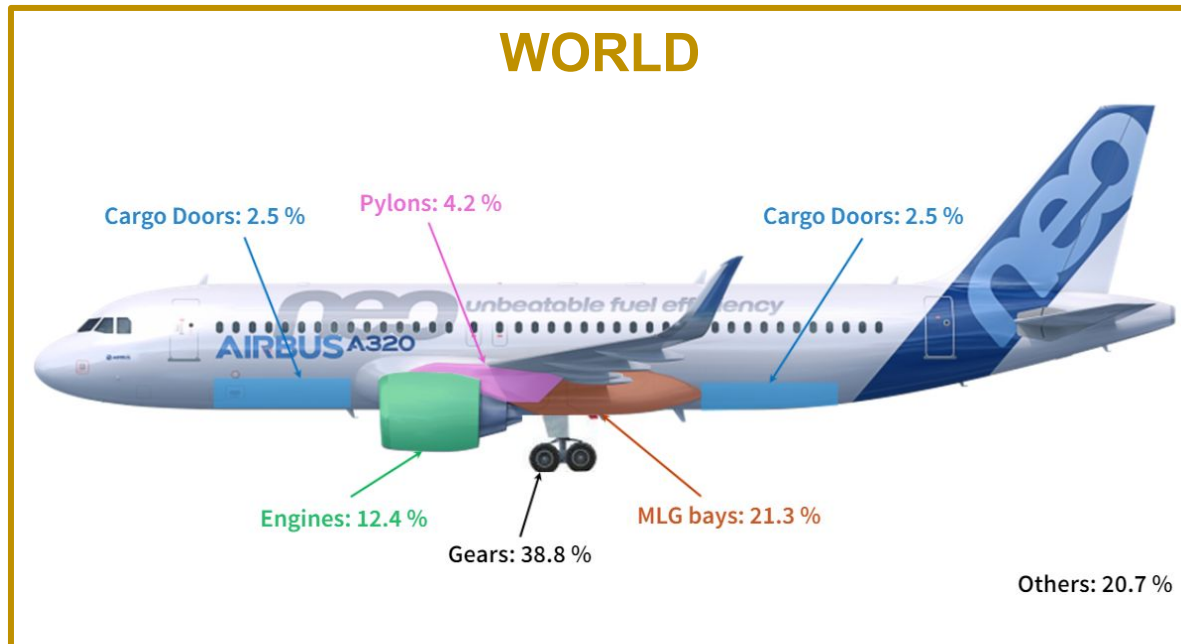
HVN Main Leak Drivers

Events reported by HVN from Jan 2021 till today



Hydraulic Leak Location (Worldwide vs HVN)

Events reported by HVN from Jan 2021 till today



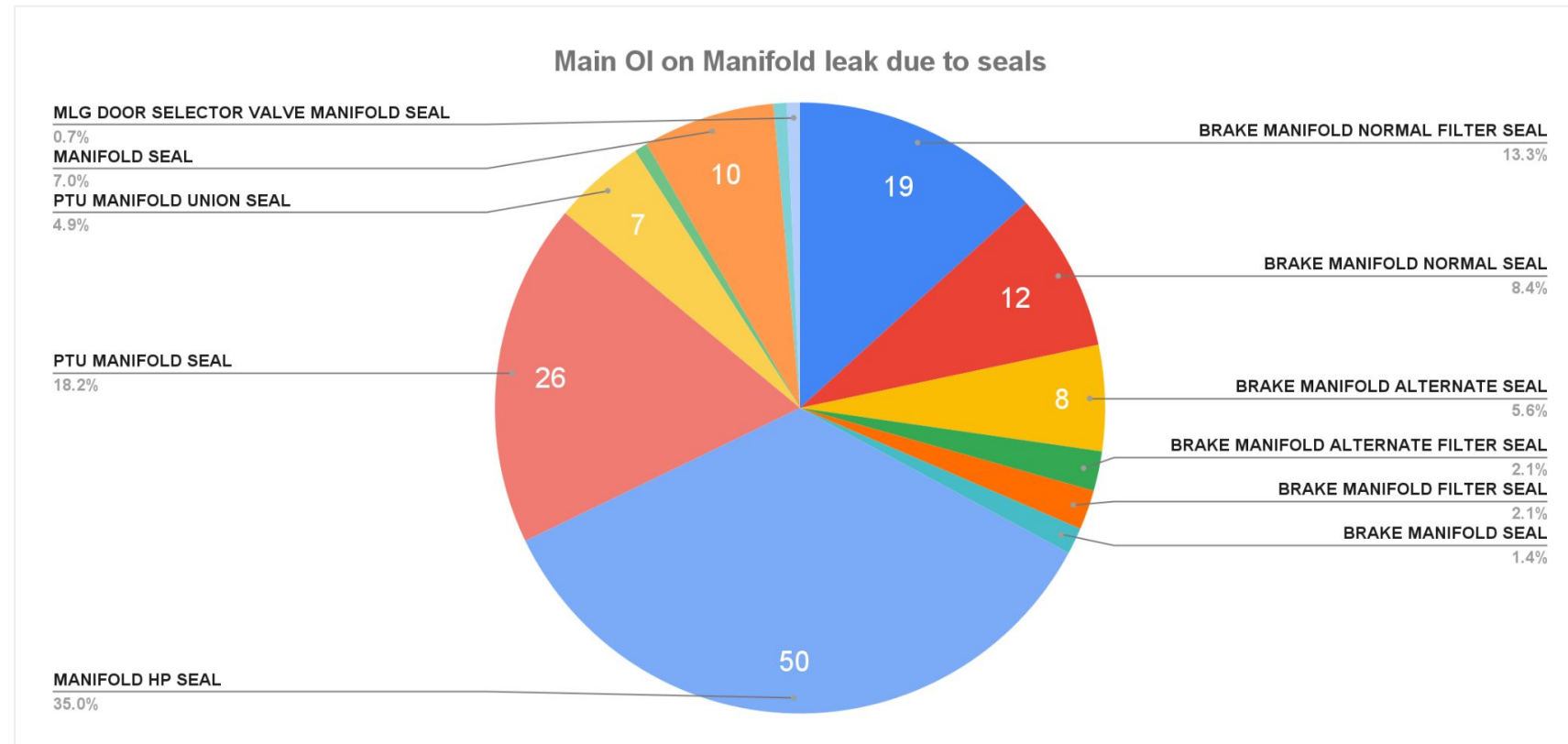
Seal Failures (Worldwide)

Events reported between Jan 21-Dec23

FAIR 18.0296
FAIR 18.0298
FAIR 11.0184

OI for Manifold leak due to seals are located :

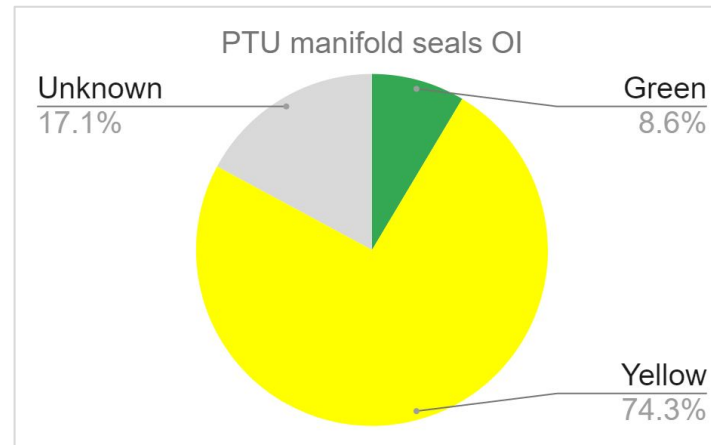
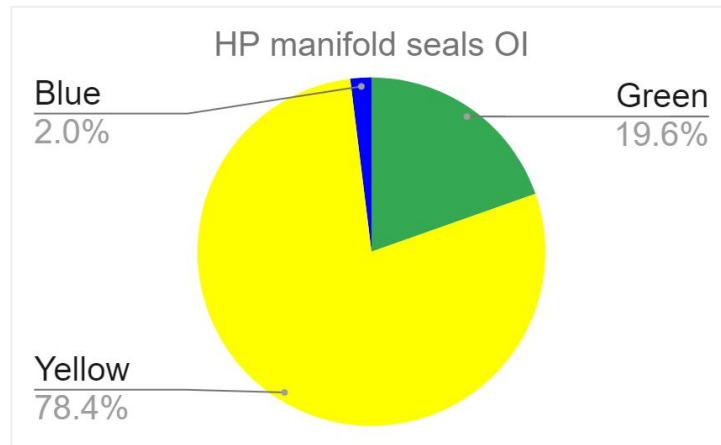
- For **35%** on HP manifold
- For **33%** on Brake manifold
- For **23%** on PTU manifold
- For **8%** on Other manifold (or not specified within OI description)
- For **1%** on MLG Door



Seal Failures (Worldwide)

Events reported between Jan 21-Dec 23

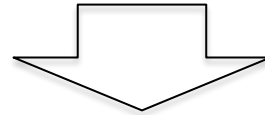
FAIR 18.0296
FAIR 18.0298
FAIR 11.0184



- 31.1% of Brake Manifold leaks due to seal issue
- 21.1% of Others Manifold leaks due to seal issue

INSPECTIONS ARE AN EFFICIENT WAY FOR:

- Detection of premature O-ring failure, Loose Connections, Damages...
- Improving experience - Learning effect



- Update Work Orders with WARNINGS to draw attention to:
 - ◆ O-ring black deposit
 - ◆ Hydraulic fluid traces
 - ◆ Red Marking misalignment
 - ◆ Clash / Low distances
- Training videos
- Class room / On-wing trainings

= REDUCTION OF OI's

Preventive maintenance

ISI 29.00.00172

For aircraft > 12 years

Sep 24 status: 33 HVN A/Cs older or equal 12Y

Quick preventive inspection of hydraulic components for leakage (ISI 29.00.00172)

High OI rate
due to leaks



QUICK
INSPECTION
Approx. 2MH



SHORTER
INTERVAL
A-CHECK



Equipments with statistically
highest number of leaks



- Manifolds
- LG Hoses

- EDP
- PRESS switch

- Short-term results monitoring
- If leak occur, seal replacement per AMM limitation.
- If no leak yet but seen black deposit, replace seal in next earliest opportunity.

Preventive inspections

MLG bay

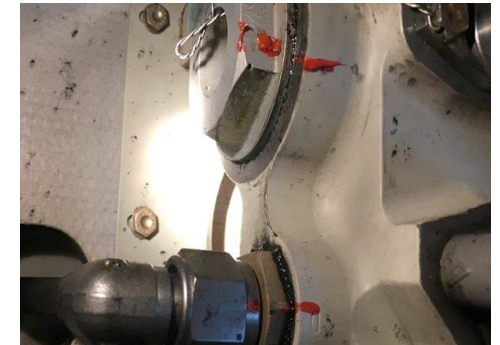
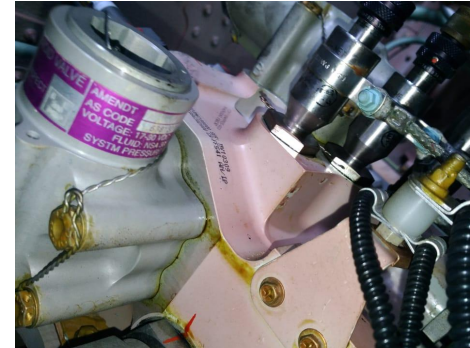
- Zonal Inspections to be performed regularly
- Specific attention to be paid at following items:

1 TRACES OF LEAKAGES / SEEPAGES

2 CONDITION OF FLEXIBLE HOSES

3 CONDITION OF RIGID PIPES

4 CONDITION OF SEALS



Preventive inspections

MLG bay

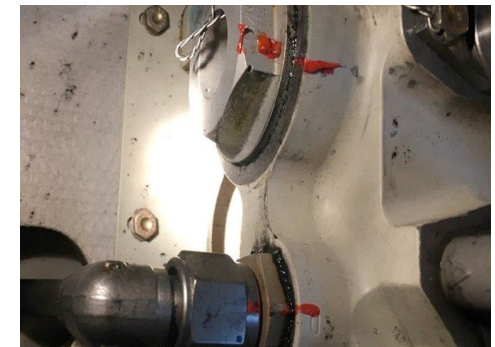
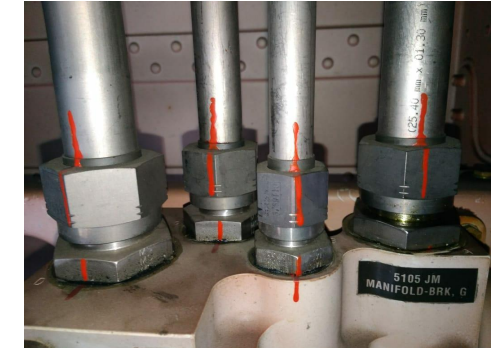
- Zonal Inspections to be performed regularly
- Specific attention to be paid at following items:

5 AIR TRAPPED IN THE SYSTEM

6 LOOSEN CONNECTIONS / RED MARKING ALIGNMENTS

7 CONDITION OF ATTACHMENTS (CLAMPS, WIRELOCK...)

8 CLEARANCES BETWEEN LINES & STRUCTURES



ISI 29.00.00172 - Quick preventive inspection of hydraulic components for leakage

1. GREEN HP MANIFOLD 1111GM
2. YELLOW HP MANIFOLD 3014GM
3. BLUE HP MANIFOLD 2111GM
4. GREEN PTU MANIFOLD 1113GM
5. YELLOW PTU MANIFOLD 3015GM
6. PTU 1088GM
7. MLG ACTUATOR HOSE LH/RH MLG LOCKSTAY ACTUATOR HOSE LH/RH
8. MLG UPLOCK
9. MLG DOOR ACTUATOR
10. EDP PRESSURE SWITCH 1074GK & 3074GD
11. EDP 1030GK & 3030GD

Airbus proposal

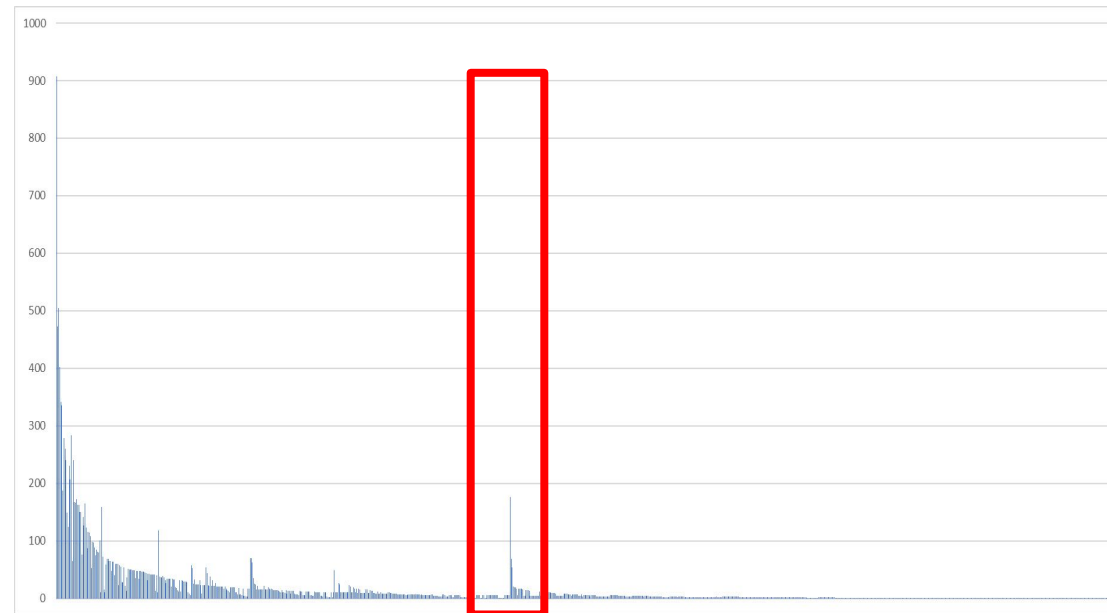
- Focus on A/C > 12Y
- Yearly inspection

O-Ring Airbus recommendations

- ✓ Check seals **shelf life limit** written on the package
- ✓ **Clean** the union thread AND component port
- ✓ NEVER re-install used o-ring, discard it
- ✓ Be careful while installing o-ring (contact with component port or tooling...)
- ✓ Quick access to dedicated **seal kit** (line maintenance use)
- ✓  **ISI 29.00.00150** for seals purchasing
- ✓ Always bleed after installation work to avoid hammer water effect

Hydraulic Pipes sales analysis

- Airbus spares consumption assessment (last 5 years)
 - 500+ pipes installed in airframe side
 - TOP 20 hydraulic pipes consumption assessment
- Improvement candidate for hydraulic pipes is also based on operator reports via Techrequest, FAIR...



Hydraulic Pipes sales analysis

PNs	5 years spare order	References	Comment
D2901005100740	908	TFU 29.10.00.061	Mid pylon chafing
D3237002303400	473	SB 29-1180	MLG bay pipe chafing
D3237003001100	505	SB 29-1180	MLG bay pipe chafing
D2901010201300	402	Damaged fire sleeve	HCU line
D2901005100800	342	TFU 29.10.00.061	Mid pylon chafing
D2901005201300	336	SB 54-1021	HCU line CFM56
D5237252003600	187	TFU 29.00.00.087	Pin hole corrosion
D2904007420000	279	No record	
D2901005100040	260	TFU 29.10.00.061	Mid pylon chafing
D2901010201200	240	Few report	HCU line IAE
D5237252003800	149	TFU 29.00.00.087	Pin hole corrosion
D3237002502100	124	2 reports	MLG uplock line
D2901005100540	231	TFU 29.10.00.061	Mid pylon chafing
D2917000902000	283	TFU 29.13.00.006	EMP broken bracket
D2917090701000	65	Some report	Crack end fitting
D2917000102100	240	TFU 29.13.00.006	EMP broken bracket
D2905602500000	167	TFU 29.00.00.091	Clearance issue in wing area
D2901010200900	166	Few report	T/R line IAE
D5237252200200	172	Few report	cargo door return line

Cargo Door pipe leakage

Description

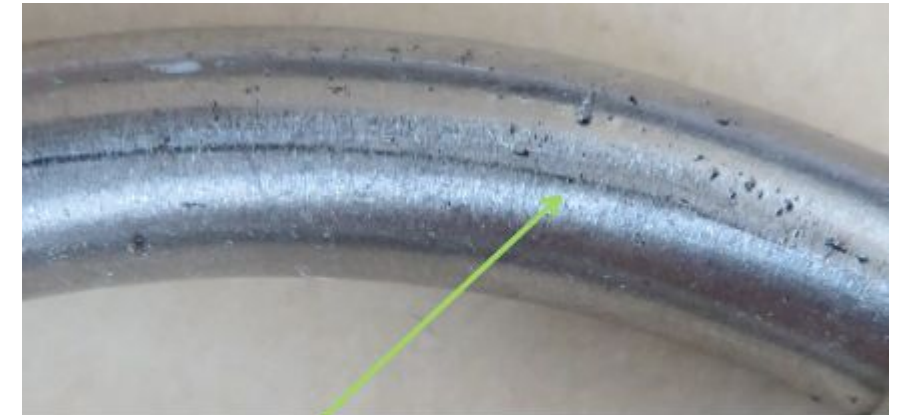
- Some operators experienced cases of hydraulic leak in the cargo door supply line
- Long delay due to replacement or repair of leaking pipe - pipe installed behind cargo lining
- As per experience from some APAC operators, pipes might start corroding after 6-7 years

Root Cause

- Pinhole corrosion of the pipe
- Leading to pinhead-size reduction of the pipe wall thickness

Solution and recommendations

- High number of pipes subject to material change from CRES to Titanium
 - 1st batch - 3 pipes: PoE CEO: MSN 8258 (April 2018) - **48 HVN AC**
PoE NEO: MSN 8256 (April 2018) and IPC for PRE MOD
 - 2nd batch - 16 pipes: PoE CEO: MSN 9185 (Sep 2019) + **17HVN AC=65AC**
PoE NEO: MSN 9189 (Nov 2019) and IPC for PRE MOD
 - 3rd batch - 5 pipes: PoE not yet available and IPC update planning to be confirmed in the TFU before the end of the year



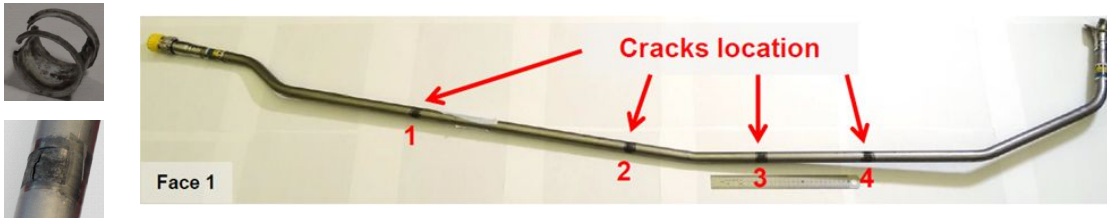
Pipe corrosion spot

Chafing of hydraulic lines in mid pylon area CEO IAE & CFM

TFU 29.10.00.061

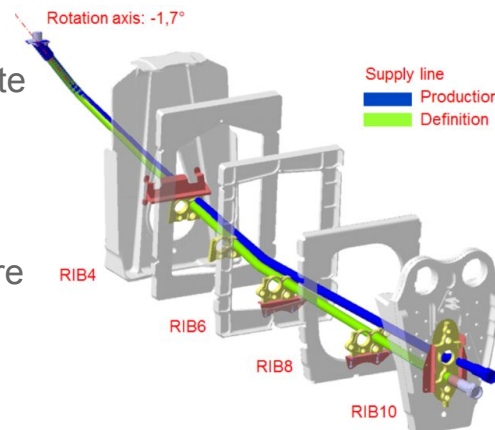
Background

Following ISI 29.11.00001 publication, few cases of **chafing** (pipe, teflon bush, clamp block) on **PN D2901005100740** (Left/Right HP line). Reported on IAE and CFM.



Investigations

- Pipe installation **under stress** despite manufacturing improvement.
- **EDP pressure pulse** and specific **engine hydraulic pipe** installation are contributors.



Mitigations

- **Visual inspection as per ISI 29.11.00001** recommended every C-check to check teflon bush and pipe wearing. Report back to Airbus the inspection results with MH & cost.
 - PN -40 improvement reduced the number of occurrences but did not fixed completely the chafing issue

Permanent solutions

- New HP hydraulic tube with new guide that correspond to the clamp block location at rib 5.
ICY 2-ways in IPC. No SB.
- **New bush material** from PTFE (teflon) to PA66 (polyamide).
SB 29-1197 released in July 24
- **AMM procedure update available since Feb-2023** to ensure better stress free installation and reduce the risk of hydraulic pipe chafing.

Case Drain Line damages NEO PW / TFU 29.00.00252

Description

Failure reported of the case drain pipe in the pylon on A320neo powered with PW1100 engines on :

- **Pipe P/N D2901583600000** (D2901584500000 can be installed as spare)
- **Pipe P/N D2901583800000**
- **Pipe P/N D2901584600000**

Impact : 27 reported events including 19 AOGs at 4 airlines, including 3 IN-FLIGHT HYD Y RSVR LO LVL (Loss System of the Yellow)

Root Cause

- Wear caused by the **friction between the clamps and the pipes in a vibratory environment**.
- The vibrations are mainly **induced by the pumps** of the HP hydraulic line and the feed line.

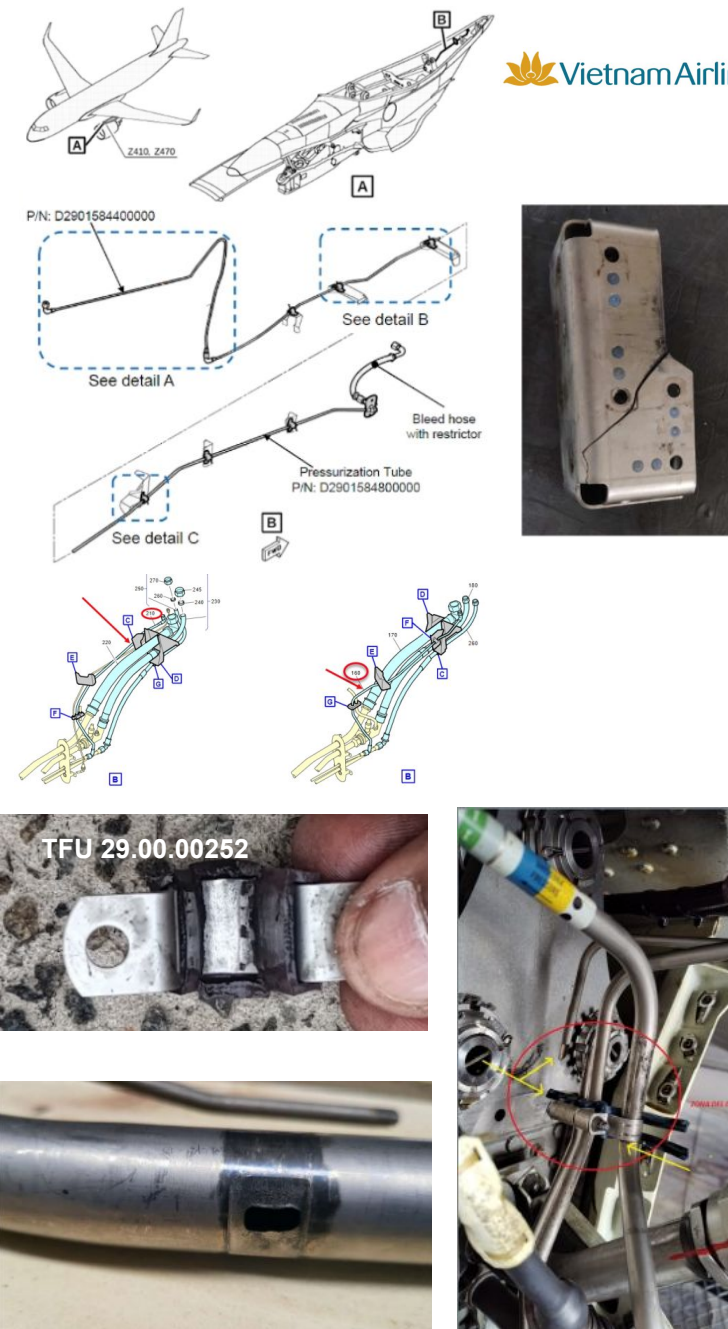
85

Mitigations

- Perform Inspection in the Pylon as per: AMM TASK 05-24-10-200-001-A General Visual Inspection of Upper Forward Pylon (EWIS) / MPD TASK: ZL-411-01 (48MO)
- Interval recommendations: C-Check or any opportunity inspection suitable to operators
- In case the pipe replacement is required:
- Install pipe in a stress-free condition as per AMM 20-23-11-911-001-A procedures

Solution

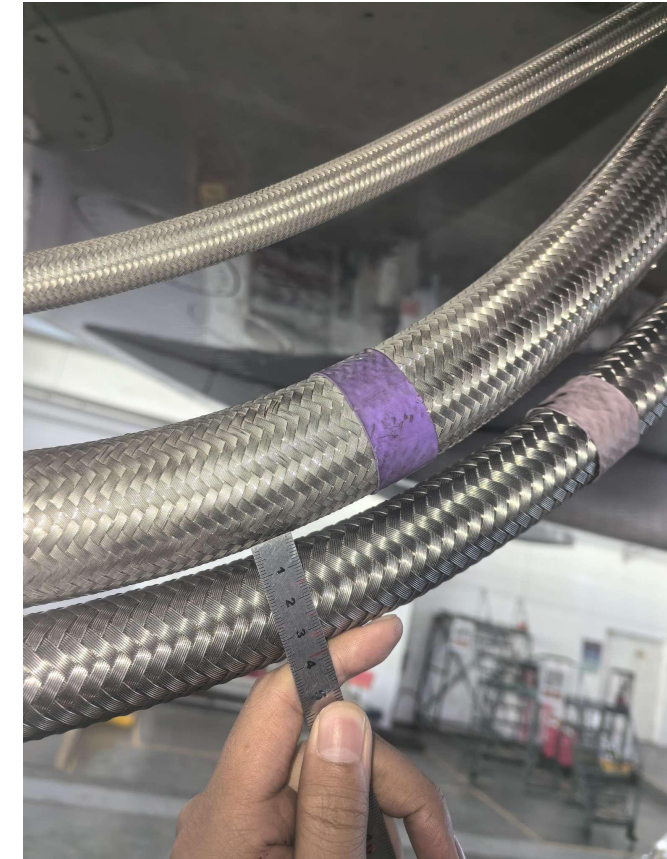
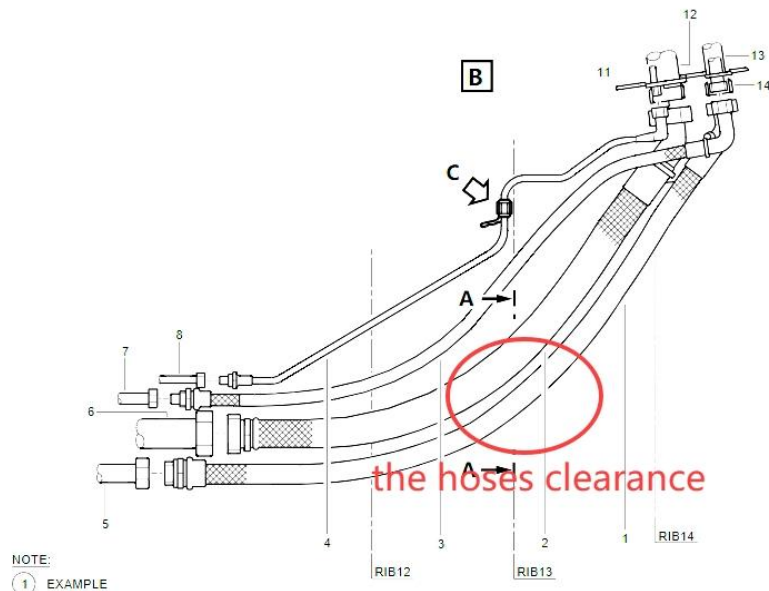
- Airbus is currently investigating the re-routing of the lines in the pylon to make sure that the impact of the vibration is reduced.



Hoses chafing in the pylon area

Description

- Safing reported on lines at the back of the pylon between the suction hose and the HP hose
- Pay special attention during inspection
- Topic under monitoring



Pipe chaffing in MLG

- MLG bay pipe chafing due to P-clamp sliding as per arrow below
- High number of leaks reported from the fleet
- Solution is to introduce an new pipe with different shape
- PoE A320CEO MSN 8116
- PoE A320NEO MSN 8155
- SB A320-29-1180/81



MLG Door actuator hose

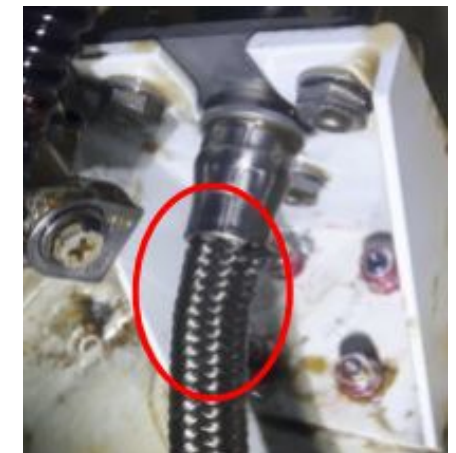
P/N AE718529-2 (Eaton)

Description

- Cases of hydraulic leakage with PN AE718529-2 have been reported
 - Average of leaks at 16000 FC based on reported failure cases
 - External surface defect as broken wires observed
- Leads to Green Hydraulic system loss which often leads to IFTB, Free Fall Extension or Operation interruption
- PN introduced in 2010 and SB 32-1371 as replacement for AE2463921G0181 and AE2463921G0097 which needed to be replaced at every 6000FC



-
- Braid failure most likely due to side loads being introduced to the hose assembly leading to fatigue failures.
 - Hose fatigue damage characteristic to hose misalignment
 - Frequent hose removal / installation due to repeat inspection of actuator fitting on the keel beam area as per SB 53-1195 (AD No 2012- 0118) may be a contributing factor



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Root Cause

- If the actuator is not correctly hold /supported during the inspection, this can cause increased strain on the hose and damage
- If the swivel joints are not correctly aligned during actuator reconnection This can decrease the hose bend radius and can leads also to the increased strain and damage

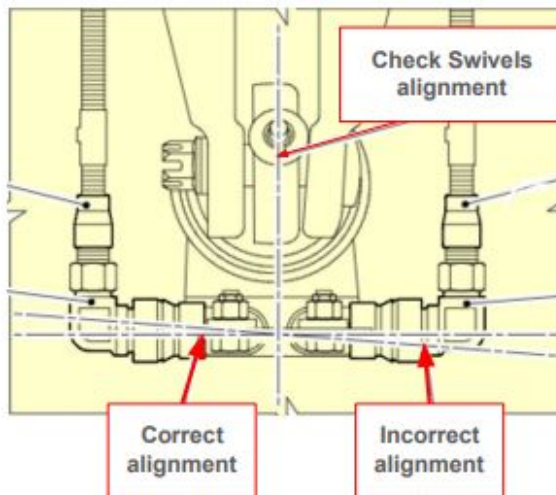
MLG Door actuator hose

P/N AE718529-2 (Eaton)

ISI 32.31.00099
FAIR 20.0196

Mitigation

- Check swivel fittings orientation as per AMM Fig 32-31-39-991-00200-00-A (Swivels fittings to be parallel to the A/C axis)
- AMM TASK 32-31-39-400-001-A – “Installation of Flexible Hose” updated to ensure that no torsional loads is introduced into the hose during assembly
- When performing repetitive inspection as per SB 53-1195 at every 2250 FC (AD No 2012- 0118),
 - Particular attention must be paid when securing the actuator. When actuator is disconnected from the A/C , the hoses shall not be overloaded by Actuator or even the swivel weight.
 - Check the proper alignment of the swivel fittings after Actuator re-installation
 - SB 53-1195 (rev 11 Nov.2023) recently updated to request swivel fitting alignment check

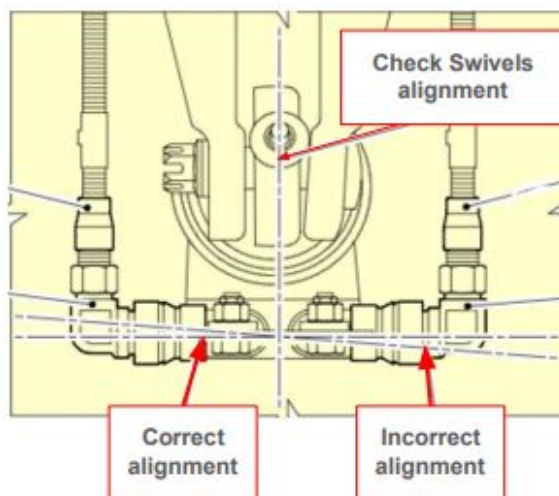


MLG Door actuator hose

P/N AE718529-2 (Eaton)

Way forward

- Continuous Fleet monitoring
- **MISP Opened in Sep 2024**



MLG Uplock Hose Leak

FAIR 20.0028

Description

- A few airlines have experienced a leakage of the MLG Uplock Hose P/N AE2463921E0183
- This issue is not included in the list of top leakage contributors
- Most A/C impacted around MSN 2000 to MSN 5000
- Less than 20 cases reported to Airbus through FAIR and TechRequest since 2020

Root Cause

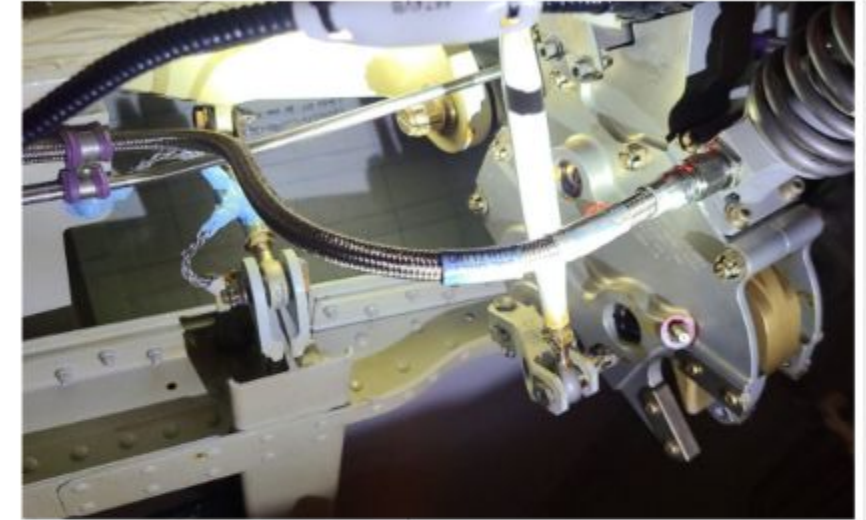
- Leakage is caused by metallic braid corrosion leading to broken wire and inner layer puncture

Mitigation

- Inspection/prevention as per ISI 29.00.00172 and 29.00.00004 : Visual inspection at A-Check

Airbus proposal for way forward

- Airbus propose to keep this item open



FAIR 20.0028

CEO/NEO Pressure switches

FAIR 23.0238

FAIR 24.0034

Description

- System pressure switch (FIN 1151GN, 2151GN, 3151GN)

Threshold:
1450 +/- 72.5 psi

High pressure available and stable
+ EDP running
+ hydraulic fluid level stable
= likely a spurious warning



- Green, blue and yellow systems: NO GO
Rationale: Used to assess the braking mode (Green and Yellow) + Nose Wheel Steering availability (Yellow) + RAT (Blue)

Maintenance and troubleshooting recommendations

Pressure transmitter will indicate that pressure is available:

→ Swapping of the faulty switch with Blue EMP switch 2074GJ (GO) to use MMEL and minimize OI impact

CEO/NEO Pressure switches

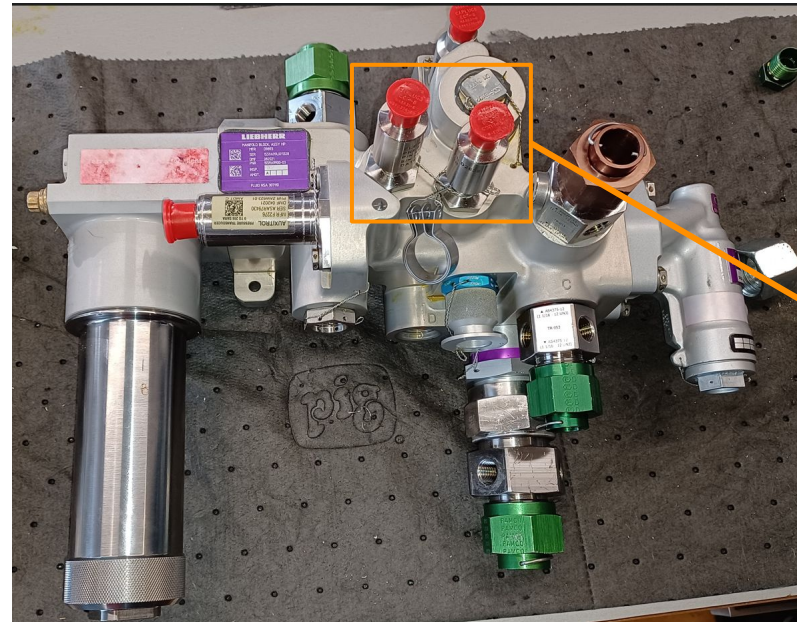
FAIR 23.0238
FAIR 24.0034

Description

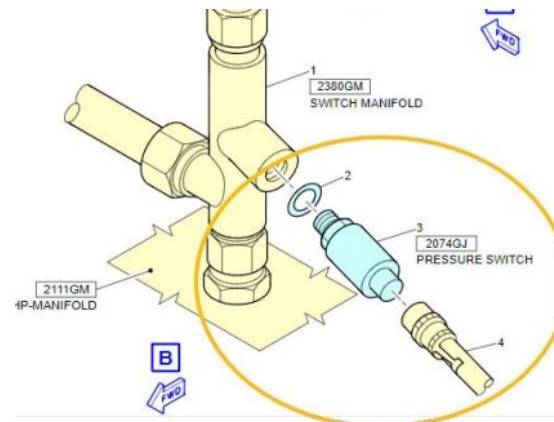
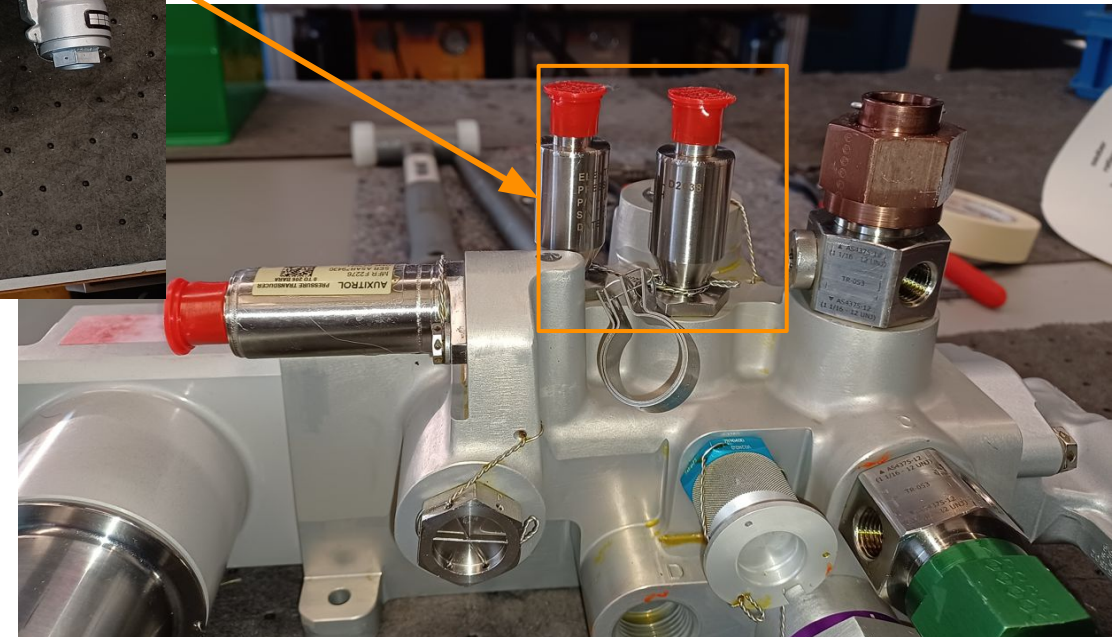
- Hydraulics:
FIN: 1151GN
FIN: 2151GN
FIN: 3151GN
- Flight controls:
FIN: 10CE1
FIN: 10CE2
FIN: 10CE3

→ Sub-component of the Green, Yellow and Blue HP manifolds

- Blue EMP pressure:
FIN: 2074GJ
- Downstream of Blue EMP



**Pressure switches P/N:
450-2-3100-00**



Supplied by Elektro-Metall (EME)

EME
Elektro-Metall

AIRBUS
[Airbus Amber]

CEO/NEO Pressure switches

Reliability figures

CONFIRMED REMOVALS OF P/N 450-2-3100-00

A320Fam worldwide

Over the last two years on all SA fleet:

- 118 removals (99 OI)
- Reliability still correct

Number of NFF unknown

Airbus follow-up on A/C less than 2YE:

- 21 removals reported (18 OI)
- 9 on aircraft less than 3MO

Maintenance and troubleshooting recommendations

Pressure transmitter will indicate that pressure is available:

- Swapping of the faulty switch with 2074GJ (GO) to use MMEL and minimize OI impact

CEO/NEO Pressure switches

FAIR 23.0238
FAIR 24.0034

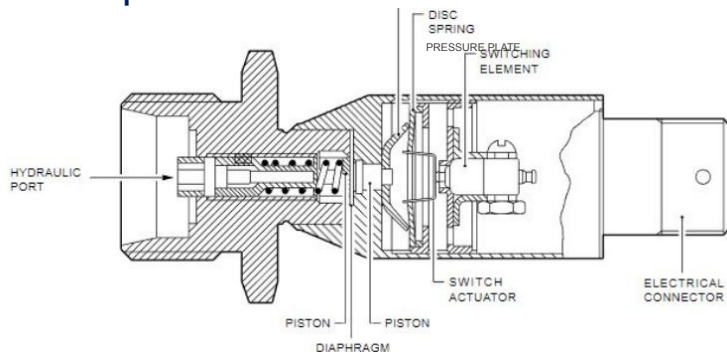
Known failure modes

- Switch contamination by hydraulic fluid
- Insulation test failure

Pressure switches are not often sent back for a deep investigation

- in-service experience limited
- deep investigation reports not available

Actions launch



- Regular meetings with EME to have some switches investigated and identify common failure modes and way forward
- Continuously monitor new cases → FAIR participation is key → **send switch to EME for investigation**
- With regards to the number of Single Aisle in operation + considering that each aircraft is equipped with 7 switches 450-2-3100-00 → Reliability of switches very high (MTBUR 1.000.000 vs GMTBUR 100.000)



BSCU/ABCU NFF

Marius OLSZEWSKI / Katia ROULLET

AIRBUS

BSCU No Fault Rate (NFF)

Background on HVN

- 25 removals between Jan23-Feb25
- BSCU NFF rate still found high (showing a NFF rate of 50%)
- Latest recommended SB/VSIL embodiment is in progress
 - E21327-32-3409 (BSCU L4.10 std): 100%
 - E21327-32-3433 (BSCU Shopzone code): Common practice (At shop discretion)
 - E21327-32-3479 (ATEC test program improvement): Applicable during return

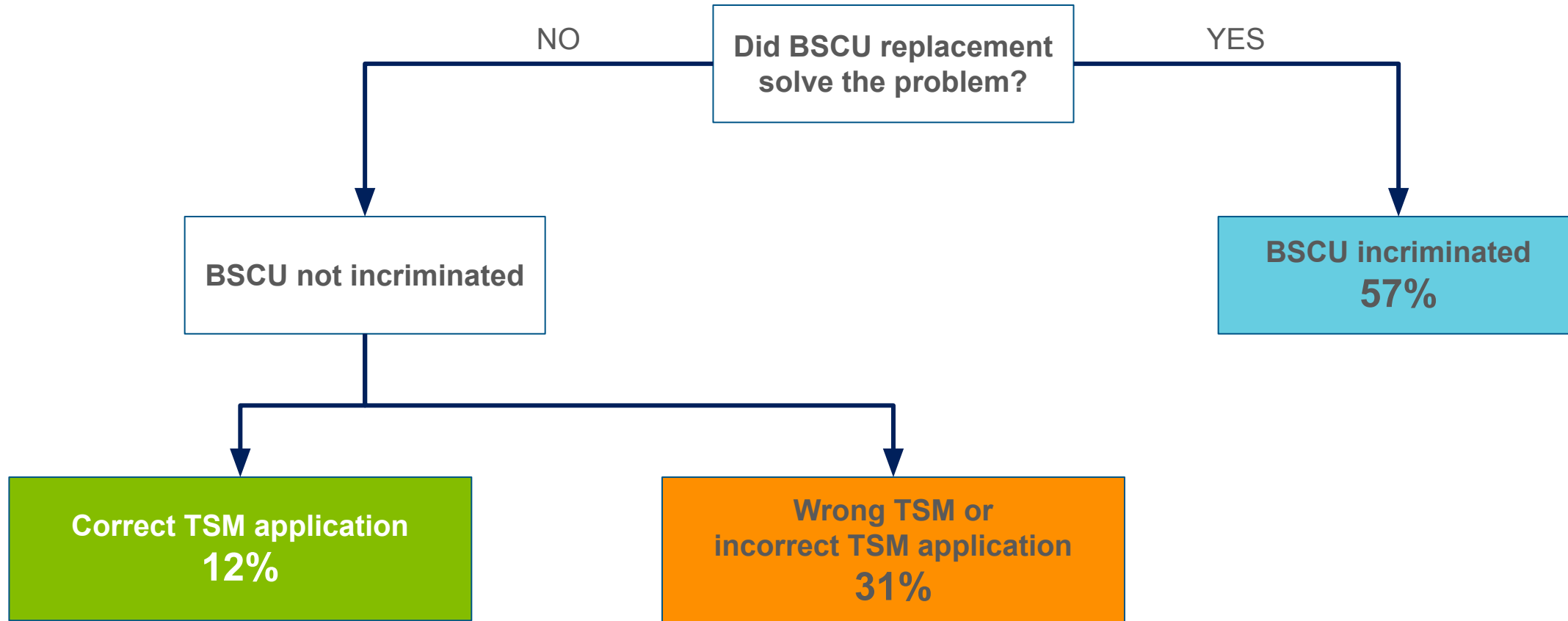
- N04971 BRAKES ALTN BRK FAULT / BRAKE N/WS MINOR FAULT (2x) leading to replacement of 2 BSCU within 1 Month - 1st NFF, 2FF
- N05340 BRAKES SYS 1 FAULT and BSCU SYS 2 FAULT leading to 2 BSCU replacements within 2 weeks - 1st FF, 2nd NFF
- N08589 BRAKE SYS 1 FAULT and BSCU SYS 2 INOP leading to 2 BSCU replacements within 2 months - 2x NFF

Background on BSCU NFF

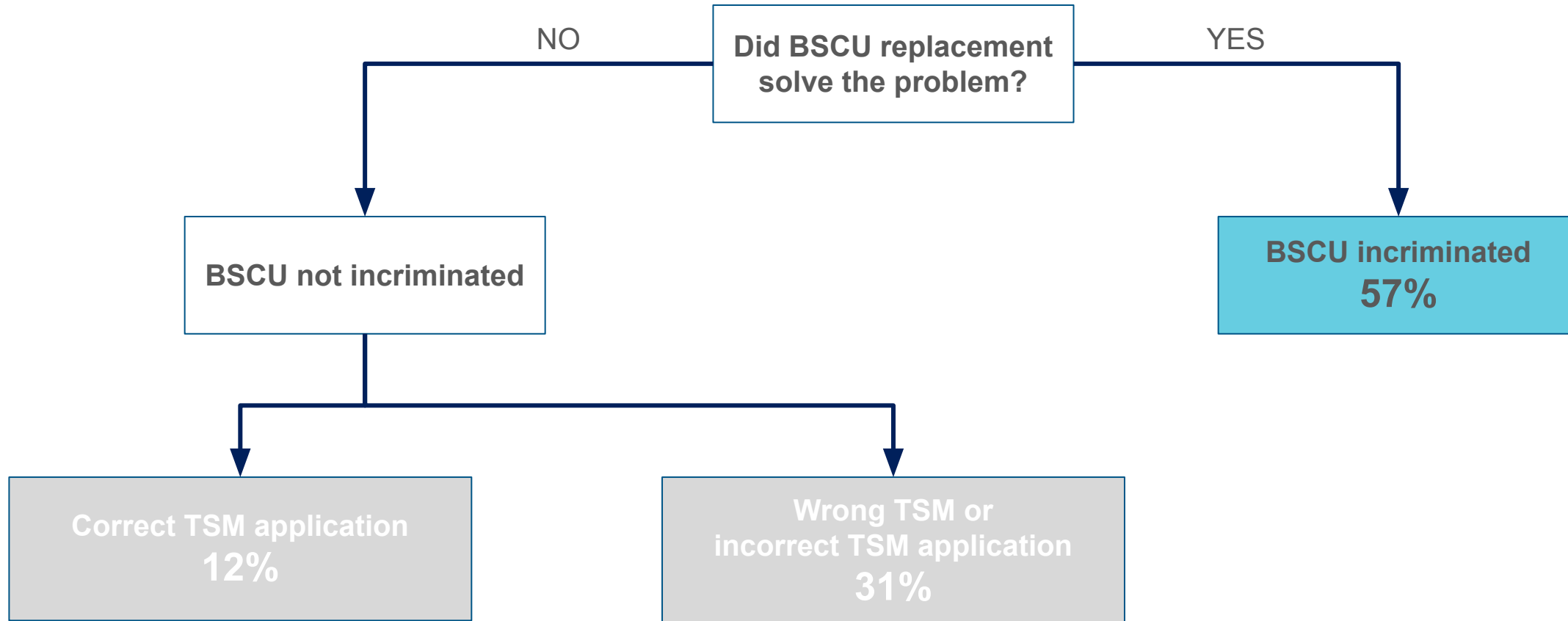
TFU 32.42.00012

- 2010 - 2017: BSCU NFF study and actions launched to improve BSCU NFF rate from 85% to 57%
 - ✓ Airbus documentation (TSM) improvement
 - ✓ Introduction of BSCU L4.10 software standard
 - ✓ Shop test modification to provide explanations and advices on BSCU shopzone faults (VSIL E21327-32-3433)
- 2019 - 2022: new BSCU NFF study launched to identify main causes

2019-2022 study - Primary causes of BSCU NFF



2019-2022 study - Primary causes of BSCU NFF

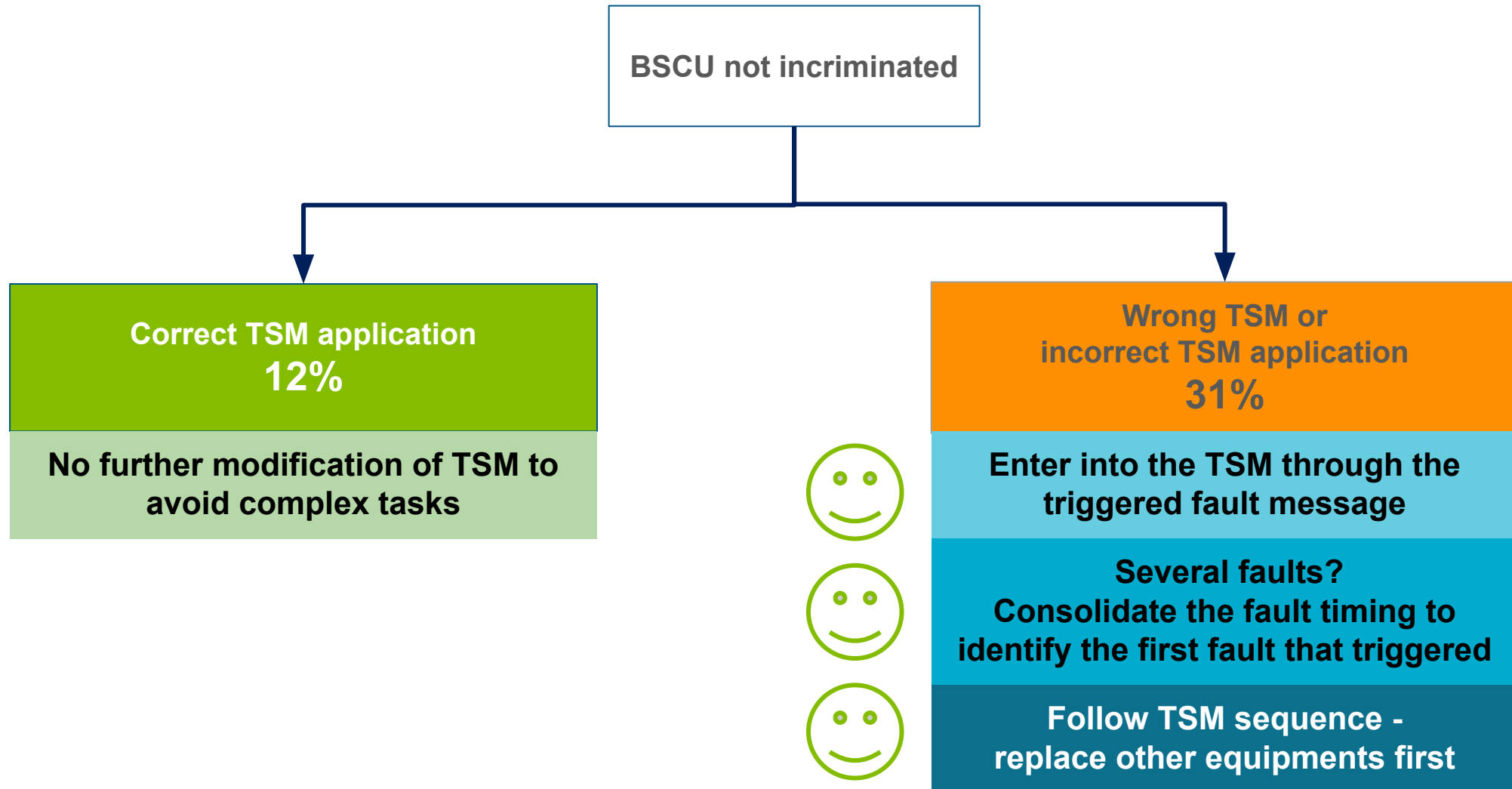


2019-2022 study - Primary causes of BSCU NFF

2nd step - Analysis of fault codes

Category	Comment	Mitigation / recommendation
Shop Zone 13%	No single root cause and solution identified.	Safran LS released in 2017 the Service Letter E21327-32-3433, providing maintenance recommendation in case of shop-zone faults (swap of cards). Common practice
Handwheel monitoring (FC 760) 13%	Discrepancy COM/MON on Steering Handwheel angle acquisition.	ATEC test coverage upgrade (as part of VSB E21327-32-3479, published in March 2023) to improve the BSCU fault detectability. Applicable during return
BPTU monitoring (FC 160 or 161) 8%	Discrepancy COM/MON on BTPU pedal deflection signal for the left or right pedal angle acquisition.	
Tachometer signal monitoring (FC 850) 8%	Rupture of tachometer drive shaft or loss of tachometer signal. <i>The BSCU is the 5th equipment to be replaced as per dedicated TSM task, which is usually done in parallel of a wiring check/correction.</i>	No dedicated recommendation required.
Various faults with less than 4% occurrence	Represents a total of 34 different fault codes.	No dedicated recommendation identified.

2019-2022 study- Primary causes of BSCU NFF



2016 recommendations



Apply the reset procedure for BRAKES SYS FAULT as per FCOM PRO-SUP-24

Consider embodiment of BSCU L4.10 with SB 32-1432 and SB 32-1442

Consider deactivating one BSCU System as per applicable MMEL Item 32-42-03 Braking/Steering Control Unit (BSCU) System 1 or 32-42-04 Braking/Steering Control Unit (BSCU) System 2

Ensure fault confirmation step of the TSM task is performed before removing equipment

Apply NFF policy

Refer to VSIL E21327-32-3433 to improve shops efficiency and facilitate BSCU repair

2022 recommendations



**BSCU shop zone fault: refer to VSIL E21327-32-3433 (swap of electronic cards)
Efficiency observed through review of several NFF BSCU cases**

**BSCU ATEC test program improvement: implement Safran Landing Systems new
ATEC test program through the VSB E21327-32-3479 (published in March 2023).
The test program includes various evolutions and extends the test coverage (e.g.
discrete input and output signals ...).**

Apply the right TSM task and to follow the procedure

Apply NFF policy and refer to VSIL E21327-32-3517 for checklist

BSCU No Fault Rate (NFF)

Background on HVN

- 25 removals between Jan23-Feb25
- BSCU NFF rate still found high (showing a NFF rate of 50%)
- Latest recommended SB/VSIL embodiment is in progress
 - E21327-32-3409 (BSCU L4.10 std): 100%
 - E21327-32-3433 (BSCU Shopzone code): Common practice (At shop discretion)
 - E21327-32-3479 (ATEC test program improvement): Applicable during return

- N04971 BRAKES ALTN BRK FAULT / BRAKE N/WS MINOR FAULT (2x) leading to replacement of 2 BSCU within 1 Month - 1st NFF, 2FF
- N05340 BRAKES SYS 1 FAULT and BSCU SYS 2 FAULT leading to 2 BSCU replacements within 2 weeks - 1st FF, 2nd NFF
- N08589 BRAKE SYS 1 FAULT and BSCU SYS 2 INOP leading to 2 BSCU replacements within 2 months - 2x NFF

Background on BSCU NFF

TFU 32.42.00012

- 2010 - 2017: BSCU NFF study and actions launched to improve BSCU NFF rate from 85% to 57%
 - ✓ Airbus documentation (TSM) improvement
 - ✓ Introduction of BSCU L4.10 software standard
 - ✓ Shop test modification to provide explanations and advices on BSCU shopzone faults (VSIL E21327-32-3433)
- 2019 - 2022: new BSCU NFF study launched to identify main causes
 - 57%: BSCU removal solved the fault even if found NFF in shop
 - 31%: Wrong TSM or incorrect TSM application
 - 12%: Correct TSM task application
- 2022: new recommendations
 - NFF policy, ATEC test program improvement
- 2023-Q4: RFW closure following BSCU NFF rate re-assessment (BSCU NFF rate far from the RFW level)

Main fault codes responsible for the BSCU removals:

- 13% due to Shop Zone code,
- 13% due to Handwheel monitoring,
- 8% due to BPTU monitoring,
- 8% due to tachometer signal monitoring

BSCU No Fault Rate (NFF)

Available improvements / recommendations

- **BSCU removal allowed to solve the problem**
 - Shop Zone fault code ⇒ VSIL E21327-32-3433
 - Steering Handwheel fault code (FC 760) ⇒ VSB E21327-32-3479
 - BPTU fault code (FC 160/161)
- **Wrong TSM task or incorrect TSM task application**
 - Apply and follow the right TSM ⇒ At customer level
 - Provide BSCU reason for removal and details ⇒ Airbus NFF policy
⇒ VSIL E21327-32-3517
- **Right TSM task application**
 - TSM tasks improvement ⇒ 2012 TSM tasks update

Wayforward proposals

- Reinforce the need to apply VSIL E21327-32-3433 in shop (impact on BSCU NFF rate confirmed)
- Monitor the application of VSB E21327-32-3479 in shop and the impact on BSCU NFF rate
- Reinforce the need to follow Airbus NFF policy and VSIL E21327-32-3517
- Share details of BSCU NFF (reason for removal, fault code) with Airbus to help identifying main fault codes or TSM that lead to BSCU removals

Key takeaways

TFU 32.42.00012 closed
BSCU NFF rate improved

+ VSB/VSIL to be monitored

+ NFF policy

ABCU No Fault Rate (NFF)

Background on HVN A320Fam EMM fleet

- 10 ABCU removals (Jan23-Dec24)
- High rate of ABCU NFF compared to A320Fam EMM fleet
- 4 MSNs impacted (over >60 A320Fam fleet): 4971 (4 removals), 4826 (3 removals), 4737 & 5427 (1 removal each)
- Recurrent and intermittent BRAKES ALTN BRK FAULT and/or BRAKES N/WS MINOR FAULT ECAM warnings
- Recurrent troubleshooting through TSM TASK 32-43-00-810-833-A - Failure of the ABCU

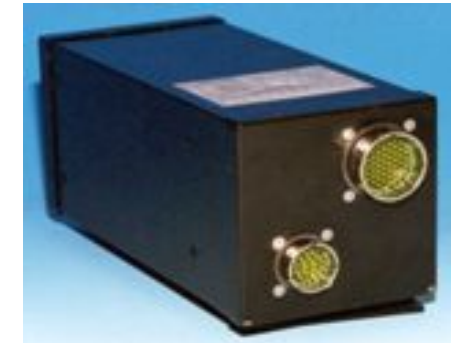
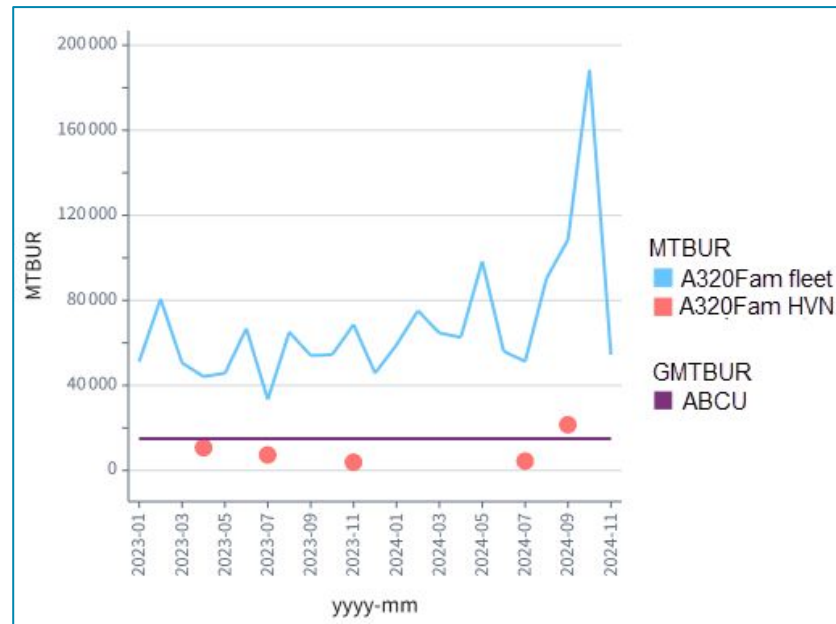
Background on A320Fam EMM fleet

Recurrent and intermittent ALT braking faults
ECAM Warnings:

- BRAKES ALTN BRK FAULT
- BRAKES ALTN L(R) RELEASED
- BRAKES-N/WS MINOR FAULT

Associated to Fault Messages:

- ABCU(95GG)
- BRK ALTN SERVVLV(xxGG)/ABCU(95GG)
- ABCU(95GG)/BRK ALTN SERVVLV(xxGG)



ABCU No Fault Rate (NFF)

TFU 92.00.00069

Root Cause

Alternate braking failures are most of the time due to:

- Corrosion on Alternate servo valves and pressure transducer connectors (water ingress).
- Wire damage due to chafing.

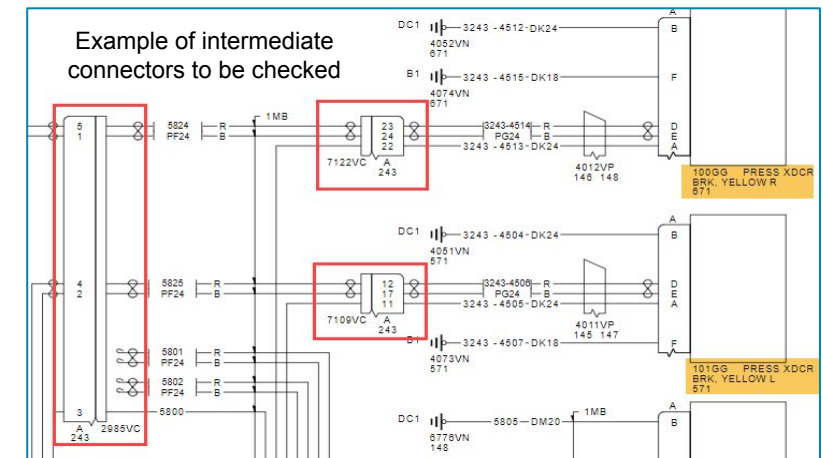
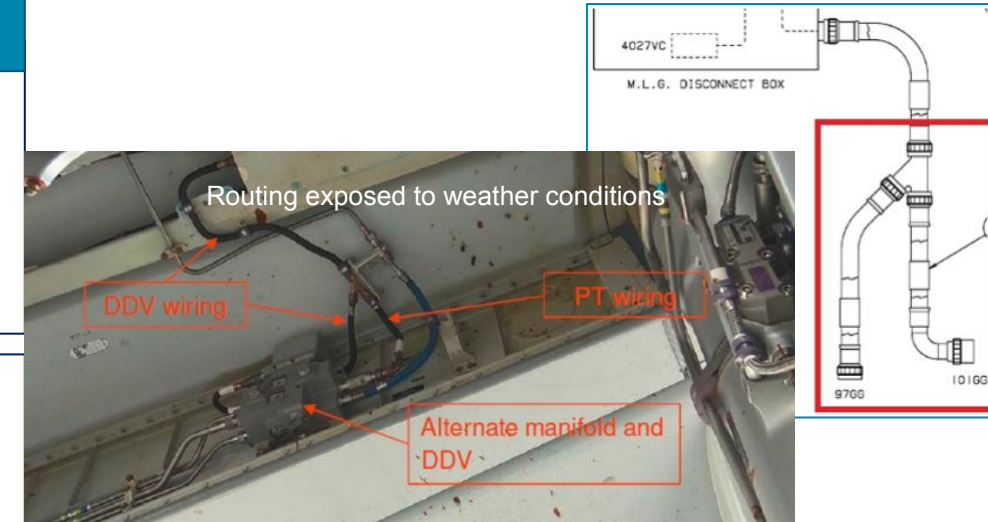
Mitigation

- Carry out the related TSM task in accordance with Fault Messages.
 - Swapping of DDV (97GG/98GG) or Pressure Transducer (101GG/100GG) has been added within TSM.
 - Identify any corrosion or water ingress issue at connector level.
 - Perform continuity and short circuit tests as per ESPM 20-52.
- pay attention to the state of **each** intermediate connectors (VC, VT, VN...)

Final Resolution

Electrical installation improvement at both Alternate Servovalves (97GG/98GG) and Pressure Transducers (100GG/101GG) is currently under review.

→ Solution is planned for **Q4 2025**





VHF, DME, ATC Antenna Erosion & Corrosion

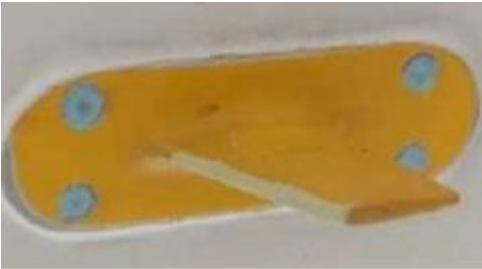
Charles MERIC & Amélie SAMUEL

AIRBUS

Problem Statement

- **A320** Family equipped with **ATC/DME Antennas P/N: 2439-89-xx** & **VHF Antennas P/N: 2438-89-xx** (Supplier: Thales Cobham) faced **erosion** and **corrosion** issues
- Findings have been reported from Airlines, corrosion issues could lead to loss of the VHF antenna blade **PDA (Part Departed Aircraft)** and could lead to AOG
- **RFW “D1502/23”** & **TFU 23.12.00024** were opened to address the **erosion** and **corrosion** issues

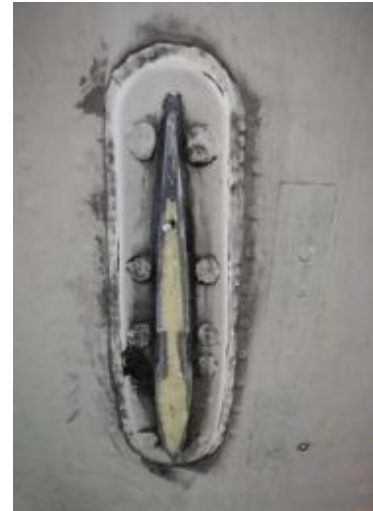
ATC/DME antennas eroded



ATC/DME antenna corroded



VHF antenna corroded



VHF antenna corroded



- Root Cause of Erosion:

- Thickness of the paint
- Paint component

- Root Cause of Corrosion:

- Lightning strike
- Material
- Coating

- Mitigations:

The following tasks give a permitted **tolerance periods** in relation to the types of damage of the **VHF & ATC/DME antennas** caused by the exposure to the environmental conditions:

- AMM TASK 23-12-11-200-801-A “Inspection of the **VHF Antenna**” (A320 Family).
- AMM TASK 34-51-11-200-801 “inspection of the **DME antenna**” (A320 family)
- AMM TASK 34-52-11-750-801-A “inspection of the **ATC antenna**” (A320 family)

Note:

- **Paint / repaint of different antenna is not allowed to avoid overcoat, masquing of erosion, ...**

In order to **mitigate** the issue related VHF & ATC/DME Antenna’s leading edge **erosion**, operators may install **an anti-erosion tape** when installing new antenna. This task is mentioned in the following AMM tasks:

- Ref to AMM TASK 23-12-11-400-001/2-A – “**Installation of the VHF Antennas**” (A320 Family) → [FIN: 4RC1, 4RC2]
- Ref to AMM TASK 23-12-11-400-002-A – “**Installation of the VHF Antennas**” (A320 Family) → [FIN: 4RC3]
- Ref to AMM 34-52-11-400-001-A – “**Installation of the ATC Antenna**” → [FIN: 7SH1, 7SH2, 7SH3, 7SH4]
- Ref to AMM 34-51-11-400-002/1A – “**Installation of the DME Antenna**” → [FIN: 3SD1, 3SD2]

- First **quick solution** to **slow-down corrosion** application of resistant primer P27-CF on aluminum protected with Surtec 650.
- **Final corrosion/erosion solution** with Corrosion and Erosion, is under study.

→ First quick solution (VHF):

- Corrosion resistant primer P27CF on aluminum protected with Surtec 650 added under the existing layers

BEFORE (2438-89-00 Amdt B)	AFTER (2438-89-00 Amdt D)
Primer P63 + A1000 polyurethane finish paint	Primer P27CF + Primer P63 + A1000 polyurethane finish paint

NOTE:

- ★ Any color can be chosen as long as it is A1000 polyurethane finish paint
- ★ Availability since July 2024 for **VHF antenna P/N: 2438-89-xx Amdt D**. First S/N available >= 60 000
- ★ For any lead time, order, *please contact Thales @: cac.customersupport@cobham.com* and ref to COBHAM SIL: 2438_2439-23-001

→ Final solution:

- “**Corrosion and erosion**” is under study (Discussion on material, paint, ...)



A320Fam CEO ATA 36 OI

Peter NG

AIRBUS

Background: CEO ENG bleed fault due to HPV, TLT, PRV, OPV

After reviewing the HVN Operational Interruption in Skywise between DEC-2023 to DEC-2024, 5 OI are linked to "**AIR ENG BLEED FAULT**" :

- 1 case leading to Sense Line #5 replacement
- 1 case leading to precooler & TCT replacement without clear root cause identified
- 1 case leading to precooler, unknown sense line, O-ring of TCT and O-ring of TLT replacement without clear root cause identified
- 1 case leading to OPV and PRV replacement without clear root cause identified
- 1 case leading to OPV, HPV and TLT without clear root cause identified

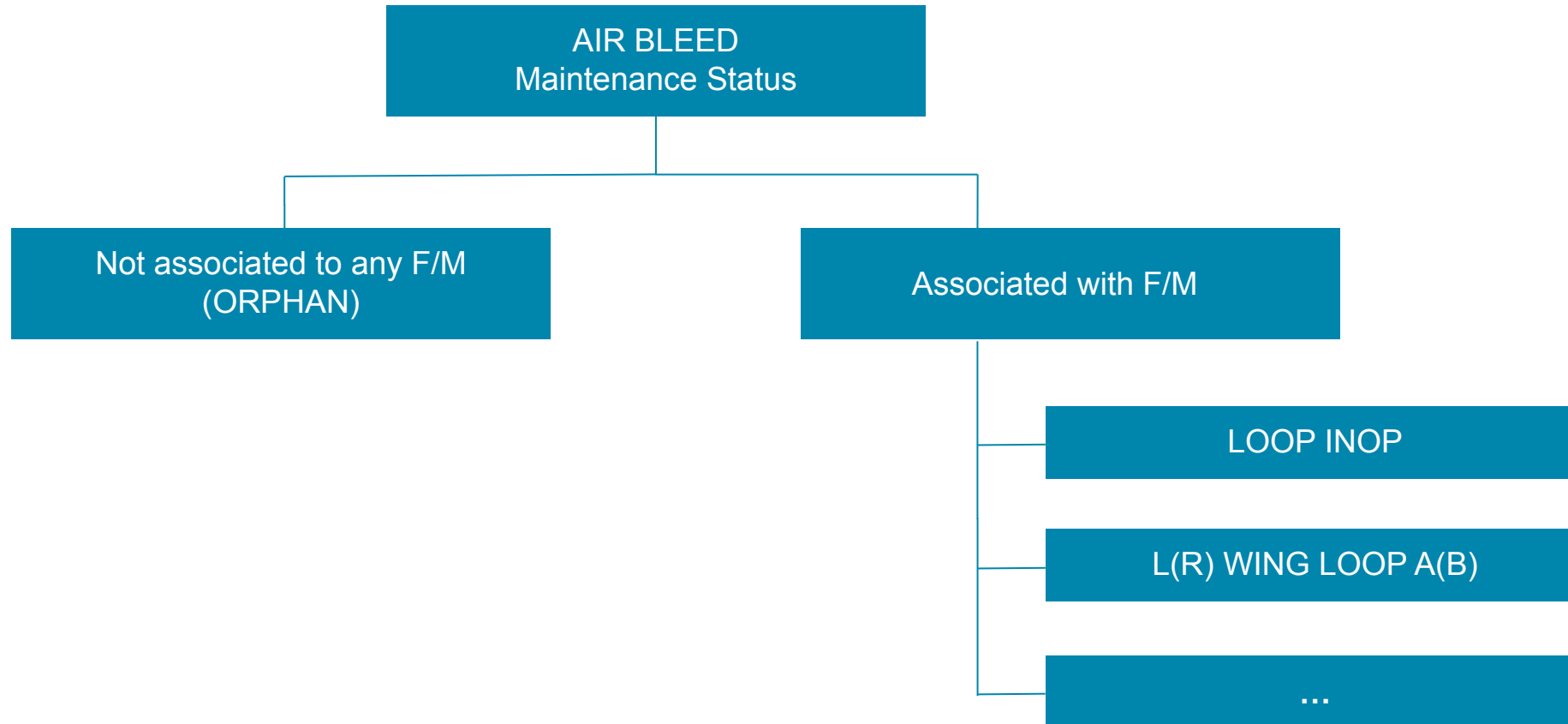
In addition, after reviewing the OI send by HVN, the following topic have been identified:

- **AIR BLEED Maintenance Status**, aligned with TFU 36.22.00.032
- **AIR ENG BLEED NOT CLSD**, TFU 36.11.00.085

A320FAM ceo - AIR BLEED Maintenance Status

DESCRIPTION:

ISI 36.22.00047



A320FAM ceo - AIR BLEED Maintenance Status

OPERATIONAL INFORMATION:

- If **“AIR BLEED” Maintenance Status** associated with **LOOP INOP F/M, without LEAK E/W**:
 - Dispatch allowed under MMEL **36-00-01B** only if RH Pylon leak detection loop affected
 - Dispatch allowed under MMEL **36-00-01C** only if LH Pylon and/or APU leak detection loop affected
- If **“AIR BLEED” Maintenance Status** associated with **“AIR APU/ENG LEAK DET FAULT” E/W**:
 - “AIR ENG 1(2) LEAK DET FAULT”: Dispatch allowed under MMEL 36-22-01
 - “AIR APU LEAK DET FAULT”: Dispatch allowed under MMEL 36-22-03
- If **“AIR BLEED” Maintenance Status** associated with **“AIR L(R) WING LEAK DET FAULT” E/W**:
 - No dispatch allowed: **NO GO**
- If **Orphan “AIR BLEED” Maintenance Status**
 - Dispatch allowed under MMEL 36-00-01A **only if no BMC fault or no LH/RH Pylon and/or APU leak detection loop affected**

A320FAM ceo - AIR BLEED Maintenance Status

ISI 36.22.00047
TFU 36.22.00.032

ASSOCIATED WITH OHDS LOOP INOP:

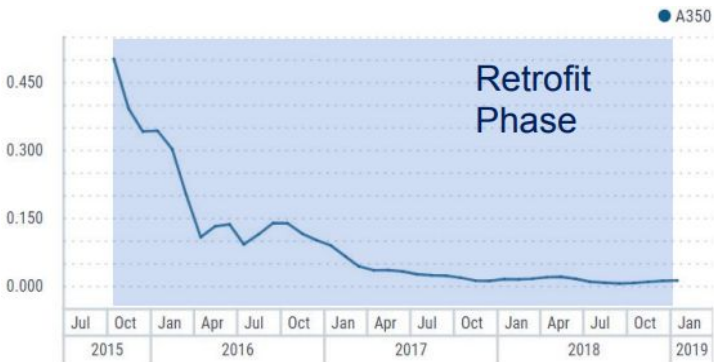
- Introduction of sensing element **FIN 73HF, 74HF, 78HF, 79HF connection re-design** at the wall panel to prevent sensing element breakage due to vibrations, available:
 - In production since **JAN-2007**
 - For In-Service aircraft as per **SB 36-1059** since **OCT-2007**
- Further enhancement of the **pylon leak detection monitoring**:
 - In case of pylon leak detection failure (Pylon loop INOP, BMC INOP or BMC failure), maintenance messages will be reclassified **Class I instead of Class II**
 - At FWC level, corresponding E/W **“AIR ENG 1(2) LEAK DET FAULT”** and **“BLEED MONIT SYS 1(2) FAULT”** are introduced, instead of Maintenance Status **“AIR BLEED”**.
 - The benefit is to identify the appropriate MEL item to be applied and the appropriate troubleshooting to perform
 - In production since **MAR-2006**
 - For In-Service aircraft as per **SB 36-1062** since **JUL-2008**

A320FAM ceo - AIR BLEED Maintenance Status

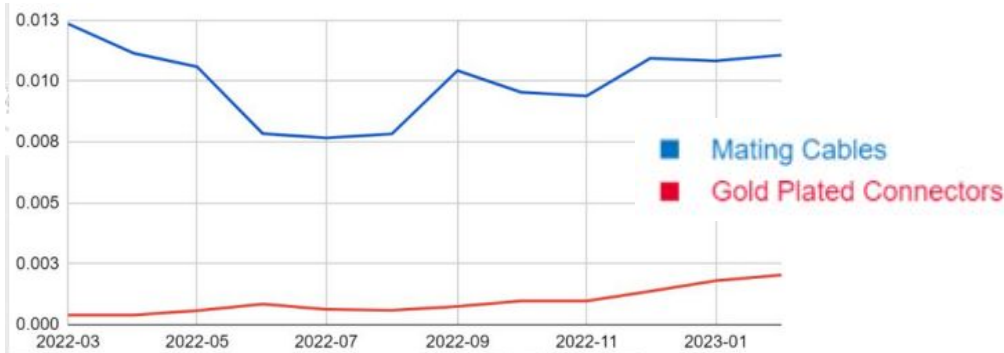
ASSOCIATED WITH OHDS LOOP INOP:

ISI 36.22.00047
TFU 36.22.00.032

- Introduction of new interconnecting cables with **gold plated connectors**:
 - For **90HF, 91HF, 92HF, 93HF** location - Length increase & last installation drawing improvement:
 - In Production available since **AUG-2018**
 - For In-Service aircraft as per **SB 36-1081 (JUN-2019)**
 - For **LH(RH) Wing Loops** (other location), **Pylon Loops** and **APU Loop** as per **SB “On-Demand” process**
 - In-Service aircraft only:
 - Pylon Loop: **SB 36-1093**, refer ISI 36.22.00055
 - LH Wing Loop (other location): **SB 36-1094**, refer ISI 36.22.00056
 - RH Wing Loop (other location): **SB 36-1096**, refer ISI 36.22.00057
 - APU Loop: **Under creation**
OR
 - **As per IPC**



A350 - ATA 36-22 Operational Interruption Trend for 100 Take-Off



A320FAMneo - ATA 36-22 Operational Interruption Trend for 100 Take-Off

- Data available in A350 & A320FAMneo
- No data available on A320FAMceo due to introduction mainly as per IPC

ORPHAN “AIR BLEED” Maintenance Status (without F/M):

- Difficulty to identify the appropriate MEL item to be applied and the appropriate troubleshooting to perform
- Investigation allow to identify that Class II F/M stored in BMC are not visible due to internal logic and Flight Leg Report is not active during this period
- To isolate a potential system misbehaviour, perform EBAS BITE Test, as per AMM TASK 36-11-00-740*
 - TSM TASK 36-00-00-810-802-A (Incorrect AIR BLEED generation) updated accordingly since MAY-2022

ASSOCIATED WITH OTHER F/M:

- Associated troubleshooting should be performed, no recurrent issue has been, currently, observed in service

A320FAM ceo - AIR ENG BLEED NOT CLSD

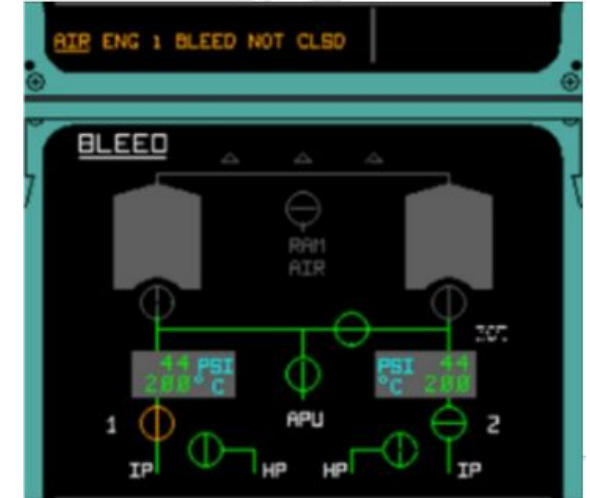
TFU 36.11.00.085

DESCRIPTION

- Since 2012, an increase trend of “AIR ENG 1(2) BLEED NOT CLSD” E/W were reported, during flight phase 2 or 9
 - Phenomenon mainly reported during Single Engine Taxi procedure
- “AIR ENG 1(2) BLEED NOT CLSD” E/W triggered if the PRV is not fully closed within 8 seconds when commanded in closed position

OPERATIONAL INFORMATION

- Dispatch allowed under MMEL **36-11-02** with PRV secured in close position and engine bleed system inoperative (MMEL 36-11-01)



ROOT CAUSE

- **PRV slow reactivity** to close when there is no pressure, exceeding the E/W triggering threshold
- PRV **micro-switch** failure

A320FAM ceo - AIR ENG BLEED NOT CLSD

TFU 36.11.00.085
ISI 36.11.00001

MITIGATION

- Operational recommendation: Select APU START – APU BLEED P/B ON – Wait 15s – Select ENG 1(2) MASTER LEVER OFF
- **BMC reset** procedure included into **FCOM** & **QRH**, since **JUN-2015**
- **Bleed Health Check** procedure to prevent unscheduled PRV removal

PERMANENT SOLUTION

- **BMC STD12** include logic changed to trigger E/W “AIR ENG 1(2) BLEED NOT CLSD”, only if **Pr > 10 psi** and the **confirmation time** was increased from 8 seconds to **12 seconds**:
 - In Production, available since **MAR-2018**
 - For In-Service aircraft as per **SB 36-1075 (MAY-2018)**
- PRV micro-switch improvement:
 - In Production available since **DEC-2016**
 - For In-Service aircraft, repair process as per **CRM 36-11-28RM**, available since **APR-2017**

ATTRACTIVE PRICE
NEGOTIATED as per VSIL
733901-36-003

A320FAM ceo - HPV Misbehavior

TFU 36.11.00.109
TFU 36.11.57.001
TFU 36.11.00.054
ISI 36.11.00001

DESCRIPTION

- Several “**AIR ENG 1(2) HP VALVE FAULT**” E/W associated with “**HP BLEED-V 4000HA1(2) OR SENSE LINE**” F/M reported:
 - Phenomenon mainly experienced during **flight phase 6** or on ground during flight phase 2 or 9
- “**AIR ENG 1(2) HP VALVE FAULT**” E/W is triggered when the HPV is abnormally closed during at least 15 seconds

OPERATIONAL INFORMATION

- Dispatch allowed under MMEL **36-11-07** with associated engine bleed system inoperative (MMEL 36-11-01)



ROOT CAUSE

- **60%** of E/W triggered on Ground when there is no bleed consumers (Pack P/B OFF)
- **40%** of E/W are mainly seen during transition from Cruise to Descent phase and on aircraft fitted with HPV solenoid

MITIGATION

- HPV Solenoid 11HA: Improved HPV solenoid **PN EV6016-1-00**, available since **FEB-2007**
- HPV Solenoid 4029KS: **Sense Line 1 (PRV, HPV, HPV Solenoid) cleaning** introduced into AMM/TSM since **NOV-2005**
- **Bleed Health Check** procedure to prevent unscheduled HPV removal

A320FAM ceo - HPV Misbehavior

TFU 36.11.00.109
TFU 36.11.57.001
TFU 36.11.00.054
ISI 36.11.00001

PERMANENT SOLUTION

- For **E/W triggered during Taxi Phase**, the Full Kit include a logic to control bleed valves in closed position when there is no bleed consumers:
 - BMC STD12, available:
 - In Production since MAR-2018
 - For In-Service aircraft as per SB 36-1075 (MAY-2018)
 - OPV G, available:
 - In Production since MAY-2017
 - For In-Service as per SB 36-1076 (OCT-2017)
 - Wiring between Pack P/B and BMC, available:
 - In Production since DEC-2017
 - For In-Service as per SB 36-1077 (ACSC Configuration - MAY-2018) or SB 36-1080 (PC/ZC Configuration - FEB-2020)
- For **E/W triggered during Cruise Phase**, AMM TASK 36-11-00-720-008* for Sense Line #1 leak check created & TSM TASK 36-11-00-810-837(838)-A update accordingly, available since AUG-2018 revision

A320FAM ceo - BMC STD12 & Full Kit

TFU 36.11.00.085
TFU 36.11.00.072
TFU 36.11.00.113
ISI 36.11 00109

DESCRIPTION:

- The Full Kit is recommended on aircraft with Pack P/B regularly in OFF position (Pack-Off/Take-Off, Pack under MMEL...)

The Full Kit is BMC STD12 + OPV G + Wiring between Pack P/B & BMC:

- **BMC STD12**, available:
 - In Production since **MAR-2018**
 - For In-Service aircraft as per **SB 36-1075 (MAY-2018)**
- **OPV G**, available:
 - In Production since **MAY-2017**
 - For In-Service as per **SB 36-1076 (OCT-2017)**
- **Wiring between Pack P/B and BMC**, available:
 - In Production since **DEC-2017**
 - For In-Service as per **SB 36-1077 (ACSC Configuration - MAY-2018)** or **SB 36-1080 (PC/ZC Configuration FEB-2020)**

ATTRACTIVE PRICE
NEGOTIATED as per VSIL
733901-36-003

ATTRACTIVE PRICE
NEGOTIATED as per VSIL
LS6740-36-02 & VSIL
LS6740-36-03

A320FAM ceo - BMC STD12 & Full Kit

TFU 36.11.00.085
TFU 36.11.00.072
TFU 36.11.00.113
ISI 36.11 00109

EFFICIENCY:

- **BMC STD12:**

- **“AIR ENG 1(2) BLEED NOT CLSD”** triggering logic improved, at engine shutdown, only when pressure is **above 10 psi** & PRV closure confirmation time increase from **8 seconds to 12 seconds**

- **Full Kit:**

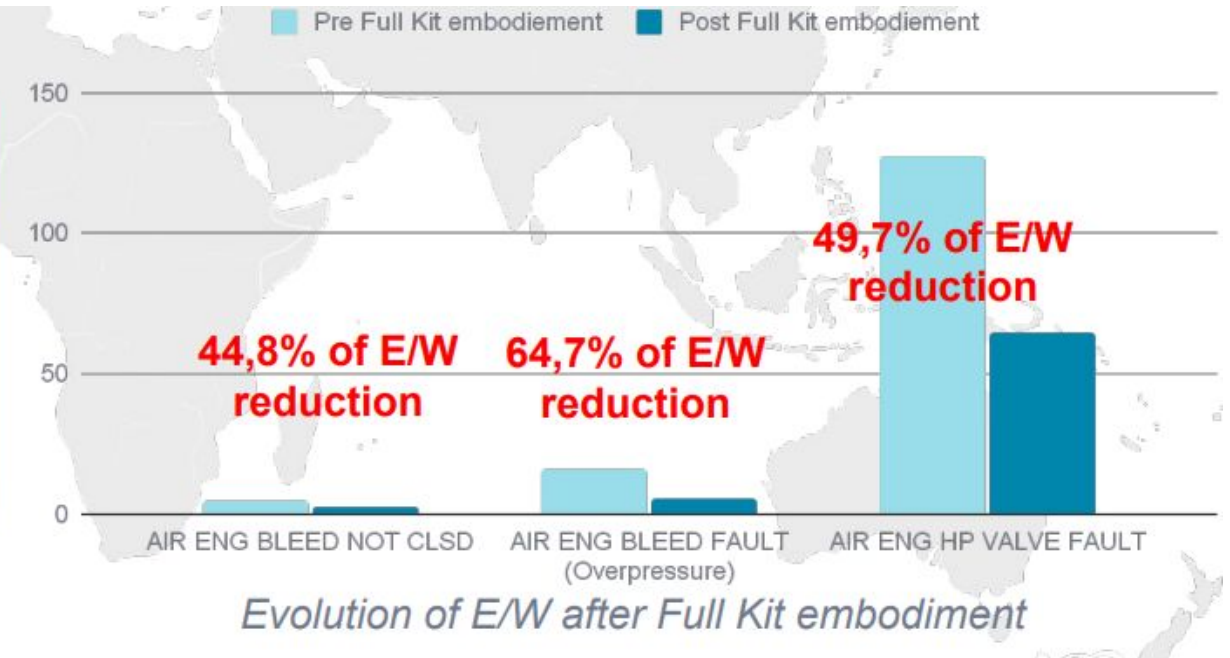
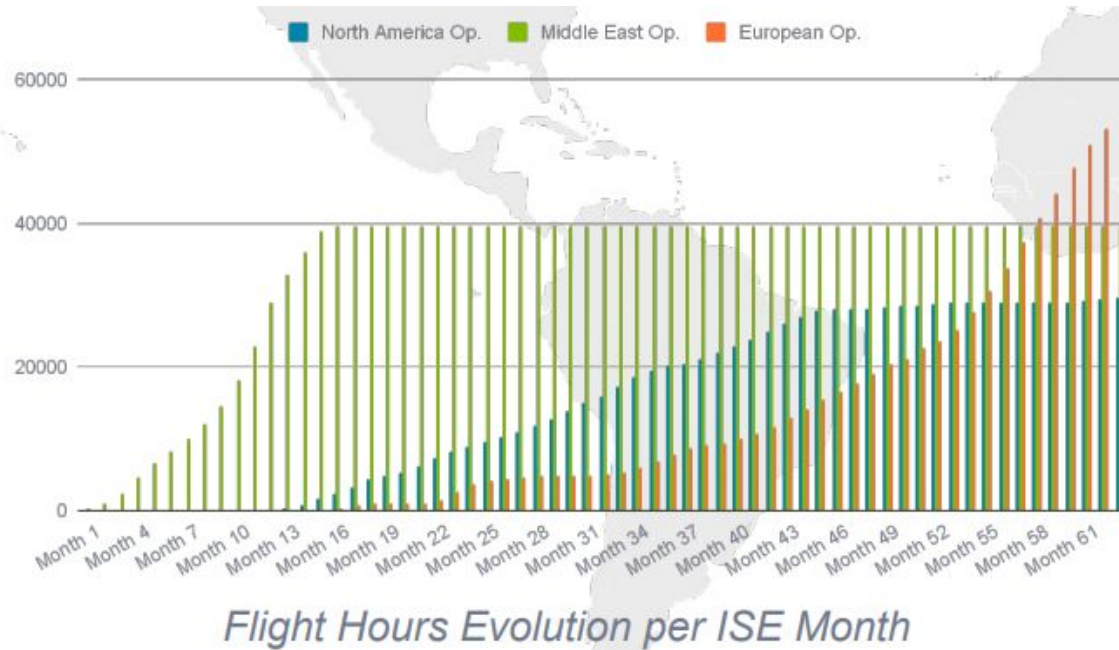
- **In absence of bleed consumers**, the **HPV** and **PRV** will be commanded to **close** position to:
 - **Reduce the overpressure events at Take-off** by accommodating the slow responsiveness of the PRV seen in-service
 - **Prevent high cycling** of PRV, HPV & TLT that occurs in-service when there is no bleed consumers

A320FAM ceo - BMC STD12 & Full Kit

IN-SERVICE EVALUATION:

- In-Service Evaluation ongoing since Q3-2018
- Conducted on 34 aircraft over the world, on recent and old aircraft
- Data as of JAN-2024

TFU 36.11.00.085
TFU 36.11.00.072
TFU 36.11.00.113
ISI 36.11 00109



Solution / Recommendation:

ISI 36.00.00021

	AIR BLEED Maintenance Status with/or OHDS Loop Inop	AIR ENG HP VALVE FAULT	AIR ENG BLEED NOT CLSD - PRV	TLT Misbehavior - 34D01*, 341E*, 341F01*	AIR BLEED Maintenance status - Orphan	Bleed Low Temperature - TCT	AIR ENG HP VALVE FAULT - Solenoid 11HA*	AIR ENG HP VALVE FAULT - Solenoid 4029KS	Bleed Overpressure at Take-Off	AIR ENG BLEED ABNORM PR - OPV 6740F01* & 6740G01*	AIR BLEED Maintenance Status - Other OHDS Defects	AIR ENG BLEED FAULT - Over Temperature
Mitigation/Solution												
- BMC STD12 + OPV G + Wiring		●	●	●					●			
<i>Note: Only if Pack-Off/Take-Off or PACK under MEL</i>			●			●			●			
- BMC STD12			●			●						●
- TLT PN 341F020000				●								●
- TCT PN 342B050000						●						●
- FAV PN 6730F010000												●
- Gold Plated Connectors	●											
- HPV Solenoid PN EV6016-100							●					
- EEC SCN 22									●			
- Sensing Elements Redesign	●								●			
- AMM/TSM Improvement		●		●	●	●		●	●	●	●	●
AIRBUS Complementary solutions												
- AIRBUS Bleed Heath Check		●	●			●			●			●
- Skywise Predictive Maintenance		●	●	●		●						●
- OHDS Pre-Summer Check	●											
LIEBHERR Complementary solutions												
- LIEBHERR Bleed Health Check		●	●			●			●			●
- Pack-Off/Take-Off SP		●	●	●					●			
- Bleed Pressure Regulation SP									●	●		
- Bleed Temperature Regulation SP						●						●
- Overhaul Solutions SP		●	●	●		●			●			●
- Modernisation		●	●	●					●			●



PW1100 FAV Assembly Damage

Peter NG

AIRBUS

PW1100 FAV Assy Damages

TFU 36.11.00104

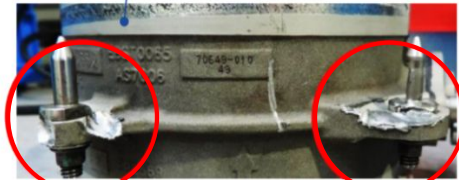
SB A320-54-1045 & MOD 164147

Description
Reports received from A320neo operators of damaged FAV in POST improved structure in pylon as per the following SB/MOD: • SB A320-54-1045 & MOD 164147 for PW engines • SB A320-54-1044 & MOD 163328 for CFM engines - It may lead to E/W AIR ENG1(2) BLEED FAULT or AIR ENG1(2) BLEED LO TEMP on the affected Engine Bleed System. - FAV PN 70649A010001 (PW) & 70649A010002 (CFM) found damaged leading to parts replacement

N08501	VN-A616	Fully embodied
N08545	VN-A617	
N08589	VN-A618	
N08649	VN-A620	
N08669	VN-A501	
N08678	VN-A619	
N08699	VN-A621	
N08703	VN-A622	
N08729	VN-A623	
N08747	VN-A624	
N08753	VN-A625	N/A
N08769	VN-A503	
N08924	VN-A505	
N08985	VN-A506	
N09051	VN-A507	
N09108	VN-A508	
N09161	VN-A509	
N09241	VN-A510	
N09242	VN-A511	
N09294	VN-A512	
N12148	VN-A513	

Fault Observation
Due to this issue on cold flow path, several symptoms can be experienced on aircraft: • Over bleed temperature triggers E/W "ENG1(2) BLEED FAULT" with "FAN AIR-V (9HAX) /SENSE LINE / PRECOOLER". • Low bleed temperature with WAI ON (Wing anti-ice) triggers E/W "AIR ENG1(2) BLEED LO TEMP" with "FAN AIR-V (9HAX)" • Low bleed temperature with WAI OFF identified by crew observation in flight looking at the ECAM BLEED PAGE. • "AIR BLEED" Maintenance status: associated with the Class 2 maintenance message (MM) "FAN AIR-V (9HAX)" or "PRECOOLER (7150HMx) / FAN AIR-V (9HAX)" • Few reports of "AIR BLEED" Maintenance status without maintenance message (Orphan) in cruise Very marginally and potentially if both engine sides are affected by a low bleed temperature, operator can notice pack <u>temperature regulation issue</u> too.

FAV typical damages:



FAV pins damaged



FAV Rod with permanent deformations

Root Cause

Root Cause / Investigation

Investigation at System, Integration and Structure levels identified 2 main root causes:

- FAV & its installation subject to **Aero-acoustic resonance phenomena**
- Effect of system **installation variability** on pylon structure

Ground and Flight tests planned between Q4 2023 and Q2 2024 to support the investigation of potential solutions.

Mitigation

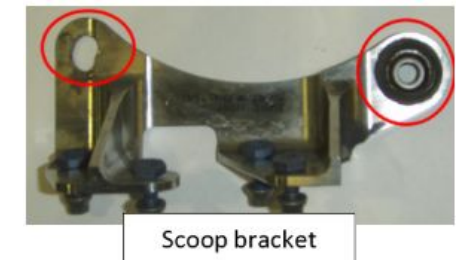
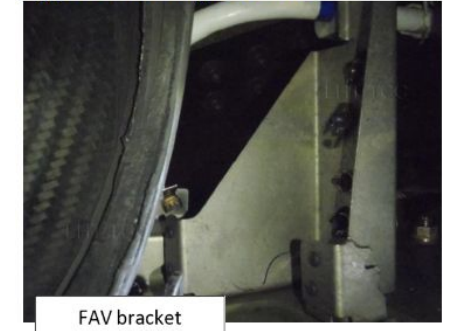
- **TSM update** released in August 2020 revision
- **AMM update for PW** released in November 2022 revision.
- ACMS report to detect low temperature scenarios with WAI OFF, linked to FAV damages
- Bleed temperature monitoring is available, through Skywise Predictive Maintenance, since June 2021

Summary

Terminating Action - Pending flight test campaign finalisation. Test currently performed on NEO Leap 1A test aircraft.



Typical findings found





CIDS DIR Obsolescence

Nurul Aida ABD. GHAFAR

AIRBUS

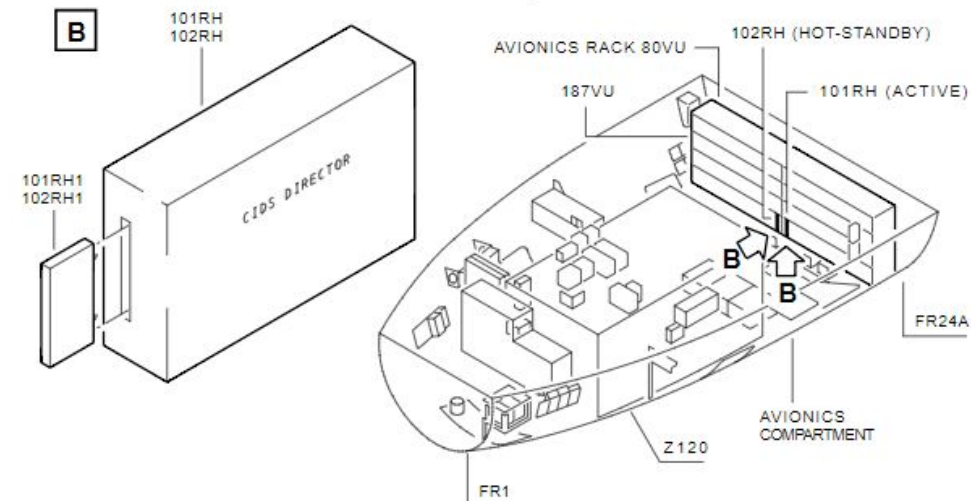
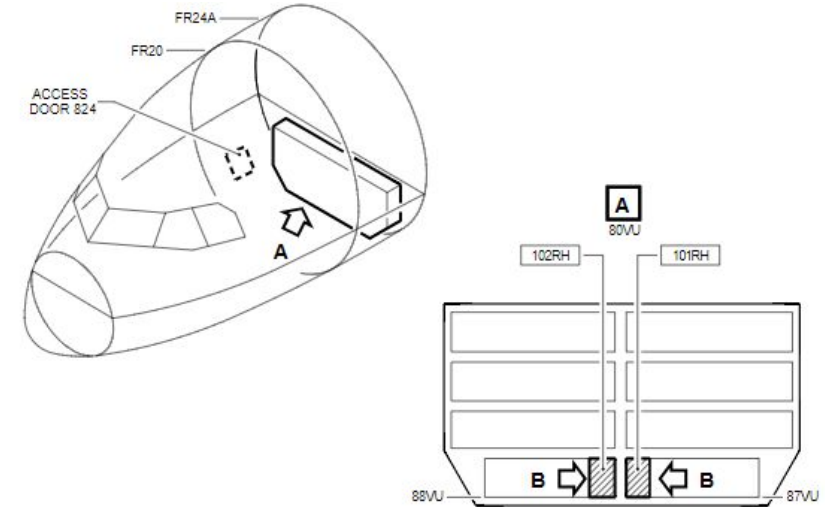
DESCRIPTION

CIDS Director are installed inside the avionic compartment at 80VU Rack

HVN fleet installed with CIDS Director;

- P/N: Z010H000521A / Z010H001521A (Classic CIDS DIR)
- **P/N: Z014H000231B**
- **P/N: Z014H000333B**
- P/N: Z014H010035B

**One aircraft installed with qty 02 CIDS DIR



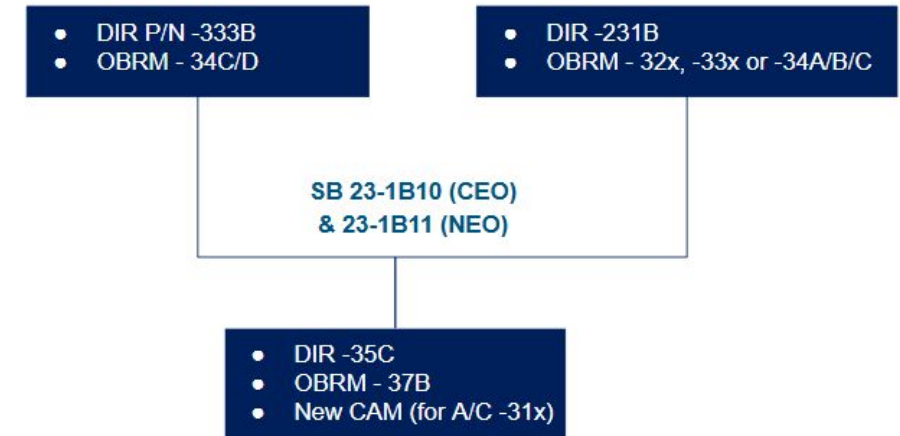
ANNOUNCEMENT

DIR P/N: Z014H000333B discontinuation

- EoP (End of Production) VSIL 23-128 published in July 2019
- Spares availability ensured until Sept 2021
- Repairs availability ensured until Sept 2024 (still possible)

DIR P/N: Z014H010035C replacement

- SB A320-23-1B10 and A320-23-1B11 published in Oct 2024
- Set of both obsolete DIR P/Ns (Z014H000333B) and OBRM can be exchanged with new DIR P/Ns (Z014H010035C) and new OBRM (Z064H000037B)
- Any information regarding the material cost and industry support terms for the equipment must be negotiated with the equipment supplier / Airbus Cabin Electronics as per TFU 23.73.00.108 attachment



* SB embodiment must be reported to Airbus

***Upgrade to DIR -35C from -333B as per SB 23-1B10/1B11 does not improve the CIDS reliability and is recommended only if spare is not available and/or repair is not possible.**

SOLUTION & RECOMMENDATION



No intermixed between DIR -333B and -35C



The replacement has to be in shipsets



SBs 23-1B10 & 23-1B11 are no subject of any Airbus Retrofit Campaign. Replacement for obsolescence reason must be negotiated with the equipment supplier.



Accomplishment of SBs 23-1B10 & 23-1B11 need to be reported to Airbus.



For A/C already fitted with DIR -35x, Airbus recommends to perform SB 23-1830 / 23-1831 to upgrade the OBRM to -37B.

A/C fitted with DIR -35x and software
-35x, -36x and -37A



Solution available (April 2024) via OBRM -37B
SB 23-1830 & 23-1831

Fix to:

- CIDS Maintenance Status
- FAP Freeze
- Smoke Lav Det Fault for -36C
- CIDS Autoreset for -36C
- **Smoke Aft & Fwd Det Fault for -36x**

Thank you

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